

Mathematics

Year 3

Teacher Guide

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YEAR 3

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TERM 1

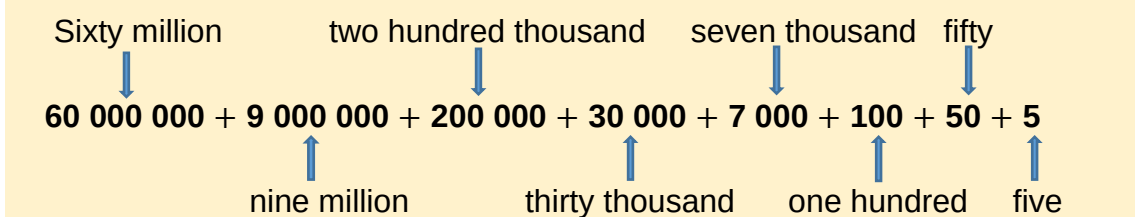
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UNIT 1

WHOLE NUMBERS: COUNTING, ORDERING, COMPARING, REPRESENTING AND PLACE VALUE

6.1 Count, recognise and represent numbers

Whole numbers are made up of parts that are called **place value parts**. For example, the number **69 237 155** is made up of the following parts and is said to be written in expanded notation.



1. Write the number symbols in expanded notation.
 - (a) Twenty-two million one hundred and thirty-seven thousand two hundred and ninety-one.
 $20\ 000\ 000 + 2\ 000\ 000 + 100\ 000 + 30\ 000 + 7\ 000 + 200 + 90 + 1$
 - (b) Twelve million seven hundred and one thousand two hundred and forty-five.
 $10\ 000\ 000 + 2\ 000\ 000 + 700\ 000 + 00\ 000 + 1\ 000 + 200 + 50 + 5$
 - (a) Forty-three million six hundred and twenty thousand five hundred and twenty-four.
 $40\ 000\ 000 + 3\ 000\ 000 + 600\ 000 + 20\ 000 + 0\ 000 + 500 + 20 + 4$
 - (b) Thirty-one million thirty-one thousand and thirty-one.
 $30\ 000\ 000 + 1\ 000\ 000 + 000\ 000 + 30\ 000 + 1\ 000 + 000 + 30 + 1$
 - (c) Seventy-nine million five hundred and forty-nine thousand four hundred and seventy-seven.
 $70\ 000\ 000 + 9\ 000\ 000 + 500\ 000 + 40\ 000 + 9\ 000 + 400 + 70 + 7$
2. The digit in the hundreds place in 12 837 465 is 4.
 - (a) Which digit is in the millions place? **2**
 - (b) Which digit is in the ten millions place? **1**
 - (c) Which digit is in the thousands place? **7**

3. Arrange the following numbers in **ascending order**.

(a) 76 896 576; 67 957 398; 73 374 376; 98 303 313

67 957 398; 73 374 376; 76 896 576; 98 303 313

(b) 57 695 836; 57 965 893; 57 596 963; 57 695 683

57 596 963; 57 695 683; 57 695 836; 57 965 893

4. Arrange the following numbers in **descending order**.

(a) 45 873 492; 45 783 492; 54 783 492; 54 873 492

54 873 492; 54 783 492; 45 873 492; 45 783 492

(b) 97 592 746; 79 592 746; 97 952 746; 79 529 746

97 952 746; 97 592 746; 79 592 746; 79 529 746

6.2 Rounding off to the nearest 1 000, 10 000 and 100 000

When rounding to the nearest 1 000, we look at the hundreds. Is the hundreds value less than, equal to or greater than 500? If less than 500, round down (the thousands value stays the same), if equal to 500 or greater than 500 round up.

When rounding to the nearest 10 000, we look at the thousands. Is the thousands value less than, equal to or greater than 5 000? If less than 5 000, round down (the ten thousands value stays the same), if equal to 5 000 or greater than 5 000 round up.

When rounding to the nearest 100 000, we look at the ten thousands. Is the ten thousands value less than, equal to or greater than 50 000? If less than 50 000, round down (the hundred thousand value stays the same), if equal to 50 000 or greater than 50 000 round up.

Example 1

3 254 rounded off to the nearest multiple of 1 000 will be 3 000 because

3 254 is between 3 000 and 4 000 and is closer to 3 000 than 4 000.

Example 2

87 943 rounded off to the nearest multiple of 10 000 will be 90 000 because the thousands digit is 7, therefore 87 943 is between 80 000 and 90 000 and is closer to 90 000 than 80 000.

If the thousands digit is 5; 6; 7; 8 or 9; we round off to the next multiple of ten thousand.

Example 3

456 943 rounded off to the nearest multiple of 100 000 will be 500 000 because the ten thousands digit is 5, therefore 456 943 is between 400 000 and 500 000 and is closer to 500 000 than 400 000.

If the ten thousands digit is 5; 6; 7; 8 or 9; we round off to the next multiple of hundred thousand.

1. Round off the following to the nearest multiple of 1 000 and 10 000.

No.	Original number	Nearest 1 000	Nearest 10 000
(a)	34 765	35 000	30 000
(b)	46 872	47 000	50 000
(c)	85 346	85 000	90 000
(d)	62 837	63 000	60 000
(e)	54 999	55 000	50 000

2. Round off the following to the nearest multiple of 100 000.

(a)	789 342	800 000
(b)	432 687	400 000
(c)	1 275 427	1 300 000
(d)	3 649 878	3 600 000
(e)	75 892	100 000

UNIT 2

WHOLE NUMBERS: ADDITION AND SUBTRACTION

2.1 Use a range of techniques to perform written calculations with whole numbers

Adding and subtracting using a calculator

Example

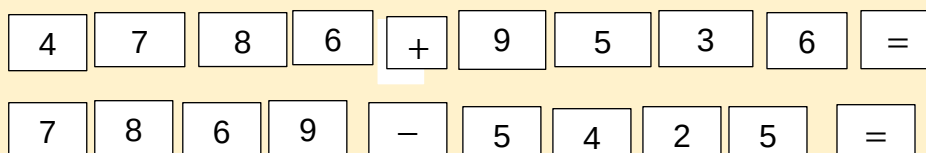
Calculate the following using a calculator

$$4\,786 + 9\,536$$

$$7\,869 - 5\,425$$



Calculator keystroke sequence



Adding using expanded column method

Example

$$57\,234\,494 + 14\,173\,245 =$$

$$\begin{array}{r} 57\,234\,494 \\ + 14\,173\,245 \\ \hline \end{array}$$

9	(4 + 5)
130	(90 + 40)
600	(400 + 200)
7 000	(4 000 + 3 000)
100 000	(30 000 + 70 000)
300 000	(200 000 + 100 000)
11 000 000	(7 000 000 + 4 000 000)
+ 60 000 000	(50 000 000 + 10 000 000)
= <u>71 407 739</u>	

Subtracting using expanded column method

Example

$$57\,274\,495 - 14\,133\,244 =$$

$$\begin{array}{r} 57\,274\,495 \\ - 14\,133\,244 \\ \hline \end{array}$$

1	(5 - 4)
50	(90 - 40)
200	(400 - 200)
1 000	(4 000 - 3 000)
40 000	(70 000 - 30 000)
100 000	(200 000 - 100 000)
3 000 000	(7 000 000 - 4 000 000)
+ 40 000 000	(50 000 000 - 10 000 000)
= 43 141 251	

Subtracting using expanded column method

Example 4

$$87\,274\,495 - 64\,136\,246 =$$

$$\begin{array}{r} 87\,274\,495 \\ - 64\,136\,246 \\ \hline \end{array}$$

9	(5 + 10 - 6)
40	(80 - 40)
200	(400 - 200)
8 000	(4 000 + 10 000 - 6 000)
30 000	(60 000 - 30 000)
100 000	(200 000 - 100 000)
3 000 000	(7 000 000 - 4 000 000)
+ 20 000 000	(80 000 000 - 60 000 000)
= 23 138 249	

1. Calculate each of the following using the column method and check your answers using a calculator.

- | | |
|-----------------------------------|--------------------|
| (a) $15\,987\,364 + 49\,790\,683$ | 65 778 047 |
| (b) $76\,263\,785 + 57\,953\,064$ | 134 216 849 |
| (c) $80\,487\,952 - 79\,956\,854$ | 531 098 |
| (d) $97\,654\,231 - 7\,493\,392$ | 90 160 839 |
| (e) $75\,643\,342 + 923\,789$ | 76 567 131 |

Addition and Subtraction in columns

Examples

Addition

$$\begin{array}{r} 81\,768\,421 + 76\,593\,094 = \\ 81\,768\,421 \\ + \underline{76\,593\,094} \\ \hline \mathbf{158\,361\,515} \end{array}$$

Subtraction

$$\begin{array}{r} 77\,473\,546 - 34\,181\,362 \\ 77\,473\,546 \\ - \underline{34\,181\,362} \\ \hline \mathbf{43\,292\,184} \end{array}$$

2. Calculate the following using the column method.

- | | |
|-----------------------------------|--------------------|
| (a) $54\,798\,456 - 17\,573\,245$ | 37 225 211 |
| (b) $26\,839\,632 - 11\,376\,489$ | 15 463 143 |
| (c) $83\,482\,563 - 49\,356\,278$ | 34 126 285 |
| (d) $13\,763\,264 + 78\,639\,652$ | 92 402 916 |
| (e) $91\,695\,481 + 21\,397\,447$ | 113 092 928 |

2.2 Problem solving

1. Calculate

- (a) A local municipality has already spent R12 102 436 of its housing budget of R85 514 559. How much money is still available?

$$R85\ 514\ 559 - R12\ 102\ 436 = R73\ 412\ 123$$

∴ R73 412 123 is still available.

- (a) 253 492 of the 292 388 voters in a district are male. How many of the voters are female?

$$292\ 388 - 253\ 492 = 38\ 896$$

∴ 38 896 voters are female

- (b) 253 476 new houses were built by the government in new settlements during a certain year. At the end of the year, there were 913 658 houses in the new settlements. How many houses were there at the beginning of that year?

$$913\ 658 - 253\ 476 = 660\ 182$$

∴ There were 660 182 houses at the beginning of that year

- (c) During a previous election there were 24 863 458 registered voters in a SA. During the next election there were 24 808 389 voters in the country. Did the number of voters increase or decrease? By how many?

$$24\ 863\ 458 - 24\ 808\ 389 = 55\ 069$$

∴ It decreased by 55 069

UNIT 3

WHOLE NUMBERS: MULTIPLICATION

3.1 Revise multiples and factors

A **factor** is a number that divides into another number without leaving the remainder.

Example 1

Factors of 100 are:

1, 2, 4, 5, 10, 20, 25, 50, 100

The factor pairs of 100 are:

- | | | | |
|----|-----------|-------|----------------------|
| 1. | 1 and 100 | | $1 \times 100 = 100$ |
| 2. | 2 and 50 | | $2 \times 50 = 100$ |
| 3. | 4 and 25 | Since | $4 \times 25 = 100$ |
| 4. | 5 and 20 | | $5 \times 20 = 100$ |
| 5. | 10 and 10 | | $10 \times 10 = 100$ |

1. Write the factors of the following numbers:

- (a) 200 $F_{200} = 1; 2; 4; 5; 10; 20; 40; 50; 100; 200$
- (b) 150 $F_{150} = 1; 2; 3; 5; 6; 10; 15; 25; 30; 50; 75; 150$
- (c) 376 $F_{376} = 1; 2; 4; 8; 47; 94; 188; 376$
- (d) 169 $F_{169} = 1; 13; 169$
- (e) 125 $F_{125} = 1; 5; 25; 125$

A **multiple** is a number that can be divided by another number without a remainder.

Multiples of 50 are 50; 100; 150; 200; 250; ...

2. Write the first five multiples of the following numbers:

- (a) 120 $M_{120} = 120; 240; 360; 480; 600$
- (b) 45 $M_{45} = 45; 90; 135; 180; 225$
- (c) 100 $M_{100} = 100; 200; 300; 400; 500$

- (d) 250 $M_{250} = 250; 500; 750; 1\ 000; 1\ 250$
 (e) 400 $M_{400} = 400; 800; 1\ 200; 1\ 600; 2\ 000$

3.2 Use a range of techniques to perform written calculations with whole numbers

Multiplication is when you take a number and add it to itself a number of times. It can be done in different ways, for example by breaking down numbers, using column method and using a calculator.

Multiplication by breaking down numbers

Example

$$\begin{aligned}
 &4231 \times 121 \\
 &= [(4000 \times 100) + (4000 \times 20) + (4000 \times 1)] + [(200 \times 100) + (200 \times 20) + (200 \times 1)] \\
 &\quad + [(30 \times 100) + (30 \times 20) + (30 \times 1)] + [(1 \times 100) + (1 \times 20) + (1 \times 1)] \\
 &= 400\ 000 + 80\ 000 + 4\ 000 + 20\ 000 + 4000 + 200 + 3\ 000 + 600 + 30 + 100 + 20 + 1 \\
 &= 400\ 000 + (80\ 000 + 20\ 000) + (4\ 000 + 4\ 000 + 3\ 000) + (200 + 600 + 100) + (30 + 20) + 1 \\
 &= 500\ 000 + 11\ 000 + 900 + 50 + 1 \\
 &= 511\ 951
 \end{aligned}$$

1. Multiply the following using breaking down method and check your answers using a calculator.

(a) 1 374 213

$$\begin{aligned}
 &= [(1000 \times 200) + (1000 \times 10) + (1000 \times 3)] + [(300 \times 200) + (300 \times 10) + (300 \times 3)] \\
 &\quad + [(70 \times 200) + (70 \times 10) + (70 \times 3)] + [(4 \times 200) + (4 \times 10) + (4 \times 3)] \\
 &= 200\ 000 + 10\ 000 + 3\ 000 + 60\ 000 + 3\ 000 + 900 + 14\ 000 + 700 + 210 + 800 + 40 + 12 \\
 &= 200\ 000 + (10\ 000 + 60\ 000 + 14\ 000) + (3\ 000 + 3\ 000) + (900 + 700 + 210 + 800) + (40 + 12) \\
 &= 200\ 000 + 84\ 000 + 6\ 000 + 2\ 610 + 52 \\
 &= 292\ 662
 \end{aligned}$$

(b) $1\,724 \times 142 = 244\,808$

(c) $2\,512 \times 327 = 821\,424$

(d) $3\,403 \times 154 = 524\,062$

(e) $4\,856 \times 309 = 1\,500\,504$

UNIT 4

WHOLE NUMBERS: DIVISION

4.1 Use a range of techniques to perform written calculations with whole numbers

Multiples and factors

Multiples of a number are the numbers obtained if the number is multiplied by natural numbers.

Factors of a number are numbers that divide exactly into a number and leave no remainder.

The number 42 can be obtained by calculating 7×6 .

We can say:

- 42 is the **product** of 6 and 7
- 42 is a **multiple** of 6
- 42 is also a multiple of 7
- 7 is a **factor** of 42
- 6 is also a factor of 42

We can also **express** (write) 42 as the product of two whole numbers, in other ways: $2 \times 21 = 42$; $3 \times 14 = 42$ and $1 \times 42 = 42$.

Note:

Multiples are bigger or equal to the number e.g. Multiples of 121: **121**; 242; 363; 484; 605; 726 ...

Factors are smaller or equal to the number e.g. Factors of 124: 1; 2; 3; 4; 31; 62 and **124**.

1. Write the first 5 multiples of the following numbers.

- | | |
|---------|--------------------------------------|
| (a) 34 | $M_{34} = 34; 68; 102; 136; 170$ |
| (b) 50 | $M_{50} = 50; 100; 150; 200; 250$ |
| (c) 122 | $M_{122} = 122; 244; 366; 488; 610$ |
| (d) 231 | $M_{231} = 231; 462; 693; 924; 1155$ |
| (e) 111 | $M_{111} = 111; 222; 333; 444; 555$ |

2. Write down the factors of:

(a) 24 $F_{24} = 1; 2; 3; 4; 6; 8; 12; 24$

(b) 36 $F_{36} = 1; 2; 3; 4; 9; 12; 18; 36$

(c) 440 $F_{440} = 1; 2; 4; 5; 8; 10; 11; 14; 44; 55; 88; 110; 220; 440$

(d) 546 $F_{546} = 1; 2; 3; 6; 7; 13; 14; 39; 42; 78; 91; 182; 273; 546$

(e) 729 $F_{729} = 1; 3; 9; 81; 243; 729$

3. Write down the multiples of 5 between 61 and 78

$= 65; 70; 75$

4. Write the factors of 64 between 3 and 17

$= 4; 16$

Division using a clue board method

When working with division, learners remember easily with multiplication facts for multiples of 10 and thereafter doubling and halving. Creating a clue board first will help as a reference to do calculations on division.

Example

$1\ 260 \div 28$

While they do not know the 28 times table, they do know 28×10 and how to use this to get multiples of 28×10 .

Learners find 28×5 by halving 28×10

Learners use doubling to find 28×2 ; 28×4 ;

Learners fill in other multiples as they need to use them.

Learners multiply and then subtract to calculate by approximation.

Clue board

$10 \times 28 = 280$
 $20 \times 28 = 560$
 $30 \times 28 = 840$
 $40 \times 28 = 1\ 120$

$5 \times 28 = 140$
 $4 \times 28 = 112$
 $6 \times 28 = 168$
 $2 \times 28 = 56$

Multiply to get an approximate answer	Subtract to find the difference
$40 \times 28 = 1\ 120$	$1260 - 1120 = 140$
$5 \times 28 = 140$	$140 - 140 = 0$

$1\ 260 \div 28 = 40 + 5 = 45$

Learners should check their calculations by multiplying:

$28 \times 45 = (28 \times 40) + (28 \times 5)$

$$= 1\,120 + 140$$

$$= 1\,260$$

5. Calculate by using a clue board

(a) $1\,176 \div 42$

Multiply to get an approximate answer	Subtract to find the difference
$20 \times 42 = 840$	$1\,176 - 840 = 336$
$6 \times 42 = 252$	$336 - 252 = 84$
$2 \times 42 = 84$	$84 - 84 = 0$

CLUE BOARD
$10 \times 42 = 420$
$20 \times 42 = 840$
$30 \times 42 = 1\,260$
$5 \times 42 = 210$
$4 \times 42 = 168$
$6 \times 42 = 252$
$2 \times 42 = 84$
$3 \times 42 = 126$

$$\therefore 1\,176 \div 42 = 20 + 6 + 2 = 28$$

(b) $3\,060 \div 36 = 85$

(c) $3\,072 \div 32 = 96$

(d) $2\,444 \div 188 = 13$

(e) $4\,205 \div 145 = 29$

(f) $6\,318 \div 234 = 27$

Long division method

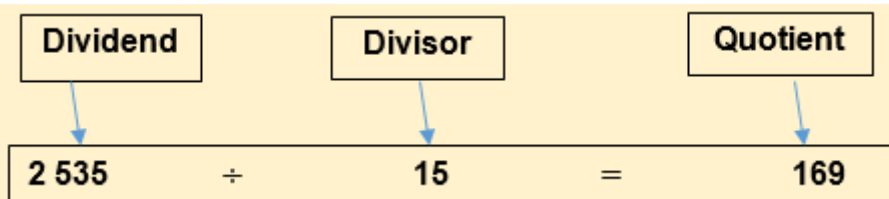
Long division is the mathematical method for dividing large numbers into smaller groups or parts. It helps to break down a problem into simple and easy steps.

There are terms that are associated with division:

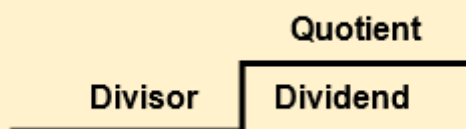
- **Divisor** is a number that divides another number (dividend)
- **Dividend** is a number to be divided by another number (divisor)
- **Quotient** is the answer you get when you are dividing
- **Remainder** is the number that is left after dividing

Examples

- $20 \div 3 = 6$ remainder 2
- $2\,535 \div 15 = 169$



The notation that is normally used for long division is:



When using long division method it is important to know multiples of the divisor and you can write them as your clue board as shown in the examples.

Example1

Use long division and a clue board to calculate $2\ 535 \div 15$

Steps to follow:

$$\begin{array}{r}
 169 \\
 15 \overline{) 2\ 535} \\
 \underline{- 1500} \\
 1035 \\
 \underline{- 90} \\
 135 \\
 \underline{- 135} \\
 0
 \end{array}$$

$\therefore 2\ 535 \div 15 = 169$

$15 \times 1 = 15$
$15 \times 2 = 30$
$15 \times 3 = 45$
$15 \times 4 = 60$
$15 \times 5 = 75$
$15 \times 6 = 90$
$15 \times 7 = 105$
$15 \times 8 = 120$
$15 \times 9 = 135$
$15 \times 10 = 150$

Example 2

Jerry wants to buy a TV game that costs R3 750. He washes cars in the neighbourhood and earns R56 for every car he washes. How many cars must he wash to earn enough money so that he can buy the TV game?

$$\begin{array}{r}
 66 \\
 56 \overline{) 3\ 750} \\
 \underline{- 336} \\
 390
 \end{array}$$

$56 \times 1 = 56$
$56 \times 2 = 112$
$56 \times 3 = 168$
$56 \times 4 = 224$
$56 \times 5 = 280$
$56 \times 6 = 336$
$56 \times 7 = 392$
$56 \times 8 = 448$

$$\begin{array}{r} - 336 \\ \hline 54 \text{ remainder} \end{array}$$

$$R3\ 750 \div 56 = 66 \text{ remainder } 54$$

$\therefore$ Jerry will have to wash 67 cars in order to have enough money to buy the TV game.

Note: It is easy to divide with the calculator. For example, to calculate $6\ 804 \div 324$, the keystroke sequence $6\ 804 \div 324 =$ gives 21.

1. Calculate using a method of your choice. Check your answers using a calculator.

(a) $6\ 804 \div 324$

Multiply to get an approximate answer	Subtract to find the difference
$20 \times 324 = 6\ 480$	$6\ 804 - 6\ 480 = 324$
$1 \times 324 = 324$	$324 - 324 = 0$

CLUE BOARD

$$\begin{aligned} 10 \times 324 &= 3\ 240 \\ 20 \times 324 &= 6\ 480 \\ 4 \times 324 &= 1\ 296 \\ 8 \times 324 &= 2\ 592 \\ 2 \times 324 &= 648 \\ 1 \times 324 &= 324 \end{aligned}$$

(b) R

$$\begin{array}{r} 234 \overline{) 6\ 318} \\ \underline{4\ 68} \\ 1\ 628 \\ \underline{1\ 628} \\ 0\ 000 \end{array}$$

$$\therefore 6\ 318 \div 234 = 27$$

CLUE BOARD

$$\begin{aligned} 234 \times 1 &= 234 \\ 234 \times 2 &= 468 \\ 234 \times 3 &= 702 \\ 234 \times 4 &= 936 \\ 234 \times 5 &= 1\ 170 \\ 234 \times 6 &= 1\ 404 \\ 234 \times 7 &= 1\ 638 \end{aligned}$$

- (c) $5\ 408 \div 13 = 416$
 (d) $7\ 353 \div 19 = 387$
 (e) $9\ 963 \div 123 = 81$

2. A businessman donates 9 396 boxes of fruits and vegetables to old age homes around the village. The boxes of fruits and vegetables are divided

equally between 54 old age homes. How many boxes of fruits and vegetables does each old age home get?

$9\,396 \div 54 = 174$ $\therefore$ each old age home gets 174 boxes of fruits and vegetables.

3. The school's drama club is putting on a play. Each club member has to sell 78 tickets. The members have to sell a total of 4 212 tickets. How many members does the drama club have?

$4\,212 \div 78 = 54$ $\therefore$ the drama club has 54 members.

4. The school has a budget of R4 600 to award learners on prize giving day. Each learner has to be awarded R44 prize. How many learners will be awarded?

$R4\,600 \div 44 = 104$ remainder R24 $\therefore$ 104 learners will be awarded

4.2 Perform multiple operations on whole numbers with or without brackets

Multiple operations require a guide that indicates the order of using the operations. To solve multiple operations:

- consider simplifying the brackets first
- then deal with multiplication and division in their order of appearance in the problem. [Work from left to right]
- addition and subtraction are dealt with last. [Work from left to right]

Examples

- $24 \div (3 \times 2 + 6) = 24 \div 12$
 $= 2$
- $150 + (25 \times 4) \div 5 = 150 + 100 \div 5$
 $= 150 + 20$
 $= 170$

1. Solve the following problems.

(a) $450 - (100 + 20 \div 2) \times 3 = 270$

(b) $2\,320 + (60 \times 2 \div 3)$ = 2 360

(c) $(4\,280 + 220) + 2 - (730 - 230)$ = 4 002

(d) $2\,896 \div 724 + 36 \div 6 \times 6$ = 40

(e) $43 \times 3 - 30 \div 3$ = 119

UNIT 5

RATIO AND RATE

Both **rate** and **ratio** are comparisons of two numbers. The main difference is that a **ratio** is a comparison of two numbers with the same unit, whereas, a **rate** is a comparison of two numbers with different units.

Ratio and rate is used for solving many everyday problems that involve comparing different numbers.

A ratio compares the size, or magnitude, of two numbers of the same kind.

4:5
4 boys to 5 girls



A ratio that compares quantities of **different types** (of measurement units) that are related to each other is called a rate.

R25 per kg



6.3 Ratio

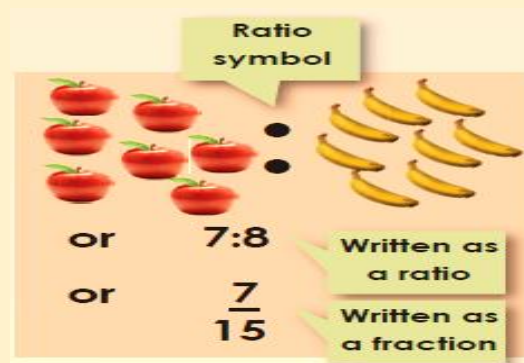
A ratio is a relationship between quantities of the same kind e.g. boys to girls as they are both human beings.

Ratio can be expressed as a fraction.

Examples

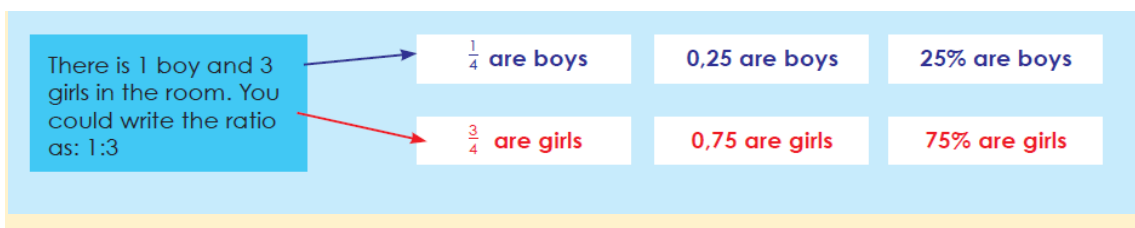
Write a ratio of apples to bananas as a fraction.

- Add the two parts to make it a whole i.e. $7 + 8 = 15$
- Write each part as part of a whole i.e. $\frac{7}{15}$ and $\frac{8}{15}$



When converting ratios to fractions they should be written in a simplified form and remain equivalent. To do this we need to ask what number can divide equally into the numerator and denominator of a fraction.

When converting fractions to percentages, we need to make use of equivalence to make the denominator to be 100.



Example

In a certain primary school there are 8 male teachers and 12 female teachers.

- What is the ratio of male teachers to female teachers?

8 males : 12 females (to simplify ÷ both sides by the common factor 4)

2 males : 3 females (always write ratio in simplified form)

- Write the ratio as a fraction and a percentage.

Fraction: Males = $\frac{8}{20} = \frac{8 \div 4}{20 \div 4} = \frac{2}{5}$

Females = $\frac{12}{20} = \frac{12 \div 4}{20 \div 4} = \frac{3}{5}$

Percentage:

Males = $\frac{8}{20} = \frac{2}{5} \times \frac{20}{20} = \frac{40}{100} = 40\%$

Females = $\frac{12}{20} = \frac{3}{5} \times \frac{20}{20} = \frac{60}{100} = 60\%$

- Write the following ratios as fractions.

(a) 3:4 = $\frac{3}{4}$

(b) 5:7 = $\frac{5}{7}$

(c) 6:8 = $\frac{3}{4}$

(d) 3:9 = $\frac{1}{3}$

(e) 1:2 = $\frac{1}{2}$

(f) 7:9 = $\frac{7}{9}$

- Write the following as percentages:

(a) 4:6 $\frac{4}{10} \times \frac{10}{10} = \frac{40}{100} = 40\%$ and $\frac{6}{10} \times \frac{10}{10} = \frac{60}{100} = 60\%$

(b) 2:8 20% and 80%

(c) 5:5 50% and 50%

(d) 12:13 48% and 52%

(e) 20:30 40% and 60%

3. There were 6 cyclists with red mountain bikes and 4 with green mountain bikes at the race.

(a) What was the ratio of red to green mountain bikes?

$$6:4 = 3:2$$

(b) Write your answer as a common fraction and a percentage

$$\text{Fraction: } \frac{6}{4} \text{ or } \frac{3}{2} \quad \text{Percentage: 60\% and 40\%}$$

6.4 Rate

Rate is the comparison of quantities of different kinds. The **unit rate** describes how many units of the type of quantity correspond to one of the second type of quantity.

Examples

- In a ranch, 100 litres of milk are consumed in 20 days. How many litres are consumed per day?

100 litres of milk in 20 days = 5 litres per day (**divide both by 20**)

1. Find the unit rate of the following:

(a) 48 orders in 6 days = 8 orders per day.

(b) 27 muffins in 3 boxes = 9 muffins per box.

(c) 24 slices from 4 bread = 6 slices per bread.

$$(d) \frac{R200}{50 \text{ kilogram}} = R4 \text{ per kg} \quad (R4/kg)$$

$$(e) \frac{120 \text{ meters}}{12 \text{ seconds}} = 10 \text{ meters per second} \quad (10m/s)$$

$$(f) \frac{R350}{7 \text{ litres}} = R50 \text{ per liter} \quad (R50/l)$$

2. Thando drove a total of 1200km and used 120 litres of petrol. What is this rate in kilometres per litre?

$$1200 \text{ km} \div 120 \text{ l} = 10 \text{ km/l}$$

3. Jane scored 8 goals in 4 netball matches. At this rate, about how many goals did she score in each game?

$$8 \div 4 = 2 \quad \therefore \text{she scored 2 goals per game}$$

4. Climbing a mountain, Noel ascended 160 metres every hour. At this rate, how many metres will he ascend in 4 hours?

$$160 \times 4 = 640 \quad \therefore \text{He will ascend 640 metres}$$

5. Autumn started and over a period of 4 hours, 120 leaves fell from a tree. At this rate, how many leaves fell in one hour?

$$120 \div 4 = 30 \quad \therefore 30 \text{ leaves fell in one hour}$$

6. R240 will be divided between David and Sally in the ratio 3 : 5. This means Sally gets R5 for every R3 David gets. How much will David and Sally each get in total?

$$3 + 5 = 8$$

David gets $\frac{3}{8}$ of 240 and Sally gets $\frac{5}{8}$ of 240

$$\frac{3}{8} \times 240 = 90 \quad \therefore \text{David gets R90}$$

$$\frac{5}{8} \times 240 = 150 \quad \therefore \text{Sally gets R150}$$

UNIT 6

COMMON FRACTIONS

Recap on Fractions

Nozuko bought a watermelon and will share it with her 3 sisters. They cut the watermelon into 4 equal parts. So each person got a quarter ($\frac{1}{4}$) of the whole.

Each person got a fraction of a whole.

1 whole



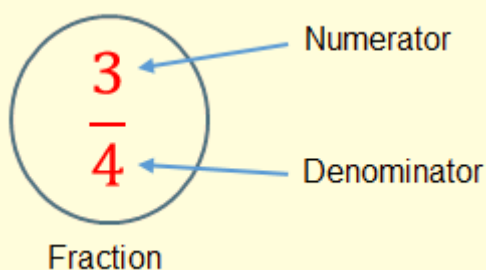
Half ($\frac{1}{2}$)



Quarter ($\frac{1}{4}$)



A fraction has two parts. The top part called the **numerator**. It shows how many equal parts of the whole or collection are taken or used. The **denominator**, written below the line, shows the total number of equal parts a whole has been divided into or the total number of equal parts which are there in a collection.

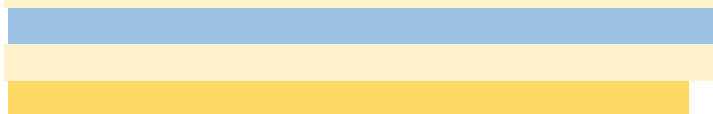


6.1 Compare and order common fractions

Example 1

Which is more, $\frac{5}{8}$ of a loaf of bread or $\frac{3}{5}$ of a loaf of bread?

You can make neat rough drawings to answer a question such as this.



From the above drawings $\frac{5}{8}$ is more.

Drawings like these are called fraction strips. A fraction strip showing eighths is called an eighths strip. A fraction strip that shows fifths is called a fifths strip.

Example 2

Compare the two fractions. Which is bigger or are they the same?

$$\frac{7}{20} \text{ and } \frac{3}{8}$$

$$\frac{7}{20} \times \frac{4}{4} = \frac{28}{80}$$

and

$$\frac{3}{8} \times \frac{10}{10} = \frac{30}{80}$$

$\therefore \frac{3}{8}$ is bigger

1. Which weighs more, or is the mass the same? You may find the diagrams useful.

(a) $\frac{6}{8}$ kg copper or $\frac{5}{7}$ kg copper [$\frac{6}{8}$ kg]

(b) $\frac{2}{7}$ kg copper or $\frac{3}{8}$ kg copper [$\frac{3}{8}$ kg]

(c) $\frac{5}{7}$ kg copper or $\frac{7}{10}$ kg copper [$\frac{7}{10}$ kg]

(d) $\frac{9}{15}$ kg copper or $\frac{3}{5}$ kg copper

Mass is the same

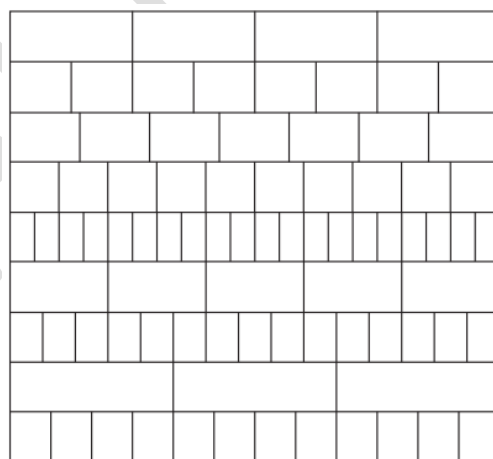
(e) $\frac{8}{12}$ kg copper or $\frac{2}{3}$ kg copper

Mass is the same

(f) $\frac{13}{15}$ kg copper or $\frac{17}{20}$ kg copper [$\frac{13}{15}$ kg]

(g) $\frac{8}{10}$ kg copper or $\frac{12}{15}$ kg copper

Mass is the same



2. Arrange the following 11 numbers from smallest to largest:

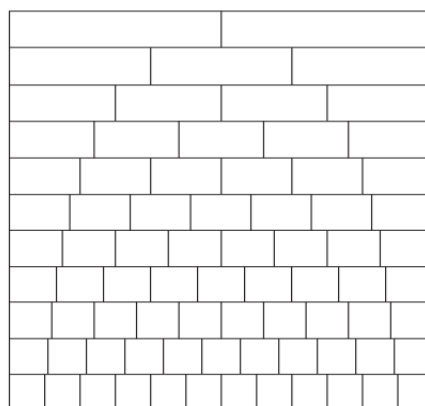
$$\frac{1}{8}, \frac{1}{6}, \frac{1}{12}, \frac{1}{9}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{2}, \frac{1}{10}, \frac{1}{7}, \frac{1}{11}$$

$$\frac{1}{12}, \frac{1}{11}, \frac{1}{10}, \frac{1}{9}, \frac{1}{8}, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}$$

3. Arrange the following 11 numbers from largest to smallest:

$$\frac{7}{8}, \frac{5}{6}, \frac{11}{12}, \frac{8}{9}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{1}{2}, \frac{9}{10}, \frac{6}{7}, \frac{10}{11}$$

$$\frac{11}{12}, \frac{10}{11}, \frac{9}{10}, \frac{8}{9}, \frac{7}{8}, \frac{6}{7}, \frac{5}{6}, \frac{4}{5}, \frac{3}{4}, \frac{2}{3}, \frac{1}{2}$$



4. Replace ■ by either <, > or =

(a) $\frac{4}{12}$ ■ $\frac{4}{10}$ <

(b) $\frac{6}{10}$ ■ $\frac{5}{100}$ >

(c) $\frac{7}{20}$ ■ $\frac{21}{50}$ <

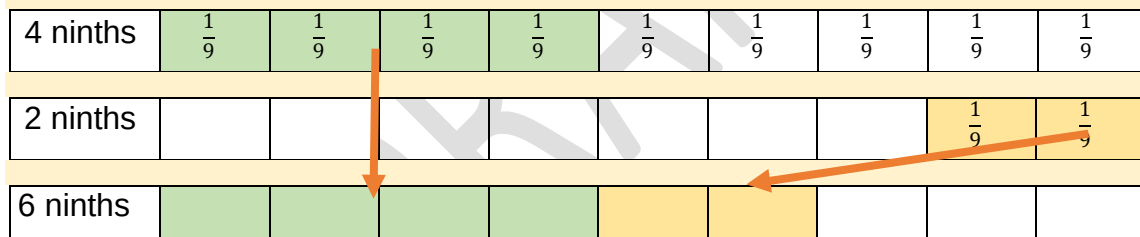
(d) $\frac{5}{25}$ ■ $\frac{10}{50}$ =

6.2 Adding and subtracting fractions

It is easy to add and subtract fractions that are expressed with the same denominator.

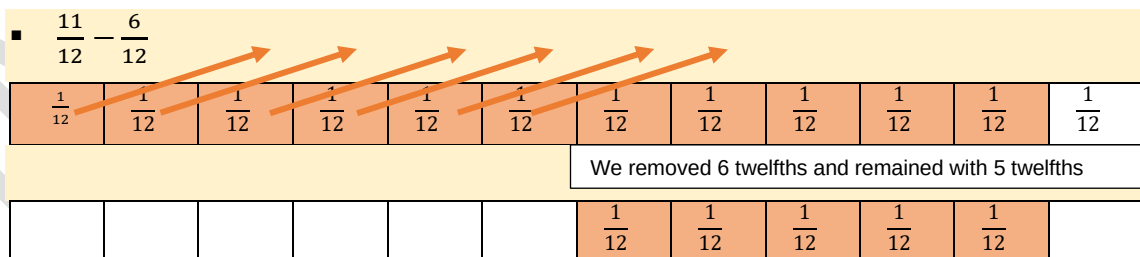
Example 1

- Use a rough sketch to add $\frac{4}{9}$ to $\frac{2}{9}$ and then calculate.



Therefore: $\frac{4}{9} + \frac{2}{9} = 4 \text{ ninths} + 2 \text{ ninths} = 6 \text{ ninths}.$

Example 2



Therefore $\frac{11}{12} - \frac{6}{12} = 11 \text{ twelfths} - 6 \text{ twelfths} = 5 \text{ twelfths} = \frac{5}{12}$

1. Calculate the following.

$$(a) \frac{3}{8} + \frac{5}{8} = \frac{8}{8} = 1$$

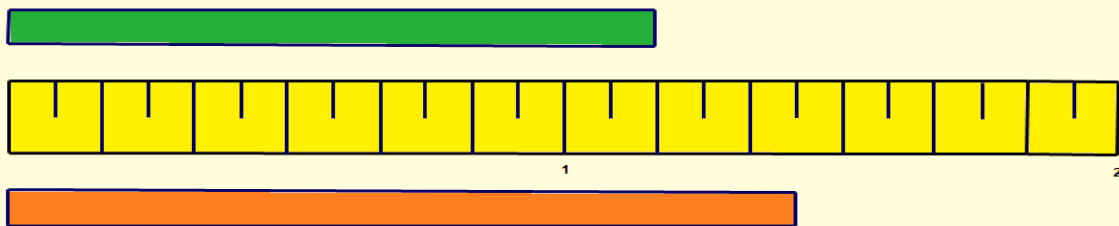
$$(b) \frac{3}{16} + \frac{6}{16} = \frac{9}{16}$$

$$(c) \frac{74}{100} - \frac{23}{100} = \frac{97}{100}$$

Example 3

1. The length of the green strip is $\frac{14}{12} = 1\frac{2}{12}$

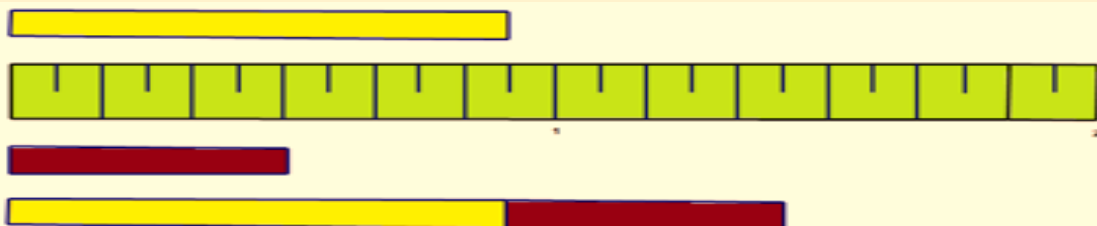
2. The length of the coral strip is $\frac{17}{12} = 1\frac{5}{12}$



Example 4

Sometimes when you add common fractions, the result is more than a whole. The answer becomes a whole number and a common fraction and we call that a **mixed number**. An example is given below.

$$\frac{6}{12} + \frac{11}{12} = \frac{17}{12} = 1\frac{5}{12}$$



2. Calculate the following

$$(a) \frac{4}{5} + \frac{3}{5} = \frac{7}{5} = 1\frac{2}{5}$$

$$(b) \frac{17}{16} - \frac{6}{16} = \frac{11}{16}$$

$$(c) 1\frac{9}{11} + \frac{7}{11} = 1\frac{16}{11} = 1 + 1\frac{5}{11} = 2\frac{5}{11}$$

$$(d) 1\frac{3}{9} - \frac{8}{9} = \frac{9}{9} + \frac{3}{9} - \frac{8}{9} = \frac{4}{9}$$

$$(e) 2\frac{3}{7} + 1\frac{2}{7} = 3\frac{5}{7}$$

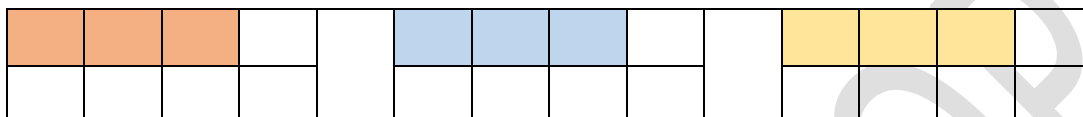
6.3 Fractions of a whole or collection

Express the following in diagrammatic form in order to assist with calculations.

Example 1

Anita ate $\frac{3}{8}$ of 3 cakes. How much cake did she really eat?

- Use diagrams to represent the 3 fractions.



$$\frac{3}{8} + \frac{3}{8} + \frac{3}{8} = \frac{3+3+3}{8} = \frac{9}{8} = 1\frac{1}{8}$$

Anita ate one whole and one eighth of the cakes.

Example 2

How much is $\frac{4}{100}$ of 250?

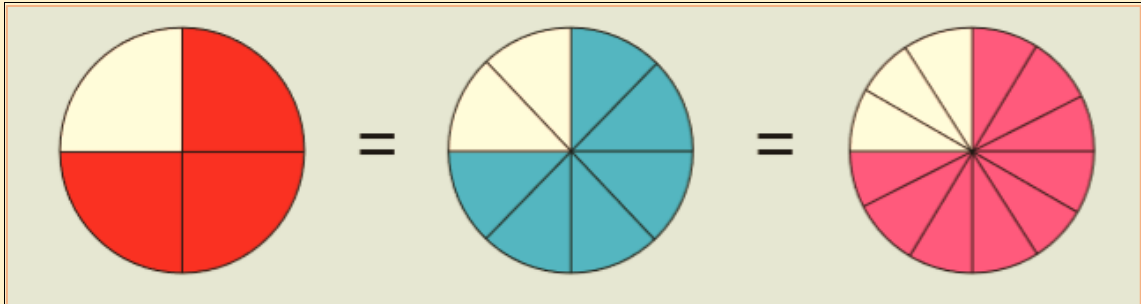
$$\frac{4}{100} \times 250 = \frac{1\,000}{100} = 10$$

- Calculate:
 - Half of 20. $= 10$
 - A quarter of 84. $= 21$
 - A fifth of fifteen. $= 3$
- 93 learners are divided equally into three classes. Each class has its own classroom.
 - What fraction of all the learners are in each of the classrooms? $\frac{1}{3}$
 - How many learners are there in each class? 31
- Ndumiso is painting the garden wall red. The wall consists of $\frac{1}{4}$ panels (divisions) of the same size.
 - On the first day, Ndumiso painted 14 of the wall. How many panels did he paint? 6
 - On the second day, he painted $\frac{1}{3}$ of the wall. What fraction of the wall was painted on the second day? $\frac{8}{12}$

(c) What fraction of the whole wall had been painted at the end of day 2? $\frac{7}{12}$

6.4 Equivalent fractions

Study the following diagrams that represent fractions. What do you notice?



The three circles are equal in size but they are separated into different sectors. The shaded areas of the circles are the same. The three fractions have different numerators and denominators but they have the same value. They represent the same proportion of the whole. We say they are **equivalent fractions** e.g.

$\frac{3}{4} = \frac{6}{8} = \frac{9}{12}$. That is also true for the unshaded areas. They are also equivalent i.e.

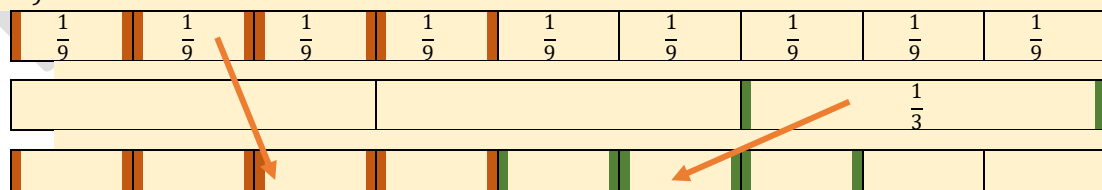
$$\frac{1}{4} = \frac{2}{8} = \frac{3}{12}.$$

Using equivalence

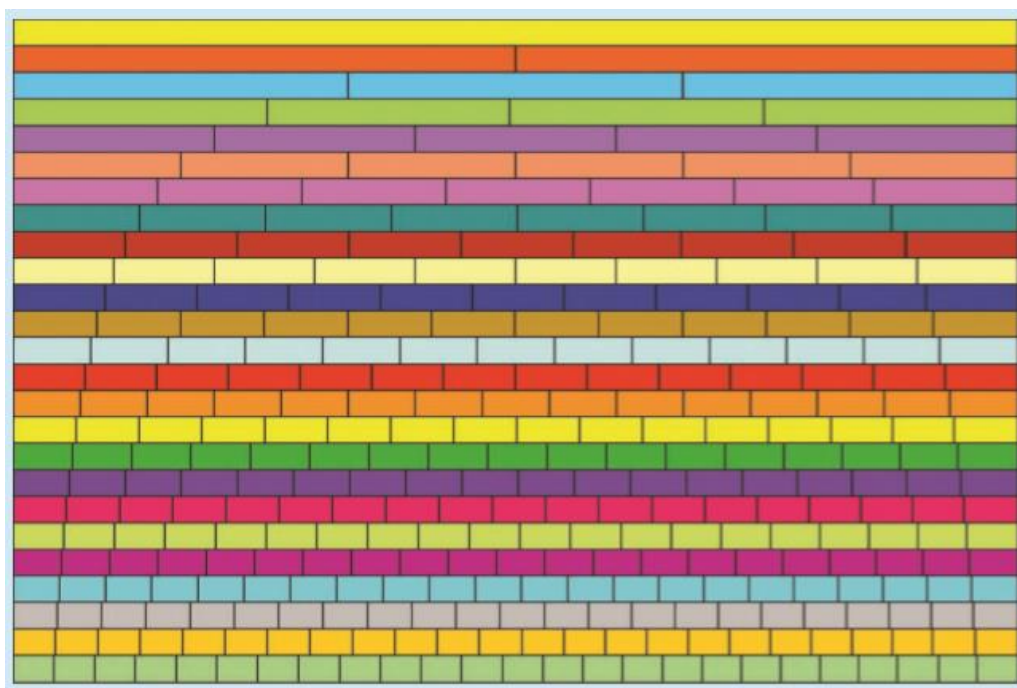
To add fractions with different denominators, we have to use equivalent fractions. For example, $\frac{4}{9} + \frac{1}{3}$: 4 ninths + 1 third is difficult to do. Therefore, to calculate $\frac{4}{9} + \frac{1}{3}$, we have to express $\frac{1}{3}$ as a ninth.

$$\begin{aligned} &= \frac{4}{9} + \frac{1}{3} \\ &= \frac{4}{9} + \frac{1 \times 3}{3 \times 3} \\ &= \frac{4}{9} + \frac{3}{9} \\ &= \frac{7}{9} \end{aligned}$$

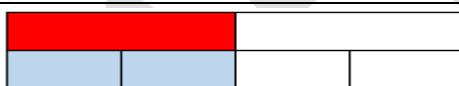
$[\frac{1}{3} \text{ is equivalent to } \frac{3}{9}]$



1. Use the fraction wall and the example provided to complete the following activity.



Example: $\frac{1}{2} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4}$



(a) $\frac{1}{3} = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$

(b) $\frac{1}{4} = \frac{1}{8} + \frac{1}{8} = \frac{2}{8}$

(c) $\frac{1}{2} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6}$

(d) $\frac{1}{2} = \frac{2}{8} + \frac{2}{8} = \frac{4}{8}$

2. Complete the following using the diagrams provided.

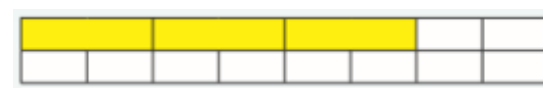
(a) $\frac{3}{4} = \frac{1}{8} + \frac{5}{8} = \frac{6}{8}$



(b) $\frac{3}{4} = \frac{2}{8} + \frac{4}{8} = \frac{6}{8}$



(c) $\frac{3}{4} = \frac{1}{2} + \frac{1}{4} = \frac{6}{8}$



6.5 Percentages

Working with hundredths.

The word **percent means hundredth**. If a person got 50 percent in a test, it means that person got 50 out of 100 or $\frac{50}{100}$. The **symbol for percent is %**.

53 % means 53 out of 100 or $\frac{53}{100}$. This is the same as $\frac{5}{10} + \frac{3}{100}$ or 0,53

Instead of saying 53 hundredths we can say 53%.

This shows that we can express 53 hundredths using different symbols or notations such as 0,53 and 53% and $\frac{53}{100}$, where

- 0,53 is a **decimal fraction** notation
- 53% is a **percentage** notation
- $\frac{53}{100}$ is a **common fraction** notation

1. Write each fraction as a percentage.

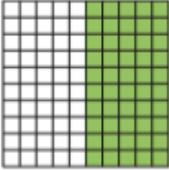
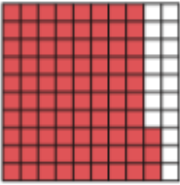
(a) $\frac{65}{100} = 65\%$

(b) $\frac{5}{10} = 50\%$

(c) Zandiswa got 21 out of 50 for her Mathematics test. What percentage did she get? $= 42\%$

2. Express each of the following shaded regions in three ways:

- in the decimal notation,
- in the percentage notation, and
- if possible, in the common fraction notation, using hundredths

Diagrammatic form	Common fraction (hundredth)	Decimal fraction	Percentage
	$\frac{50}{100}$	0,5	50%
	$\frac{83}{100}$	0,83	83%
	$\frac{6}{8}$	0,75	75%
	$\frac{12}{16}$	0,75	75%

Percentages of whole numbers

Calculating a percentage of a whole number is the same as calculating a fraction of a whole number. Therefore, a percentage is a fraction written in a different notation. So, to determine 5% of 65 is the same as determining $\frac{5}{100}$ of 65. We simply divide by 100, which is a skill we have been practicing and can do mentally. We can also do them using our calculators.

3. Use your calculator to do the following:

(a) 10% of 85

$$= \frac{10}{100} \times 85$$

$$= 8,5$$

(b) 20% of 400

$$= \frac{20}{100} \times 400 = 80$$

(c) 15% of 450

$$= \frac{15}{100} \times 450 = 67,5$$

$$\begin{aligned} \text{(d) 25\% of 444} &= \frac{25}{100} \times 444 = 111 \\ \text{(e) 3\% of R600} &= \frac{3}{100} \times \text{R600} = \text{R18} \\ \text{(f) 14\% of R1 500} &= \frac{14}{100} \times \text{R1 500} = \text{R210} \end{aligned}$$

4. A clothing shop has a 25% off sale. This means that the clothes on sale will sell for 25% less than the normal price. Calculate what the sale price will be if the original price is:

(a) R100

$$\frac{25}{100} \times \text{R100} = \text{R25}$$

$$\therefore \text{The sale price will be } \text{R100} - \text{R25} = \text{R75}$$

(b) R400

$$\frac{25}{100} \times \text{R400} = \text{R100}$$

$$\therefore \text{The sale price will be } \text{R400} - \text{R100} = \text{R300}$$

(c) R600

$$\frac{25}{100} \times \text{R600} = \text{R150}$$

$$\therefore \text{The sale price will be } \text{R600} - \text{R150} = \text{R450}$$

(d) R 1 200

$$\frac{25}{100} \times \text{R1 200} = \text{R300}$$

$$\therefore \text{The sale price will be } \text{R1 200} - \text{R300} = \text{R900}$$

5. Last year, there were 20 girls in a class. This year, there are 10% more girls. How many girls are in the class this year?

$$10\% \text{ of } 20 \text{ is } 2 \text{ therefore there are } 2 \text{ more girls in class which is } 22$$

6. The population of the town was 18 000, it then increased over the years by 25%. What is the population of the town now?

$$\frac{25}{100} \times 18\,000 = 4\,500$$

$$\therefore \text{The population is now } 18\,000 + 4\,500 = 22\,500$$



UNIT 7

DECIMAL FRACTIONS

7.1 Representation of decimals on a place value column

To convert common fractions to decimal fractions we use a place value column which will be demonstrating the position of the digits from the fractions with denominators of 10 or 100.

Start by rewriting and converting tenths and hundredths in common fraction form to decimal fractions. Where denominators of other fractions are factors of 10 e.g. 2, 5 or factors of 100 e.g. 2, 4, 25, 20, 50 learners

Note:

- 0,1 is another way to write $\frac{1}{10}$
- 0,01 is another way to write $\frac{1}{100}$
- 0,1 and $\frac{1}{10}$ are different notations for the same number. $\frac{1}{10}$ is called the (common) fraction notation and 0,1 is called the decimal notation

Example

Represent the following as decimal fractions:

(a) $\frac{2}{10} = 0,2$

(b) $\frac{2}{5} \times \frac{2}{2} = \frac{4}{10} = 0,4$

(c) $\frac{25}{100} = 0,25$

(d) $\frac{3}{4} \times \frac{25}{25} = \frac{75}{100} = 0,75$

decimal comma			$\frac{1}{10}$	$\frac{1}{100}$
H	T	O	t	h
		0	2	
		0	4	
		0	2	5
		0	7	5

Example

Study the following diagram



(a) What part of the rectangle is coloured yellow?

$$\frac{10}{100}$$

(b) What part of the rectangle is red?

$$\frac{1}{100}$$

(c) What part is blue?

$$\frac{30}{100}$$

(d) What part is green?

$$\frac{2}{100}$$

(e) What part is not coloured?

$$\frac{57}{100}$$

1. Write the following in decimal notation

(a) $\frac{1}{4} = \frac{1}{4} \times \frac{25}{25} = \frac{25}{100} = 0,25$

(b) $\frac{1}{5} = 0,2$

(c) $\frac{4}{5} = 0,8$

(d) $\frac{4}{10} = 0,24$

(e) $\frac{37}{100} = 0,37$

(f) $\frac{28}{100} = 0,28$

(g) $\frac{56}{100} = 0,56$

7.2 Count, order and compare decimal fractions

Place value of digits

2,935 in can be expressed in expanded notation as $2,935 = 2 + 0,9 + 0,03 + 0,005$. These parts are named as follows:

- the 2 represents the **ones**
- the 9 represents the **tenths**
- the 3 represents the **hundredths**
- the 5 represents the **thousandths**

The **value** of the 9 is nine tenths but the **place value** of the 9 is **tenths**, because any digit in that **place** will represent the number of **tenths**.

Example

In 4,857 – the **value** of the five is 0,05 and its **place value** is **hundredths**, because the **value of the place** where it stands is **hundredths**.

We can also write 2,935 in words as:

2 ones + 9 tenths + 3 hundredths + thousandths

1. Write each of the following as one number:

- (a) $2 + 0,5 + 0,07 = 2,57$
(b) $2 + 0,5 + 0,007 = 2,507$
(c) $5 + 0,4 + 0,03 = 5,43$
(d) $5 + 0,04 + 0,003 + 0,1 = 5,143$

2. Write the **value** and the **place value** of each of the underlined digits.

- | | | |
|------------|----------------------|----------------------------------|
| (a) 2,34 | value is <u>0,3</u> | place value is tenths |
| (a) 4,67 | value is <u>0,07</u> | place value is hundredths |
| (b) 1,95 | value is <u>0,05</u> | place value is hundredths |
| (c) 34,856 | value is <u>0,8</u> | place value is tenths |
| (d) 564,34 | value is <u>0,04</u> | place value is hundredths |
| (e) 0,987 | value is <u>0,9</u> | place value is tenths |

3. Write the following in words:

- (a) 2,54 two comma five four
(b) 5,37 five comma three seven
(c) 3,589 three comma five eight nine
(d) 7,036 seven comma zero three six

Compare and order decimal fractions

Decimal fractions can be written from smallest to largest or largest to smallest. Note that tenths are bigger than the hundredths and thousandths are smaller than the tenths e.g. 0,3 is bigger than 0,07

Arrange 0,545 ; 0,654 ; 0,454 ; 0,554 in descending order.

Hint: Check the ones first then tenths, then hundredths, then thousandths

- Ones are the same
- On tenths **6** in 0,654 is bigger than 5 and 4 therefore 0,654 will be the bigger one followed by 0,554 since 4 in tenths (0,454) is smaller than 5 in 0,554; followed by 0,545 and the last one will be 0,454.

[0,654; 0,554; 0,545; 0,454]

$0,654 > 0,554$

$$0,454 < 0,545$$

1. Fill in $>$, $<$ or $=$

- (a) $0,4 > 0,04$
- (b) $0,05 > 0,005$
- (c) $0,1 = 0,10$
- (d) $0,62 > 0,26$
- (e) $0,58 < 0,85$
- (f) $0,37 < 0,73$

2. Write the following in ascending order:

- (a) $0,8 ; 0,05 ; 0,5 ; 0,15 ; 0,465$ $0,05 ; 0,15 ; 0,465 ; 0,5 ; 0,8$
- (b) $0,5 ; 0,75 ; 0,4 ; 0,62$ $0,4 ; 0,55 ; 0,62 ; 0,75$

3. Write the following in descending order:

- (a) $0,753 ; 0,8 ; 0,82$ $0,82 ; 0,8 ; 0,753$
- (b) $0,67 ; 0,007 ; 0,06$ $0,67 ; 0,06 ; 0,007$

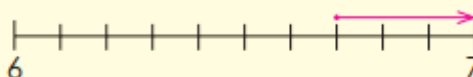
7.3 Rounding off decimal fractions

Just as whole numbers can be rounded off to the nearest multiple of 10, 100 or 1 000, decimal fractions can be rounded off to the nearest whole number or to one, two, three digits after the comma.

A decimal fraction is rounded off to the number whose value is closest to it. Therefore 12,24 rounded off to one decimal place is 12,2 and 11,37 rounded off to one decimal place is 11,4. A decimal ending in a 5 is an equal distance from the two numbers to which it can be rounded off. Such decimals are rounded off to the biggest number, so 10,45 rounded off to one decimal place becomes 10,5.

Round off to the nearest unit

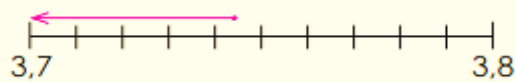
Example: $6,7$
 ≈ 7



Round off to the nearest tenth

Example: 3,745

$\approx 3,7$



1. Round each of the following numbers off to the nearest whole number:

- (a) 7,6 ≈ 8
- (b) 18,3 ≈ 18
- (c) 204,5 ≈ 205
- (d) 1,89 ≈ 2
- (e) 34,7 ≈ 35

2. Round each of the following numbers off to one decimal place:

- (a) 7,68 $\approx 7,7$
- (b) 18,93 $\approx 18,9$
- (c) 21,47 $\approx 21,5$
- (d) 0,643 $\approx 0,64$
- (e) 0,938 $\approx 0,94$

3. Round each of the following numbers off to two decimal places:

- (a) 3,432 $\approx 3,43$
- (b) 54,117 $\approx 54,12$
- (c) 4,809 $\approx 4,81$
- (d) 3,762 $\approx 3,76$
- (e) 4,258 $\approx 4,26$

7.4 Calculations with decimal fractions:

Addition and subtraction of decimal fractions

When you add or subtract decimal fractions, you can change them to common fractions to make the calculation easier.

Example

- $0,4 + 0,5 = \frac{4}{10} + \frac{5}{10} = \frac{9}{10} = 0,9$
- $0,6 - 0,2 = \frac{6}{10} - \frac{2}{10} = \frac{4}{10} = 0,4$
- $0,12 + 0,23 = \frac{12}{100} + \frac{23}{100} = \frac{35}{100} = 0,35$

NOTE: Place value is important when adding or subtracting decimal fractions. fractions. Make sure the commas are under each other.

Example

- Add length and width from the following diagram.



Method 1: Expanded notation

$$\begin{aligned} 4,7 + 3,5 &= (4 + 3) + (0,7 + 0,5) \\ &= 7 + 1,2 \\ &= 8,2 \end{aligned}$$

Method 2: Column method

$$\begin{array}{r} 4,7 \\ + 3,5 \text{ (comma under each other)} \\ \hline 8,2 \end{array}$$

- Calculate $2,37 + 4,53 - 3,88$

Column method

$$\begin{array}{r} 2,37 \\ + 4,53 \\ \hline 6,90 \\ - 3,88 \\ \hline 3,02 \end{array}$$

Expanded notation

$$\begin{aligned} &2,37 + 4,53 - 3,88 \\ &= (2 + 4 - 3) + (0,3 + 0,5 - 0,8) + (0,07 + 0,03 - 0,08) \\ &= 3 + 0 + 0,02 \\ &= 3,02 \end{aligned}$$

1. Calculate the following by using any method:

- | | |
|--------------------------|----------|
| (a) $3,45 + 4,67$ | $= 8,12$ |
| (b) $2,15 + 3,24$ | $= 5,39$ |
| (c) $4,46 - 2,23$ | $= 2,23$ |
| (d) $5,85 - 3,36$ | $= 2,49$ |
| (e) $3,24 + 3,35 - 5,36$ | $= 1,23$ |
| (f) $4,76 + 6,11 - 3,52$ | $= 7,34$ |
| (g) $6,89 + 9,10 - 5,19$ | $= 10,8$ |

2. The owner of an internet café looks at her bank statement at the end of the day. She finds the following amounts paid into her account: R281,45; R39,81; R104,54 and R9,80. How much money was paid into her account on that day?

An amount of R435,59 was paid into her account.

Multiply by 10 and 100

When multiplying a decimal number by **10**, we shift the digits **1 place** to the left.

e.g. $2,54 \times 10 = 25,4$

When multiplying a decimal number by **100**, we shift the digits **2 places** to the left. e.g. $3,456 \times 100 = 345,6$

Example

The length of a ribbon is 3,45 m. What is the length of 100 ribbons?

Total length = $3,45 \times 100 = 345$ m.

3. Calculate:

- (a) $0,5 \times 10 = 5$
- (b) $0,3 \times 100 = 30$
- (c) $0,42 \times 10 = 4,2$
- (d) $0,675 \times 100 = 67,5$
- (e) $1,123 \times 10 = 11,23$

4. Lebo buys a packet of minced meat. It weighs 0,215 kg. How much will 10 packets weigh?

$$0,215 \text{ kg} \times 10 = 2,15 \text{ kg}$$

Ten packets will weigh 2,15 kg.

5. A dressmaker needs to sew a wedding dress for a customer. To make this dress she will need 100m of the fabric and the fabric costs R23,45 per meter. How much did she pay for the fabric?

$$R23,45 \times 100 = R2\ 345,00$$

She will pay R2 345 for the fabric.

6. What is the total mass of the sugar shown in the following diagram?



$10 \times 1,5 \text{ kg} = 15 \text{ kg}$ The total mass will be 15 kilogram

7.5 Recognize equivalence between common fraction, decimal fraction and percentage forms of the same number

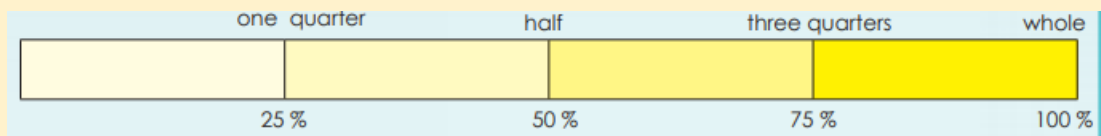
$$\frac{1}{2} \times \frac{50}{50} = \frac{50}{100} = 0,5 \rightarrow \text{Decimal fraction}$$

Use of equivalence → Percentage

From common fractions use equivalence to have a denominator of 100

Percentage is a number expressed as a fraction of 100: $\frac{50}{100} = 50\%$

A percentage can be expressed in the form of a common fraction and a decimal fraction e.g. $50\% = \frac{50}{100} = 0,5$



1. Express each of the following in three ways:

Number	Decimal fraction	Percentage	Common Fraction
(a) 20 hundredths	0,2	20%	$\frac{20}{100}$
(b) 25 hundredths	0,25	25%	$\frac{25}{100}$
(b) 75 hundredths	0,75	75%	$\frac{75}{100}$

2. There are 120 Year 3 students.

(a) 50% of the students are males. How many males are in class?

There are 60 males.

(b) 25% of students like strawberry ice cream. How many students like strawberry ice cream?

30 students like strawberry ice cream

(c) What percentage of students like other flavoured ice creams? How many students like other flavoured ice creams?

90 students like other flavoured ice creams

TERM 2

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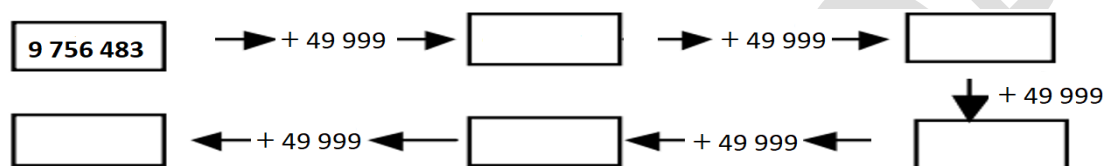
CONTENTS

UNIT 1

WHOLE NUMBERS: COUNTING, ORDERING, COMPARING, REPRESENTING AND PLACE VALUE

1.1 Counting, ordering, comparing numbers

1. Complete the following by **counting** forwards and backwards.



2. Complete the table by filling in the values of A, B and C.

4 324 300 4 326 775 4 329 250 4 331 725 4 334 200 4 336 675 4 339 150

3. Complete the table by filling in the values of E and F.

27 356 801 27 303 701 27 250 601 27 197 501 27 144 401

4. Order the following numbers in ascending order.

(a) 295 059; 295 095; 295 950; 295 509; 295 905

295 059; 295 095; 295 509; 295 905; 295 950

(b) 2 543 334; 2 544 434; 2 543 343; 2 543 443; 2 544 334

2 543 334; 2 543 343; 2 543 443; 2 544 334; 2 544 434

(c) 88 680 008; 88 670 008; 88 660 008; 88 650 008

88 650 008; 88 660 008; 88 670 008; 88 680 008

5. Compare these numbers. Use $<$; $>$ or $=$

(a) 407 652 * 407 625

407 652 $>$ 407 625

(b) 1 664 446 * 1 664 646

1 664 446 $<$ 1 664 646

(c) 22 841 787 * 23 814 787

22 841 787 $<$ 23 814 787



1.2 Recognise the place value of digits and represent numbers

1. What is the value of 4 in each of the following numbers?

- (a) 978 435 400
(b) 3 490 841 400 000
(c) 47 481 032 40 000 000

2. Complete the table below:

Number	Expanded Notation	Words
(a) 970 895	900 000 + 70 000 + 800 + 90 + 5	nine hundred and seventy thousand eight hundred and ninety-five
(b) 7 324 006	7 000 000 + 300 000 + 20 000 + 4 000 + 6	Seven million three hundred and twenty-four thousand and six

1.3 Rounding off to the nearest multiple of 1 000; 10 000; 100 000

1. Complete the following:

Number	Nearest 1000	Nearest 10 000	Nearest 100 000
(a) 834 792	835 000	830 000	800 000
(b) 5 208 147	5 208 000	5 210 000	5 200 000
(c) 62 987 504	62 988 000	62 990 000	63 000 000

UNIT 2

FINANCE

2.1 Solve problems that involve whole numbers, percentages and decimal fractions in financial contexts.

Profit

A profit is an amount of money you gain when you are paid more for something than its cost.

Loss

A loss is the negative difference between the cost price and the selling price.

For an example, if a product is sold at a price less than its cost price then the seller makes a loss.

Profit	Loss
Mr Kunene bought a television set at R1 000 and sold it at R1 200. Calculate the profit and the percentage profit.	Mrs. Smith bought a hand bag at R 2999 and sold it at R2 499. Calculate the loss and the percentage loss.
(a) Profit = selling price – cost price = R1 200 – R1 000 = R 200	(a) Loss = cost price – selling price = R2 999 – R2 599 = R 400
(b) Percentage Profit = $\frac{\text{profit}}{\text{cost price}} \times 100$ = $\frac{R200}{R1000} \times 100$ = 20%	(b) Percentage Loss = $\frac{\text{loss}}{\text{selling price}} \times 100$ = $\frac{R400}{R2599} \times 100$ = 15%

1. Copy this table into your book. Fill in whether each situation is a **profit** or **loss**. Calculate the **percentage profit** or **percentage loss**.

No.	Original Price	Selling price	Profit or Loss?	Percentage profit or loss
(a)	R600	R540	Loss	10%
(b)	R410	R360	Profit	13,8%
(c)	R2 500	R950	Loss	10%
(d)	R24 000	R19 000	Profit	43,3%

2. Find the profits and losses on these amounts:

- (a) 30% profit on R 6 000 = R180
- (b) 20 % loss of R36 000 = R720
- (c) 40% profit on R 300 = R120
- (d) 15% loss on R230 = R34,50

3. Solve the word problems

- (a) Mzimasi bought a bicycle for R1 450 and sold it three years later at a loss of 20%. How much did he sell the bicycle for?

He sold the bicycle for R1 160

- (b) Sam bought eggs from a chicken farm and sold them at the market. He paid R 49,99 for the eggs and sold them at 20 % profit. How much money did Sam make? What was the new selling price?

He made R10. The new selling price was R60.00.

- (c) A shirt costs R175. I got 25% discount. How much did I pay for it?

I paid R131,25 for the shirt.

Discount

A discount is an amount that is subtracted from the cost or regular price of an item. It is usually calculated in percentages.

- How to calculate discount in percentages?
 - Subtract the final price from the original price.
 - Divide this number by the original price.
 - Finally, multiply the results by 100.

Examples

In a video store a DVD that costs R360 is marked down to R229.

- (a) What is the discount price?

$$\begin{aligned}\text{Discount} &= \text{cost price} - \text{selling price} \\ &= R360 - R229 \\ &= R131\end{aligned}$$

(b) What is the percentage discount of the DVD?

$$\begin{aligned}\% \text{ discount} &= \frac{\text{discount price}}{\text{cost price}} \times 100 \\ &= \frac{R131}{R360} \times 100 \\ &= 36,39\%\end{aligned}$$

The percentage discount is 36,39%.

(c) In a candy store, a R12 jar of candy is labelled 20% discount.

What is the discount price?

$$\text{Discount} = \frac{20}{100} \times R12 = R2,40$$

$$\text{Discounted price} = R12 - R2,40 = R9,60$$

4. Study the following advertisement. ITEMS ON SALE:

The advertisement features a central yellow starburst that reads "Sale Up to 75% Off the Original Price". Surrounding this are several items, each with a blue starburst indicating a discount percentage and its original price:

- Calculator:** 5% off, R60
- Television:** 25% off, R1 500
- Bicycle:** 40% off, R650
- Watch:** 30% off, R160
- Takkies (Shoes):** 75% off, R430
- T-shirt:** 10% off, R220

Copy and complete the table. Show your working in your book.

No.	Item	Cost	Discount	Sale price
E.g.	Watch	R160	$\frac{20}{100} \times R160 = R48$	$R160 - R48 = R112$
(a)	Calculator	R60	$\frac{5}{100} \times R60 = R3$	$R60 - R3 = R53$
(b)	Bicycle	R650	$\frac{40}{100} \times R650 = R260$	$R650 - R260 = R390$
(c)	Takkies	R430	$\frac{75}{100} \times R430 = R322,50$	$R430 - R322,50 = R107,50$
(d)	Television	R1 500	$\frac{25}{100} \times R1\ 500 = R375$	$R1\ 500 - R375 = R1125$
(e)	T-shirt	R220	$\frac{10}{100} \times R220 = R22$	$R220 - R22 = R198$

5. Solve the following word problems.

- (a) The original price of a bowl of tomato soup is R3,50. How much will Nancy pay if she buys it during sale when it is labelled “25% off”?

Nancy will pay R2,63.

- (b) In a retail store a R299 dress is marked, “save 10%”. Calculate the discount price and the sale price of the dress.

Discount price is R29,90 and the sale price is R269,10

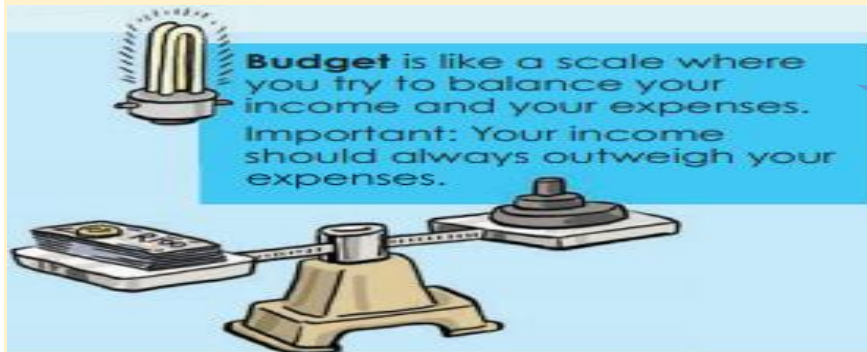
- (c) If you bought your bicycle for R1300 and you are selling it for R 1500. Calculate the percentage profit.

The percentage profit is 15,38 %.

Budget

A budget is a plan that helps you to manage your money effectively. It shows how much money you receive (income) and how you plan to spend it (expenditure). Creating a budget is the most important step in controlling your money.

The first rule to budgeting is: **spend less than you earn!**



Grade 7 DBE Workbook
11TH Edition. Page 22.

THE CHILD'S PERSONAL BUDGET FOR THE MONTH OF SEPTEMBER

BUDGET: BONGY LOOTS

SEPTEMBER 2022

Income

Pocket money	R100
Money from chores	R 50

Total R150

Expenses

Tuck shop	R 50
Charity	R 10
Gift – friend	R 70
Savings	R 20

Total R150

Income and expenses

There are different types of Income or Expenses such as fixed, variable and occasional.

- **Fixed** means it does not change with time.
- **Variable** means it changes over time, according to the situation.
- **Occasional** means it occurs from time to time.

The following table shows some examples of Income and Expenses

INCOME	EXPENSES
▪ Grant: Child grant, old-age pension, disability grant ▪ Pension: Government and private ▪ Old-age grant ▪ Salary: Monthly earnings from an employer ▪ Wages: Weekly earnings from an employer ▪ Commission: Money earned from selling ▪ Profit: Extra money gained on sales of goods and services ▪ Gift ▪ Financial assistance ▪ Rental income for a property	▪ Living expenses ▪ Accounts ▪ Telephone ▪ Insurance ▪ Personal taxes ▪ Loan payments ▪ Savings ▪ Salaries and wages ▪ Business running expenses

Note: There are several things you should aim for in your personal budget:

- It should be realistic, so that you can stick to it.
- It should list all of the items that are needed and should try to anticipate unforeseen expenses.
- It should focus on the high priority items (essential items such as food and health care). If too much of the income is spent on non-essential items and not on savings, your budget is going to become problematic in the future.

- It should be balanced. If your income is less than your expenses, then you need to revise it until the two sides balance. If your income is more than your expenses, then you should plan to save the extra money.
- An ideal budget should include a plan to save money for the future, or to pay off debts to allow for savings in the following months.

Worked example: Study the table with income and expenditure of a certain family.

Income:

- an Old Age Grant of R1 985,
- Child Grant of R480
- a Disability Grant of R1 985 and a
- salary of R5 250.

Expenses:

- rent R2 500
- transport R670
- cell phone R299
- pre-paid electricity R800
- water bill R350
- TV contract R250
- loan repayment R310
- furniture store account R570
- clothing store account R455
- groceries R3 200
- medical expenses R120
- This month, one of the children falls ill and they have additional medical expenses of R500 for doctor's visits and medication.

6. Draw up an income and expense statement for the household for this month.

FAMILY MONTHLY BUDGET

ITEM	INCOME	EXPENCES	TYPE
Old Age Grant	R1 985		Fixed
Disability grant	R1 985		Fixed
Child Grant	R480		
Salary	R5 250		
Rent		R2 500	
Transport		R670	

Cell phone		R299	
Prepaid electricity		R800	
Water bill		R350	
TV contract		R250	
Loan repayment		R310	
Furniture store account		R570	
Clothing store account		R455	
Groceries		R3 200	
Medical expenses		(R120 + R500) = R620	
TOTAL	R9 700	R10 024	

(a) What is the total difference between the income and expenses?

Difference = Income – Expenditure

= R9 700 – R10 024

= (–R324)

[The amount is negative which means that the income does not cover all the expenses]

(b) Which costs could be reduced in their budget?

This will depend on the needs of the family.

(c) If those costs were reduced, would the family have enough money to cover their expenses?

It could because they are over with a small amount. If they reduce some expenses, their income would cover all expenses.

7. Study the family budget for the month of December 2013, for the Chile family.

There are two adults and two school children in the family.

ITEM	INCOME	EXPENDITURE	
		FIXED	VARIABLE
Mrs. Chile's salary	R9 500		
Mr Chile's salary	(a) R10 000		
Other income	(b) R2 500		
Bond repayment		(c) R5 550	
Food			(d) R2 750
Clothing account		(e) R 896	
School fees		(f) R 2 400	
Transport			(g) R 1555
Entertainment			(h) R 975

Savings		(i) R475	
Car repayment		R1 300	
Municipal rates		(j) R 975	
Electricity bill		R200	(k)
Cell phone contract		R700	
TOTAL	R22 000	R12 496	R5 280
Surplus or Deficit?	Surplus of R4 224		

Complete the Chile family budget by using the following information.

Refer to the table for answers. (a – k)

- Mr Chile's income: He works 20 days per month at a rate of R500 per day.
- Additional income: Mr Chile owns additional property which he rents out to people at a fixed charge of R2 500 per month.
- The monthly bond repayments are fixed at R5 550 per month.
- The average amount spent on food each month comes to R2 500. Mrs. Chile believes that this should be increased by 10% due to recent food price increases.
- Mr Chile pays a clothing account an amount of R800 per month. However, since he bought his children their school uniforms on account, he estimates that this amount will increase by a further 12%.
- The school fees are R1 200 per child per month.
- Transport costs are as follows:
 - For the children, taxi fare per child = R5,00 per trip to school and another R5,00 each for the trip home. There are 20 school days in a month.
 - Mr and Mrs. Chile spend R 1 155
 Calculate the total amount that should be budgeted for transport.
- The amount budgeted for entertainment is estimated at 5% of the combined income of Mr and Mrs. Chile.
- Savings are currently 5% of Mrs. Chile's income.
- The amount budgeted for municipal rates is 5% of the total income earned by the Chile household.
- Is the Chile family within budget? Explain your answer.

2.2 Personal and household financial documents

Accounts

What is the meaning and definition of accounts?

An account is a record in an accounting system that tracks the financial activities of a person or a business. These records increase and decrease as the business events occur throughout the accounting period.

Accounts allow people to buy items even when they do not have the full amount of money needed to pay for those items immediately. There is a **limit** on the amount of money that people may spend on an account and an arrangement for when they must make **repayments**.

Account holders receive a **statement** of the account at the end of each month. The statement details how much money is spent on the account, any repayments that have been made, the due date and the amount for the next payment.

It can be:

- an arrangement with a bank to hold money and keep records of transactions.
- an arrangement for regular dealings with a business.
- an arrangement in which a person uses the Internet or email services of a particular company.

Shop Account Statements

At some clothing and food stores, it is possible to open an account, **buy goods on credit** and pay off what you owe the store on a monthly basis. Monthly payments are called **monthly instalments**.

Example

Mr Bear has an Actex store account. He receives the account statement shown below.

BISHOP

STATEMENT TAX INVOICE

You can use your store card in the following shops:
 Palm Stores
 Ginger Tree
 Spray Alert
 Sun-Berry Outlet



MR B. BEAR
 1041 BREAD STREET
 EAST LONDON
 5200

Page 1 of 1

ENCLOSE THE UPPER PORTION WITH POSTAL PAYMENT

0013010000011370690

DATE	REFERENCE	DESCRIPTION	AMOUNT
PAYMENTS THIS STATEMENT			400,00 CR
12 MTHS REVOLVING			OPENING BALANCE
29 MAR EX KING WILLIAMS TOWN			2 350,86
12 APR			PAYMENT
INSTALMENT			400,00 CR
345,00 DUE			INTEREST
290,00			35,03
			CLOSING BALANCE
			1 985,89

STATEMENT PERIOD - 13/03/2022 - 12/04/2022

STATEMENT DATE - 12/04/2022

ACCOUNT NUMBER - 0013010000011370690

INTEREST RATE CHANGE
 From 25 March 2022, the interest rate applicable to your Actex Account will increase from 18.80% to 19.35%

HOW TO PAY YOUR ACCOUNT

1. At any bishop store
2. Bank Trasfer: Enquire at the shop
3. Online: Pay using your card or account no. on <https://actex.com/pay> now.
4. Easypay: Scan the barcode at any Palm stores, Ginger Tree, Sun-Berry Outlet or Spray Alert

My current credit allocation	What can I spend?	What is my instalment?	How much am I in arrears by?	What is my balance?
3 398,00	1 412,00	345,00	0,00	1 985,89

CURRENT	30 DAYS	60 DAYS	90 DAYS +	TOTAL DUE (BY 1ST)
290,00	0,00	0,00	0,00	290,00

Your interest rate is 19.35%

- At which five stores can Mr Bear use his store card?
 Bishop; Palm Stores; Ginger Tree; Spray Alert and Sun-Berry Outlet
- What is the Statement period? April 2022
- How much did he pay in March 2022? R400
- How much must he pay on his account? R290
 - When is this amount due? On the 1st of May
 - Will he be charged extra if he pays the due amount late? Yes
- What is the current balance for this statement? R1 985.89
- How much credit does he still have available? R1 412
- What is the difference between "credit available" and "credit limit"? R1 986
- How much is the balance brought forward from the previous statement?
 R2 350.86
- Does the amount due include interest? How much is the interest? No
- How much did Mr Bear spend at Actex in the month of April 2022? R0

1. Study the following account statement and answer questions.

LGZ STORES		LGZ	
LGZ Stores; PO Box 12345 Cape Town 8000 Telephone 0860 21 31 55 Fax 093 907 985 LGZ Stores (Pty) Ltd Reg. no. 1999/008237/07 A registered credit provider NCRZG23 Email: queries@lgz.co.za			
Ms NH Nodudo PO Box 1456 Cape Town 8000		STATEMENT DATE	15/04/2022
		PAYMENT DUE DATE	30/04/2022
		ACCOUNT NUMBER	123 478 9094
		INSTALMENT FREQUENCY	MONTHLY
STORE CARD STATEMENT			
YOUR TRANSACTION DETAILS			
DATE	DESCRIPTION	AMOUNT	
31/03/2022	Opening balance	3 579.00	
25/03/2022	Cash payment. Thank you.	350.00	
25/03/2022	Prepaid airtime	55.00	
15/04/2022	Interest due	52.95	
	Closing balance	3 336.95	
MINIMUM PAYMENT	280.00	30 DAYS	00.00
PAYMENT DUE DATE	30/05/2022	CREDIT LIMIT	6 500.00
		60 DAYS	00.00
		CREDIT AVAILABLE	3 163.05
Banking Details: LGZ Bank Branch Code: 2435 Account Number: 2436778			
Beneficiary Reference: Please use your LGZ account number as it appears on your LGZ Store Card			

- (a) What is Ms. Nodudo's opening balance? **R3 579,00**
- (b) What is Ms. Nodudo's closing balance? **R3 336,95**
- (c) What is the name of the store in which Ms. Nodudo buys? **LGZ stores.**
- (d) What is her account number? **1234789094**
- (e) How much is due on this statement? **R280**
- (f) How much is her credit limit? **R6 500**

Bank Accounts

Banks offer different types of accounts and services.

Savings account: A bank account that earns interest. You can use a savings account for short term savings.

Cheque or current account: A bank account that is used to deposit and withdraw money by visiting the bank branch, using an ATM or Internet banking or by writing a cheque. These are usually available to people who earn a regular income.

Fixed deposit account: This account is aimed at those who have a lump sum they want to invest over a fixed period of time (i.e. a medium or long-term saving). Interest is also earned on the investment.

Credit account (with credit card): An account either with a store or bank, that allows the account holder to purchase items now and pay for them later.

Debit account (with debit card): Debit cards can be used to pay for purchases. When it's swiped, money is deducted from the account. Credit is not available on this account.

Bank Statement

A bank statement is **an official list of everything paid into and out of a bank account.**

The statement includes **deposits, charges, withdrawals, as well as the beginning and ending balance for the period.** Account-holders generally review their bank statements every month to help keep track of expenses and spending, as well as monitor for any fraudulent charges or mistakes.

What must a bank statement show?

- Bank name and address
- Client's personal information: Address, Name and Surname etc.
- The statement date, as well as the beginning and ending balance of the account.
- Details of each transaction: the amount, date, and payment that took place in the bank account, the period on which the deposits, withdrawals, cheques paid, and any service charges. **Debit transaction:** Amount of money paid out of an account. **Credit transaction:** Amount of money deposited into an account

How are bank statements sent?

- Some banks email statements to customers as an attachment.
- Some bank automatic teller machines (ATMs) offer the option to print a summarized version of a bank statement, called a transaction history.

- Some institutions charge for paper statements, while many online-only banks require digital delivery.

Example

Savings Account Statement			 EASIMONY 		
MRS PG STONE 27 FROG STREET PINETOWN 3600		Tax Invoice VAT Registration Number 4680173	Easimony Bank 16 Point Road Stellenbosch 7600		
Posting Date	Transaction Date	Description	Money In (R)	Money Out (R)	Balance (R)
26/03/2022	26/03/2022	SMS Notification Fee: 1 notification(s)		-0.25	427.55
27/03/2022	24/03/2022	Karabo Parking Managem Durban (Card 5333)		-7.00	420.55
30/03/2022	26/03/2022	Nando's Ballito (Card 5333)		-367.00	53.55
31/03/2022	31/03/2022	Interest Received	0.99		54.54
31/03/2022	31/03/2022	Monthly Account Admin Fee		-6.50	48.04
03/04/2022	03/04/2022	Card Purchase Insufficient Funds Fee: Kfc Old Main Road Un Za		-0.40	47.64
08/04/2022	08/04/2022	Payment Received T Zama	600.00		647.64
08/04/2022	08/04/2022	SMS Notification Fee: 1 notification(s)		-0.25	647.39
09/04/2022	09/04/2022	SMS Notification Fee: 1 notification(s)		-0.25	647.14
13/04/2022	09/04/2022	Engen Pinelands (Card 5333)		-600.05	47.09
16/04/2022	15/04/2022	Payment Received T Zama	300.00		347.09
16/04/2022	16/04/2022	SMS Notification Fee: 1 notification(s)		-0.25	346.84
17/04/2022	17/04/2022	ATM Cash Withdrawal: Atm Germiston Gw 77 Germiston Gtza		-300.00	46.84
17/04/2022	17/04/2022	ATM Cash Withdrawal Fee		-8.00	38.84
17/04/2022	17/04/2022	SMS Notification Fee: 1 notification(s)		-0.25	38.59

- Zola receives the following statement from her bank, detailing her transactions from 25 January 2013 to 20 February 2013. Study the statement and answer the questions that follow.

Date	Description	Amount	Amount	Balance
25/01/2013	Salary	8 000,00		8 050,50
27/01/2013	Car insurance		-100,00	7 950,50
01/02/2013	Electronic transfer Mr. Serei (RENT)		-3 000,00	4 950,50
01/02/2013	Debit order Healthsaver medical aid		-500,00	4 450,50
02/02/2013	Debit order Mobi contract		-250,00	4 200,50
03/02/2013	Debit order Supa Fashion Store		-300,00	3 900,50
05/02/2013	Purchase at Shop 'n Save		-2 000,00	1 900,50
14/02/2013	PAYMENT Mrs. S Khumalo	500,00		2 400,50
20/02/2013	Automechanix		-1 000,00	1 400,50
Total remaining:				1 400,50

- (a) How can you tell the difference between the debits and the credits in this statement?

The debit is indicated by a negative sign.

- (b) List Zola's debits and credits for the month of January 2013.

Debit = R100 Credit = R 8 000

- (c) In the first line of this statement, Zola receives a salary of R 8 000. Look at the balance and work out what she had in the account before the payment was made.

R50,50

- (d) Zola receives some birthday money as well as her salary this month. Identify and write down the birthday transaction.

R 500

- (e) How much money would she have been left with on 20 February 2013, if she hadn't received money for her birthday?

R1 950,50

- (f) Zola wants to save 15% of her remaining money this month. How much can she save?

R 210.

Cell Phone Bill

Cell phone companies offer different cell phone packages. They are either pre-paid or contract accounts. You need to decide on your own needs, and then work out which account would be best for you.

Presented below is an example of a Cell phone bill.

TalkMore

Computer Generated

Tax invoice

Mrs CT Bunn
1277 Sitt Close
ARTC
40241

Account number: 91676-8
Cellular number: Combined
Invoice number: H5-XE8J2
Invoice date: 01/02/2022
Due date: 28/02/2022

Your VAT registration number:

Cellular Number : 099 893 1290

Description		Amount	Discount	Subtotal	Total
Subscription Charges					
Airtime Subs - RED Flexi 175 Line Rental	February	152.17		152.17	
Handset Installment 15	February	281.74		281.74	
Total Subscription Charges					433.91
Recharges					
Airtime Advance		29.58		29.58	
Total Recharges					29.58

Cellular Number : 099 899 3457

Description		Amount	Discount	Subtotal	Total
Subscription Charges					
30GB Data Top Up	February	346.96		346.96	
Bonus Social Ticket 10GB 24 Months	February	0.00		0.00	
Bonus Video Ticket 10GB 24 Months	February	0.00		0.00	
Handset Installment 15	February	34.78		34.78	
Promotional My Gig 20 - 24Months	February	0.00		0.00	
Promotional My Gig 30 - 24Months	February	0.00		0.00	
Reduced Subscription Discount Proration-New	February	-86.96		-86.96	
Router and Modem Insurance (R0-R750)	February	17.39		17.39	
Total Subscription Charges					312.17

Should you have any queries regarding the contents of your bill, please contact Customer Services.

Invoice total**775.66****Terms of Payment**

Always remember that your account is due by the last day of each month. If your account is not paid by this date, your line may be locked.

- (a) To whom does the cell phone bill belong? **It belongs to Mrs. C.N. Bunn**
- (b) What is the cellphone company providing the service? **Talk More**
- (c) How many contracts does he/she have with the cellphone company? Explain.
Two Contracts. There are two cellphone numbers belonging to two contracts on the bill.
- (d) What is the due date of payment on the contract(s)?
End of the month of each month.
- (e) How much discount did she receive on the bill? **R86.96**
- (f) How much does she pay for a Cell phone gadget? **Cellphone 1: R281.74**
- (g) How much data does she receive per month? **30GB**
- (h) What is the amount paid on the insurance monthly? **R17.39**
- (i) If Mrs. Bunn wanted to query the account, what references may she quote?
Name 4 references.

Cellphone numbers x 2; Invoice number; Account number; ID number

(j) What is the amount due in the month of February on the account? **R775.66**

Municipality Bill

A household must pay municipal charges for rates, electricity; water & sanitation, cleansing & solid waste as well as sewage disposal.

Tax Invoice					
Tax Invoice No. : 5874322/02/01 Mrs JG Zulu PO Box 333 Durban 4000		THE METRO BILL REVENUE DEPARTMENT PO Box 828, Durban, 4000 Tel: (031)324 5000 http://correspondence.durban.gov.za:200/ Web: www.durban.gov.za Council VAT Registration No.: 488 019 3505			
		 ETHEKWINI MUNICIPALITY			
		 Post Office Allocation code 0018			
		 58743			
		 1 1350 35 8743 your payment partner			
Your Bill Details					
Date	Account Number	VAT Number	Guarantee (R)	Deposit (R)	
2022/01/19	83013358743	N/A	0.00	4,657.71	
Reference	Details	Amount (R)			
2021/12/21-ND	Balance brought forward	26,246.69Cr			
	Payment - Thank you (D/Delay "01")	5,000.00Cr			
	Sub-total	31,246.69Cr			
	Current month's charges (from detailed invoices)	3,650.58			
	VAT	412.78			
	Total current month's charges	4,063.36			
	Total	27,183.33Cr			
Please register to receive Metro statements via e-mail. Follow https://eservices.durban.gov.za/v2/ eThekweni Municipality will launch its 4th general valuation roll on 11/02/2022. To view your property valuation, go to www.durban.gov.za . Section 49 notices will be posted to all property owners. Closing date for objections is 31 March 2022.					
Rates					
Property Category	Market Value	Ratable Value	General Randage	Special Rating Area Randage	Deferred Rates
Residential	940,000	820,000	1.3152 c/R	0.00 c/R	0.00
Description	Units	Rate (R)	Amount (R)		
General Monthly Rates-Jan 2022	-	-	898.72		
Water and Sanitation					
Meter No.	Previous Meter Reading	Current Meter Reading	Usage		
	Date	Reading	Date	Reading	Units
CLD721	2021/12/02	784.00	2022/01/07	819.00	35.00 kl
Description	Units	Rate (R)	Amount (R)		
Consumption charge Between 0 and 200 litres per day for 36 days	7.20 kl	25.41 /kl	182.95*		
Consumption charge Between 201 and 833 litres per day for 36 days	22.80 kl	30.05 /kl	685.14*		
Consumption charge Between 834 and 1000 litres per day for 36 days	5.00 kl	40.04 /kl	200.20*		
	35.00 kl		1,068.29		
Water Loss Levy	0.00000	0.0000	14.07*		

3. Study the municipal bill received by Mrs. JG Zulu for services and answer questions.

(a) Name the five services she is paying for.

rates, electricity; water & sanitation; cleansing & solid waste as well as sewage disposal.

(b) How much is she paying for each?

Rates = R898,72; water & sanitation = R1068,29; Sewage disposal = R 200,77; Electricity = R 412,78

(c) What is the opening balance for this invoice? R26 246,69

(d) What is the closing balance for the month? R27 183,33

(e) According to the bill, when did Mrs. JG Zulu last make a payment to the municipality? 21 December 2021

(f) How much was her last payment? R 5 000

service. All receipts must have the date of purchase on them. It acts as a proof of payment for both your business and the customer.

Receipts, however, should be issued any time a payment is received from the customer. If you accept a payment that is made immediately upon providing the goods or services, you don't need to issue an invoice, but can offer a receipt as proof of payment.

According to the IRS, a business should keep their receipts for **three (3) years**. If a business claimed a loss for any tax year, then the receipts for that year must be kept for **seven (7) years**. The **different types** of receipts are for Cash payment, Cleaning, Car/Vehicle, Credit card, Donation, Delivery, Hotel etc. By law, South African till slips or receipts must include what is indicated in the labels below.

Example:

Name & address of the shop	SUNSHINE COFFEE SHOP HIGHVELD 9035																																		
Vat number	VAT No: 868048110	INV33711	Invoice number																																
Date & time of sale	Tax Invoice Customer: C 11/04/2022 10:33:36 Rep 1 Rep 1																																		
Description of item bought	<table border="1"><thead><tr><th>Code</th><th>Description</th><th>Qty</th><th>Total</th></tr></thead><tbody><tr><td>0204</td><td>CAFE LATTEXX</td><td>2.00</td><td>50.00</td></tr><tr><td></td><td>Discount</td><td></td><td>0.00</td></tr><tr><td></td><td>Vat</td><td></td><td>0.00</td></tr><tr><td></td><td>Total</td><td></td><td>50.00</td></tr><tr><td></td><td>Tendered</td><td></td><td>100.00</td></tr><tr><td></td><td>CT 100.00</td><td></td><td></td></tr><tr><td></td><td>Change</td><td></td><td>50.00</td></tr></tbody></table>	Code	Description	Qty	Total	0204	CAFE LATTEXX	2.00	50.00		Discount		0.00		Vat		0.00		Total		50.00		Tendered		100.00		CT 100.00				Change		50.00		VAT amount
Code	Description	Qty	Total																																
0204	CAFE LATTEXX	2.00	50.00																																
	Discount		0.00																																
	Vat		0.00																																
	Total		50.00																																
	Tendered		100.00																																
	CT 100.00																																		
	Change		50.00																																
	Cashier 20 NICKY		Total amount paid																																

4. Study the following till slip that Patty received when she bought groceries at the MZ Carriers and answer questions.

- (a) Where is MZ Carriers located?
Centurion
- (b) On which date did Patty go to MZ Carriers? At what time?
14 March 2022 at 09:41
- (c) How did Patty pay for her shopping?
Tendered cash
- (d) How many items did Patty buy?
- (e) How much does 750 ml of Bonaqua Berry cost at MZ Carriers? **R10,99**
- (f) What is the total cost of the 5 items she bought? **R95,95**
- (g) How much did she pay? **R100**
- (h) Why do you think the amount is slightly different to the balance due?
The amount has been rounded down to the advantage of the customer.
- (i) Who served Patty at the till? At which till? What is her number?
Junitha; at till 004; 47
- (j) What is the invoice number of the till slip? **8 033**

MZ CARRIERS			
CENTURION			
HIGHVELD SHOPPING CENTRE			
CENTURION, PRETORIA			
TEL NO: (012) 665-14			
TAX INVOICE			
WAT NO: 4660249			
LIQUOR LICENSE NO: GAU/20080			
			R
SIMBA SALT & VIN	120GR	16.99	A
LILLETS SUPER MD	20'S	24.99	A
NECTARINE TUB	PREPACK		
1 @	28.99	28.99	*
BONAQUA BERRY	750ML	10.99	A
POWER PLAY AP/KI	440ML	13.99	A
TOTAL FOR 5 ITEMS		95.95	
TENDERED Cash		100.00	
CHANGE Cash		4.10	
ROUNDING		0.05	
ROUNDED TOTAL		95.90	
----- TAX INVOICE -----			
VAT rate	excl.	TAX	incl.
0.00%	28.99	0.00	28.99 *
15.00%	58.23	8.73	66.96 A

SLIP / TILL / CASHIER / DATE / TIME			
8033 / 004 / 47 / 14.03.22 / 09:41			
CASHIER NAME: Junitha			
THANK YOU FOR SHOPPING WITH US			

2.3 Workplace and small business finance

Payslips

A payslip is a piece of paper given to an employee by an employer at the end of each week or month, which states how much money has been earned and how much tax has been taken off?



MOONLIGHTCOURIERS

PO Box 76123, BLOEMFONTEN, 9300;
Enquiries: info@moonlightcouriers.com;
Tel: 085 877 9994

PAYSLEIPS FOR THE MONTH OF MARCH 2022

Employee Name	Mr GN Couver	Paid Days	30
Gender	Female	LOP	0

OT: Over time; LOP: Leave without pay; PAYE: Pay As You Earn

EARNINGS		AMOUNT	DEDUCTIONS	AMOUNT
Basic		10 000,00	Salary advance	1 000,00
OT Hours	50		PAYE	1 065,25
Rate	75			
OT Payment		3 750,00		
Total payment		13 750,00	Total Deductions	2 065,25
NETT PAY		11 684,75		

What must a payslip show? - Total pay before deductions ('gross amount'). - Total pay after deductions ('net amount'). - Amounts of any 'variable deductions'.	What can you use a payslip for? - Applying for credit like a home loan or a phone contract. - Negotiating your package with a prospective employer. - Going into a rental agreement for a place to stay etc.
When should pay slips be issued? - Payslips must be provided on or before payday. - Your payslips can be used as proof of your earnings. - Tax paid and any pension contributions. - Employers can choose whether they provide printed or electronic (online) pay slips.	

5. Study the pay slip provided and answer questions.

PAYSLIP NO. 1 OF 2022			
 MOONLIGHT COURIERS		PO Box 76123, BLOEMFONTEN, 9300; Enquiries: info@moonlightcouriers.com ; Tel: 085 877 9994	
PAYSLEIPS FOR THE MONTH OF JANUARY 2022			
Employee Name	Mr GN Couver	Paid Days	30
Gender	Female	LOP	0
OT: Over time; LOP: Leave without pay; PAYE: Pay as You Earn			31 JANUARY 2022

EARNINGS		AMOUNT	DEDUCTIONS	AMOUNT
Basic		10 000,00	Salary advance	500,00
OT Hours	38	(a) R3 230	PAYE	1 145,00
Rate	R85 p/h		Medical Aid	799,00
OT Payment				
Total Income		(b) R13 230	Total Deductions	(c) R 2 444
NETT PAY (INCOME – DEDUCTIONS)		(d) R10 786		

(a) How many days did Ms. Noom work? **She worked for 30 days**

(b) Calculate the missing values in order to complete the payslip.

See (a – d) in the payslip.

Quotations

A **quotation** is a document that a seller provides to a buyer to offer goods or services at a stated price, under specified conditions. **It represents your business identity and professionalism directly.** If it appears to be unpleasant and not well-structured, your clients will assume that your business runs the same way. Contrary to this, a good quote exhibits your attention to detail and the value of service you are pricing

		QUOTATION	
Box 78123, KIMBERLEY, 8301 ; Enquiries: simon@ssrverspreiders.com ; Tel: 085 877 9984		Date	26 April 2011
		NOTE: This quotation is only valid for 30 days.	
Customer Name	Mr GN Couver		
Address	1267 32rd Avenue; Clernaville; 3602		
Contact number	089 909 9909		
Description	QTY	UNIT	TOTAL
Basin new install	1	1 700	1 700
Toilet new install	1	1 900	1 900
Shower new install and shower door	1	4 800	4 800
Geyser new install	1	2 400	2 400
Sewer line	480 m	250	12 000
Water supply inlet new install	38 m	60	2 280
	MATERIAL		25 080
	LABOUR		21 450
	TOTAL DUE		46 530
NOTE: SSR Verspreiders is not a VAT vendor. Banking Details: EasiMony Bank; Branch Code: 2435; Account Number: 2436778; Please use your invoice number as a reference.			
_____ Signature		_____ Date	

6. Answer the following questions.

- (a) How much will the client pay for materials? **R25 080**
- (b) How much will the client pay for labour? **R21 450**
- (c) What is the total amount due? **R46 530**
- (d) Will the quotation still be valid on the 5th of June? **No, it will not be valid**

Loans

A **loan** is an agreement of lending (borrowing) money with interest and a plan to repay it.

Types of loans: Personal Loan, Business Loan, Home Loan, Rental Deposit Loan etc.

When someone lends money to someone else, the borrower usually pays a fee to the lender. This fee is called '**interest**'. There are two kinds of interest. '**simple**' and '**compound**'. **Simple** or '**Flat rate**' interest is usually paid each year as the

fixed percentage of the amount borrowed or lent at the start. With **'compound'** interest you also pay **interest on the interest**.

7. The following is an extract from a letter from FCS loans to Ms. Dube. It shows the amounts that are available on instant loan from FCS loans and the repayment involved.

Dear Ms. Dube

As a valued customer, we are pleased to offer you a personal loan at the following rates.

Loan amount	24 months	36 months	48 months	60 months
R4 000	R229	R174	R147	R131
R8 000	R448	R338	R285	R253
R16 000	R864	R643	R534	R470
R25 000	R1 344	R1 000	R830	R730

These loan repayments include a monthly premium of R3,95 per R1 000 of the loan and a monthly administration fee of R9,50 for your optional personal protection plan.

Regards

Secretariat

Answer the following questions.

If Ms Dube chooses to borrow R16 000 from FCS loans, calculate how much he will finally repay if he takes the loan over:

- (a) 24 months **R1744,80 + R16 000 + R17 744 80**
(b) 36 months **R18 617, 20**
(c) 60 months. **R20 362**
(d) Do you advise her to borrow for a longer or shorter time?

A shorter time period is advisable.

Invoices

An invoice is a document given to the buyer by the seller to collect payment. It includes the cost of the products purchased or services rendered to the buyer. Invoices can also serve as legal records, if they contain the names of the seller and client, description and price of goods or services, and the terms of payment.

There are different types of invoices such as, electronic, online, mobile and hand-written invoices. A good, legally-sound invoice should contain all the information the customer will need to make payments and provide an avenue to address any questions.

Elements of an Invoice

- The word "**Invoice**" and the company logo must be prominent on the document so that they can be identified easily.
- Every invoice must have a unique identifier like an **invoice number**, purchase order number, or reference number to distinguish it from the rest of the seller's invoices.
- Recording the **date of service rendered** or date of sale can help simplify cataloguing.
- The **date of sending the invoice** must be included to ensure payments are made on time.
- The invoice should clearly show the **seller's name and contact information** to ensure the customer can contact you for any queries.
- The **client's contact information** must also be included to ensure legality of the document.
- All the **company's terms and conditions**, including due dates and penalties for non-payment or partial payments, must be clearly included in the invoice.
- The invoice must **describe each product or type of service rendered**.
- The **cost of each individual item or service** must be included.
- The **taxable amount** for the products or services must be included.
- The **total amount due** for payment must be clearly included in the invoice.
- Invoices **can be personalized** by including a customized thank-you note.

Tips for invoicing

Here are some tips to help you make the best use of your invoices and ensure that you get paid promptly:

- Early invoice payments and automated invoicing
- Make the invoice and payment process more convenient for customers
- Use recurring invoicing, if applicable
- Negotiate and establish payment terms in writing
- Get upfront payments
- Work on creating closer relationships with clients
- Hold the work or stop projects if a client doesn't pay on time

 P.O. Box 76123, KIMBERLEY, 8301 ; Enquiries: simon@ssrverspreiders.com ; Tel: 085 877 9994		TAX INVOICE	
		Invoice Date	
		Invoice Number	
		Due Date	
Customer Name			
Address			
Contact number			
Description	Price	Quantity	Amount
	Sub Total		
	VAT		
	Amount Due		
NOTE: SSR Verspreiders is not a VAT vendor. Banking Details: EasiMony Bank; Branch Code: 2435; Account Number: 2436778; Please use your invoice number as a reference.			
Signature		Date	
<i>Dankie daj jy by ons inkoopies doen!</i>		<i>Thank you for shopping with us!</i>	

UNIT 3

NUMBER SENTENCES (ALGEBRAIC LANGUAGE)

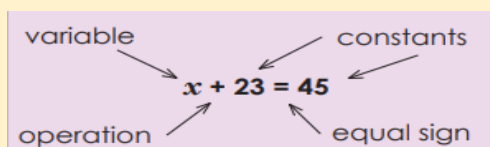
3.1 Identify variables and constants in given formulae

Words	Flow diagram	Expression
Multiply a number by two and add six to the answer.		$2 \times x + 6$

In number sentences we usually write a box to represent the unknown but in algebraic expressions, instead of writing a box we may write a single letter symbol like x or y .

An algebraic expression indicates a sequence of calculations that can also be described in **words** or by means of a **flow diagram**. The flow diagram illustrates the order in which the calculations must be done.

In algebraic language, **the multiplication sign is usually omitted**. So, we write $2x$ instead of $2 \times x$. We also write $x \times 2$ as $2x$.



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$x + 23$

[expression]

$x + 23 = 45$

[equation]

In the **expression**, $x + 23$, the letter symbol x can be replaced by many different input numbers. The symbol x represents a variable. If, however, the expression $x + 23$ is equal to 45 as in the number sentence $x + 23 = 45$, the symbol x represents only one value and that is 22.

In the expression $x + 23$, the number 23 is a constant. In the number sentence $x + 23 = 45$, x is an unknown value.

- Complete the following table:

	Words	Flow diagram	Expression
(a)	Add three to a number and then multiply the answer by two.		$2(x + 3)$
(b)	Add 7 to a product of a number and 4		$7 + 4 \times x$

2. Write the following expressions in “normal” algebraic language:

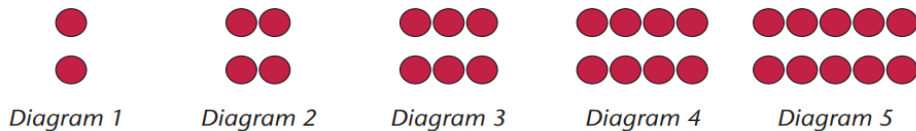
(a) $2 \times a + b = 2a + b$

(b) $a^2 = 2a$

(c) $y + 4 \times x = 4x + y$

3. The diagrams below represent arrangements of small circles. In every arrangement there are two rows of circles.

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(a) What remains the same in the diagrams above? **The multiplier – 2**

(b) What changes in the diagrams? In other words, what are the variable quantities in the situation? **Diagram number and number of circles.**

(c) How many circles will diagram 11 have if the pattern is extended? Explain.

Diagram 11 = $2 \times$ Diagram number 11 = 22 circles.

(d) Write the rule for the pattern in terms of x

(e) What does the letter symbol x represent in the rule $2 \times x$?

Diagram number.

3.2 Determine the numerical value of an expression by substitution.

1. Say whether it is an expression or an equation

(a) $9 + 7 = 16$ **Equation**

(b) $11 + 2$ **Expression**

(c) $x + 4 = 12$ **Equation**

(d) $14 + x$ **Expression**

2. Identify **variable(s)** and **constant(s)** and **coefficient(s)** in the following expression in $2x^3 + 5x - 6$

Constant(s): -6

Coefficients: 2 and 5

Variable(s): x

3. Write number sentences for each of the following statements.

(a) Three plus a certain number equals 0. $x + 3 = 0$

(b) Two times a certain number p decreased by eight is equal to twelve.

$$2p - 8 = 12$$

(c) Thabo has R50 with which to buy chocolates for friends. Each chocolate costs R6. How many chocolates can he buy for that amount? Let x represent the number of chocolates.

$$6x = 50$$

$x = 8,310$. He can buy 8 chocolates.

4. Solve for x and check your answer by substitution

(a) $x + 5 = 9$ $x = 4$ because $4 + 5 = 9$

(b) $x + 7 = 19$ $x = 12$ because $12 + 7 = 19$

(c) $x + 18 = 34$ $x = 16$ because $16 + 18 = 34$

(d) $23 + x = 43$ $x = 20$ because $20 + 23 = 43$

UNIT 4

NUMERIC AND GEOMETRIC PATTERNS

7.1 Investigate, extend and describe numeric and geometric patterns

1. What are the next three numbers in each of these sequences?

3; 6; 9; 12; 15; **18; 21; 24**

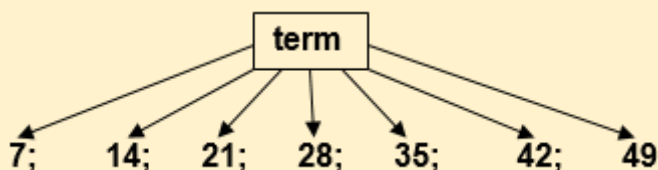
3; 6; 12; 24; 48; **96; 192; 384**

3; 6; 12; 21; 33; **48; 66; 87**

It is important to give at least 3 or 4 terms in each pattern in order to be able to continue with the pattern.

Numeric Patterns

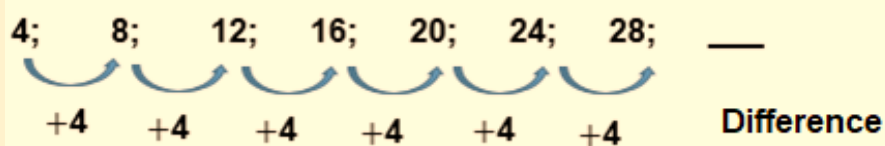
Each number in the given sequences can be formed from the previous number by performing the same or a similar action. We say there is a **pattern** in the sequence. This is called a **numeric pattern**. Each number in a pattern is called a **term**.



Patterns with a common difference

A sequence can be formed by repeatedly adding or subtracting the same number. In this case the difference between one term and the next is constant. In the following example, the common difference is $+4$.

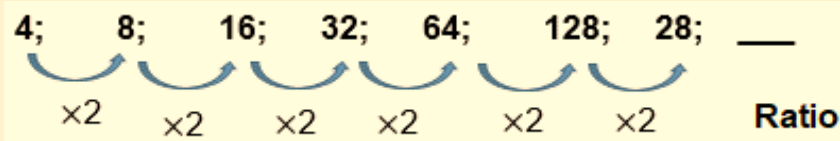
Example:



Patterns with a constant ratio

A sequence can be formed by repeatedly multiplying or dividing by the same number. In this case the ratio between one term and the next is constant. In the following example, the constant ratio is $\times 2$.

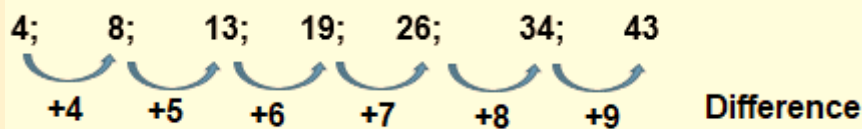
Example:



Patterns with neither common difference or constant ratio

A sequence can also be formed in such a way that neither the difference nor the ratio between one term and the next is constant. In the following example, there is, a pattern in the differences between the terms, which makes it possible to extend the sequence. Consecutive numbers, starting with 4, are added to form the next term.

Example:



2. Extend the following patterns and specify the relationship between consecutive terms.
- (a) 80; 85; 90; 95; 100; 105 (add 5)
 - (b) 200; 175; 150; 125; 100; 75 (subtract 25)
 - (c) 20; 25; 35; 50; 70; 95 (add consecutive multiples of 5 to get the term)
 - (d) 5; 20; 80; 320; 1 280; 5 120 (multiply previous term by 4)
 - (e) 1; 5; 3; 10; 7; 15; 13; 20
 - (f) 625; 125; 25; 5; 1; 0,2 (divide by 5)
 - (g) 6; 18; 54; 162; 486; 1458 (multiply previous term by 3)
 - (h) 0,9; 0,7; 0,5; 0,3; 0,1; - 0,1 (subtract 0,2)
 - (i) 1; 1; 2; 3; 5; 8; 13; 21 (to get the next term add the two previous consecutive terms)
 - (j) $\frac{1}{4}$; $\frac{1}{2}$; $\frac{3}{4}$; 1; $1\frac{1}{4}$; $1\frac{1}{2}$ (add $\frac{1}{4}$)

Geometric Patterns

A **geometric pattern** is the arrangement of objects, shapes, drawings, diagrams into a certain order or rule as shown in the following figure. It is very important not

to translate the geometric pattern into a numeric pattern but to rather structure and interpret it vertically and horizontally. Drawing the next diagram makes it easier for one to see how the pattern extends.

To get the next diagram, one needs to add one circle horizontally and one circle on both arms.

Example:



Diagram 1

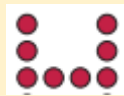


Diagram 2



Diagram 3



Diagram 4

3. Study the following geometric patterns and draw figure 5 and describe how each pattern grows or extends.

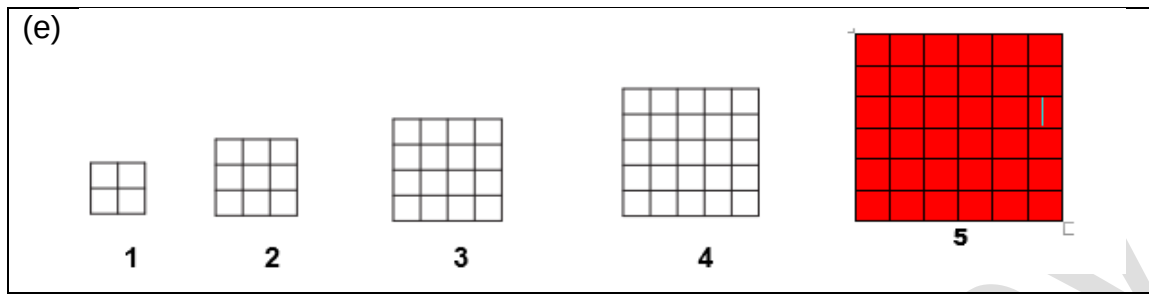
(a) 1 2 3 4 5

(b) Figure 1 Figure 2 Figure 3 Figure 4 Figure 5

(c) 1 2 3 4 5

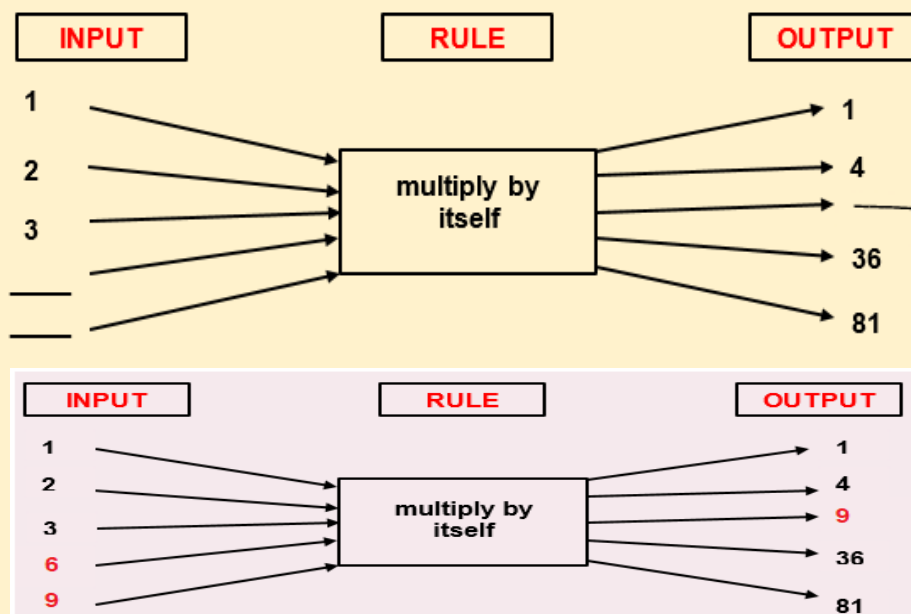
(d) 1 2 3 4 5

Answer

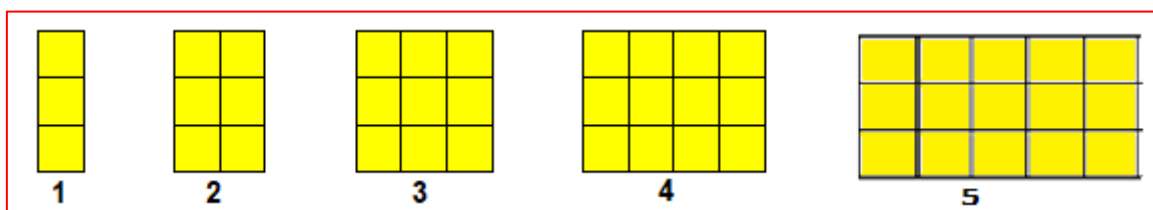


7.2 Patterns given in flow diagrams: input and output diagrams

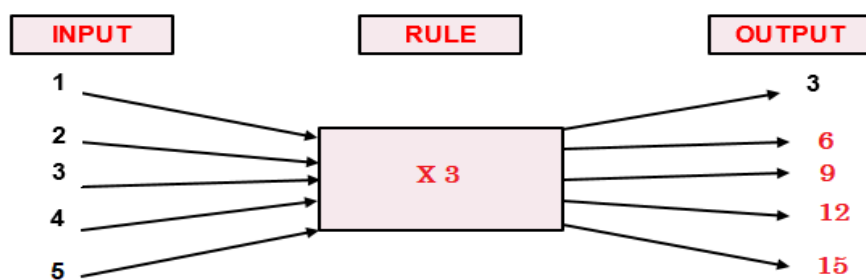
Flow diagrams are another way of displaying or representing patterns. A relationship between two quantities can be shown with a flow diagram. A flow diagram has numbers that are called inputs and outputs. Each **input number** in a flow diagram has a corresponding **output number**. The first (top) input number corresponds to the first output number. The second input number corresponds to the second output number, and so on. We operate the input with the rule to get the output.



- Study the following pattern.
Draw the 5th diagram.



(a) Translate the pattern into a flow diagram.

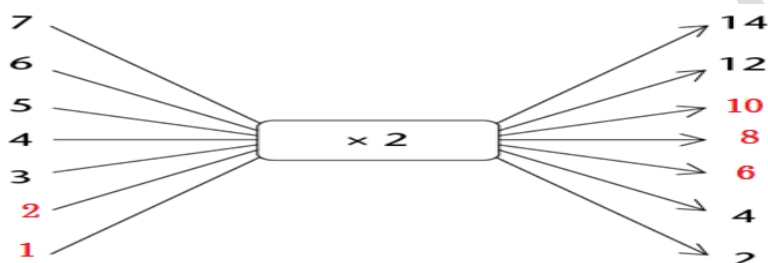


(b) What rule did you use to complete the pattern?

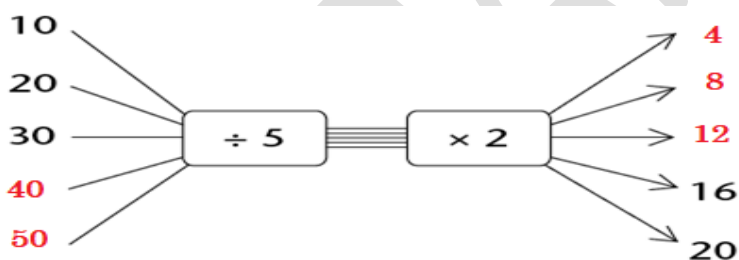
Multiply the input by 3

2. Complete the flow diagrams below.

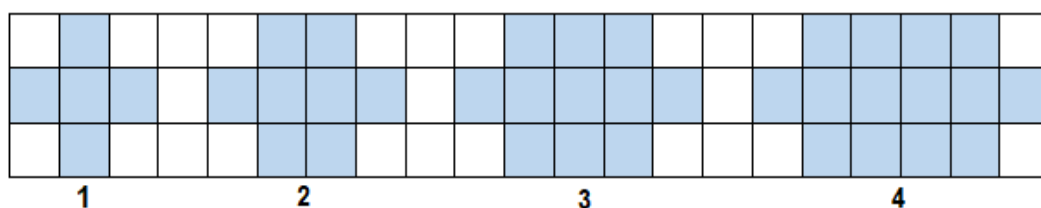
(a)



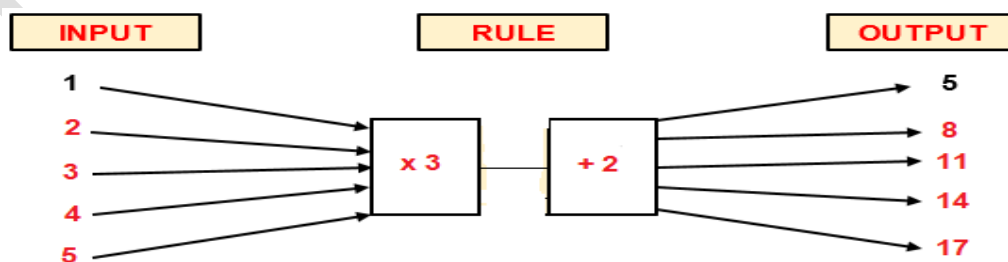
(b)



3. Study the following pattern.



(a) Translate the information into a flow diagram.



(b) What rule did you use to complete the flow diagram?

$$\text{Input} \times 3 + 2$$

7.3 Patterns given in tables

Patterns can also be represented using tables which is a useful way to record patterns. The first row of a table displays **input values (term number or position)** and the bottom row(s) display(s) **output numbers or terms**. The rule is used to generate the output values.

Term number (Position)	1	2	3	4	5
Term	Term1(T ₁)	Term 2(T ₂)	Term 3(T ₃)	Term 4(T ₄)	Term 5(T ₅)

1. Complete the following tables using the given rules.

(a) Add 37

Term number (Position)	1	2	3	4	5	6	7	8	9	10
Term	38	39	40	41	42	43	44	45	46	47

(b) Multiply by 7

Term number (Position)	1	2	3	4	5	6	7	8	9	10
Term	7	14	21	28	35	42	49	56	63	70

(c) Divide by 3

Term number (Position)	1	2	3	4	5	6	7	8	9	10
Term	0,33	0,67	1	1,33	1,67	2	2,33	2,67	3	3,33

(d) Multiply by 3 and add 5

Term number (Position)	1	2	3	4	5	6	7	8	9	10
Term	8	11	14	17	20	23	26	29	32	35

(e) Multiply by 3 and subtract 5

Term number (Position)	1	2	3	4	5	6	7	8	9	10
Term	-2	1	4	7	10	13	16	19	22	25

Analysing diagrams patterns

Example

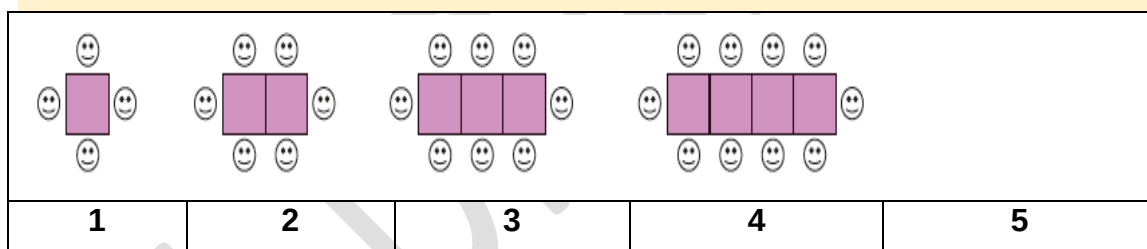
In a restaurant, one table sits 4 people, two tables sit 6 people, three tables sit 8 people, etc.

(a) Determine how many people would sit in 4 tables.

Note: In each diagram there are two people sitting opposite each other on the table and two more at the tables on the edges (one at the head and one at the foot).

In **diagram 4**, we have **four pairs** of people sitting opposite each other on the **4 tables**. There is also one at the **head** and **foot** of the table.

Answer: $4 \times 2 + 2 = 10$



(b) Determine how many people would sit in 5 tables.

Hint: Draw the 5th diagram and analyse the diagram.

In **diagram 5**, we have should have **five pairs** of people sitting opposite each other on the **5 tables**. There is also one at the **head** and **foot** of the table.

Answer: $5 \times 2 + 2 = 12$

(c) Determine how many people would sit in 15 tables.

Hint: Drawing will take a long time and it is important to come up with another strategy. Let us introduce a table.

Looking at the previous analysis, in diagram 15, we should have **fifteen pairs** of people sitting opposite each other on the **15 tables**. There is also one at the **head** and **foot** of the table.

Answer: $15 \times 2 + 2 = 32$

(d) Determine how many people would sit in any number (**n**) of tables.

Looking at the previous analysis, in diagram **n**, we should have **n pairs** of people sitting opposite each other on **n tables**. There is also one at the **head** and **foot** of the table

Answer: $n \times 2 + 2 = 2n + 2$

Using the table

Example

In a restaurant, one table sits 4 people, two tables sit 6 people, three tables sit 8 people, etc.

Term number (Position)	1	2	3	4	5		15	n
Term	4	6	8	10	12		32	$2n + 2$

Common difference

+ 2 + 2 + 2

Term 1 (T_1):	$1 \times 2 + 2 = 4$	The difference between the terms is 2. We therefore call 2 the common difference. We multiply the term number by the common difference and then add 2.
Term 2 (T_2):	$2 \times 2 + 2 = 6$	
Term 3 (T_3):	$3 \times 2 + 2 = 8$	
Term 4 (T_4):	$4 \times 2 + 2 = 10$	
Term 5 (T_5):	$5 \times 2 + 2 = 12$	
Term 15 (T_{15}):	$15 \times 2 + 2 = 32$	
Term n (T_n):	$n \times 2 + 2 = 2n + 2$	

$\therefore$ Term n (T_n) = $2n + 2$ meaning we can use this formula to get any term number.

2. A pattern with matches is shown below.

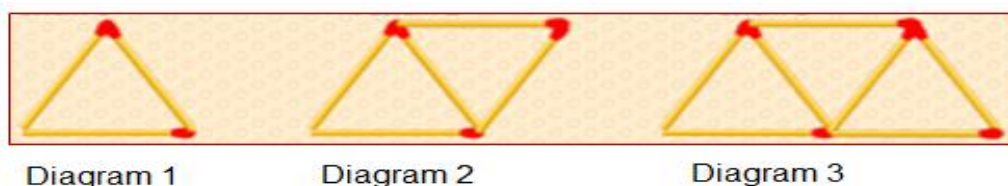


Diagram 1

Diagram 2

Diagram 3

(a) Explain how the pattern is formed.

The pattern is formed by joining 3 matchsticks to form a triangle and joining 2 matchsticks to previous pattern to get the next pattern join two matchsticks

(b) Copy and complete the table.

Term number (Position)	1	2	3	4	5		9	17	n
Number of matchsticks	3	5	7	9	11		19	35	$2n + 1$

(c) What rule did you use to complete the table?

To complete the table, add 2 to the previous term

(d) How many matches are needed to form Figure number 9 or term 9?

19

(e) How many matches are needed to form Figure number 17? Explain

$35 = 2 \times 17 + 1$, two multiplied by the position 17 plus 1

(f) What rule did you use to calculate the number of matches in question (e)?

$T_n = 2n + 1$

(g) Copy the flow diagram below and use it to write down the rule that you can use to calculate the number of matches needed for any figure in the pattern.



3. Another pattern with matches is shown below.



Figure 1



Figure 2



Figure 3

(a) Explain how the pattern is formed.

Pattern formed by forming a square with 4 match sticks and joining 3 matchsticks to the previous pattern to get the next pattern.

(b) Copy and complete the table

Term number (Position)	1	2	3	4	5		9	20	n
Number of matchsticks	4	7	10	13	16		28	61	$3n + 1$

(c) What rule did you use to complete the table?

Add 3 matchsticks to the previous term to get the next term.

(d) How many matches are needed for Figure 9 or term 9? 28

(e) How many matches are needed for Figure 20 or term 20? 61

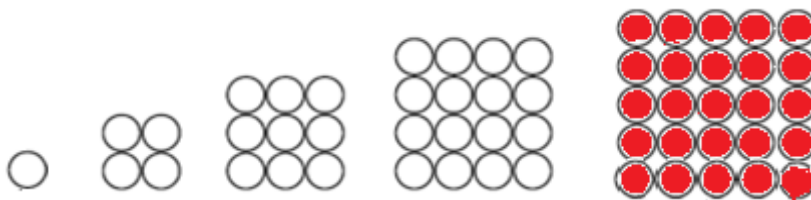
(f) What rule did you use to calculate the number of matches in question (e)?

$$T_n = 3n + 1$$

(g) Copy the flow diagram below and use it to write down the rule that you can use to calculate the number of matches needed for any figure in the pattern.



4. Study the pattern below.



(a) Draw the 5th diagram

(b) Explain how the pattern is formed. Multiply term number by itself

(c) Copy and complete the table using the pattern in 4.

Term number (Position)	1	2	3	4	5		9	20	n
No. of circles	1	4	9	16	25		81	400	n^2 $n \times n$

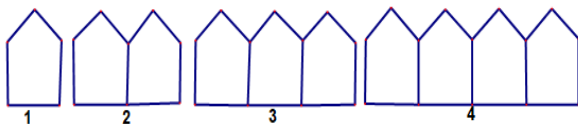
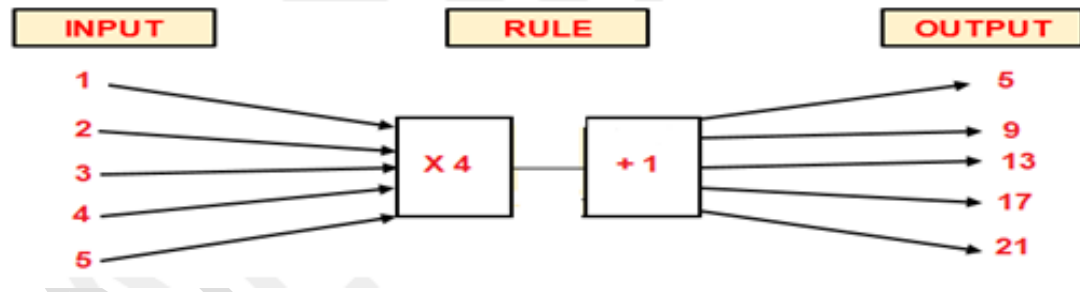
(d) Explain how the pattern will extend.

Square the term number / position

7.4 Equivalent forms: Verbally, Flow diagrams, Tables and Equations

When patterns of different forms give the same results, we say they are equivalent forms of patterns. Different forms of patterns can be presented in tables, flow diagrams, diagrams and verbally.

1. Use the geometric pattern to develop a table, flow diagram and describe how the pattern extends. Write a matching numeric pattern for the geometric pattern.

GEOMETRIC PATTERN						NUMERIC PATTERN		
						5; 9; 13; 17		
VERBAL DESCRIPTION						VERBAL DESCRIPTION		
Add 4 sticks to get the next pattern						Multiply the term number by 4 and add 1		
FLOW DIAGRAM						FLOW DIAGRAM		
								
TABLE								
Term number (Position)	1	2	3	4	5		20	n
Term	5	9	13	17	21		81	4n +1
GENERAL RULE OR EQUATION								
Term n (T _n) = 4n +1								

TERM 3

UNIT 1

MEASUREMENT: Length, Mass and Capacity

6.5 Length

Measuring a length means measuring of the length of objects using measuring instruments such as a ruler, meter stick, tape measure, trundle wheel, and a builder's tape. The length of a pencil can be measured in centimetres and millimeter using a ruler. The height of a person can be measured in metres using a tape measure. Examples of measuring instruments are shown below.



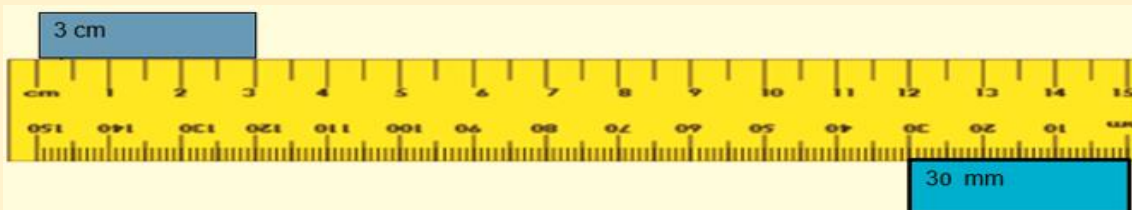
Measurement of length is recorded in units of length such as kilometres (km), metres (m), centimetres (cm) and millimetres (mm). Length can also be expressed in fractions and decimal fractions.

We can convert between units of length as shown in the following examples.

- $\frac{1}{2}$ cm arrow is the same as 5 mm



- 3 cm strip is the same length as the 30 mm strip.



- $2\frac{1}{2}$ cm can be written in decimal form as 2,5 cm or 25 mm.



- Write the following in meters and centimetres.

Example: 7 300 mm = 7 m and 30cm

- 63 m = 63 m and 0 cm
- 698 cm = 6 m and 98 cm
- 741 cm = 7 m and 41 cm

- Express the following measurements in metres (m).

- $\frac{1}{2}$ cm = 0,05 m
- $6\frac{1}{2}$ km = 6 500 m
- 5,6 mm = 0,0056 m

- Complete the table by filling in the missing units.

No.	Kilometres (km)	METRES (m)	Centimetres (cm)	Millimetres (mm)
(a)	3 km	3 000 m	30 000 cm	300 000 mm
(b)	0,010 km	110 m	1 000 cm	10 000 mm

Solve problems in context involving length.

Common fractions and decimal fractions form part of length:

1 km is equal to 1 000 m: $\frac{1}{2}$ kilometre = 500 m or 0,5 km

$\frac{1}{4}$ of a kilometre = 250 m or 0,25 km

First work through the examples and then solve the problems using fractions and decimal fractions:

- I bought 4 200 mm and then 3 300 mm of string.

- How much string did I buy?

7 500 mm

- Write your answer in mm and cm and then in m.

750 cm and 7,5 m

- Nelly had 4 800 metres hosepipe, she cut a 2 550 metres piece and used it for gardening

- How many meters does she have left?

2 300 m

(b) Write the answer in kilometers.

2,3 km

(c) Write the answer in (b) as a common fraction.

$2\frac{3}{10}$ km

3. Lebo works for 8 hours per week in the garden and Thabo works for 6 hours in the garden. They are paid R 566 in total.

(a) How many hours did they work altogether?

14 hours

(b) How much does each of them earn in one hour?

R40,42

(c) How much did Lebo earn more than Thabo?

R80,90

6.6 Mass

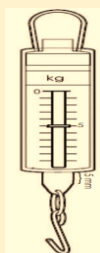
We use scales to measure mass and weight. Most people use both mass and weight as if they are the same, though they are different. The **mass** of an object is how much **matter** it contains and it does not change. The weight of an object changes according to gravity. An object weighs six times more on earth than on the moon, and weighs nothing in empty space, even though its mass remains the same everywhere.

A balance scale measures mass. A spring scale measures weight. On earth both types of scale give us the same approximate reading, so for everyday practical activities we can also use spring scale and balance scale to measure mass. Examples of measuring instruments are shown below.

Kitchen Scale



Spring balance



Bathroom Scale



Balance Scale



1. Complete the following table

No.	Object	Appropriate unit of measurement	Appropriate measuring instrument
(a)		Grams	Balancing scale
(b)		Kilograms	Bathroom scale
(c)		milligrams	Kitchen Scale

2. Convert the kilograms to grams

- (a) 4,73 kg 4 730 g
 (b) 5,4 kg 5 400 g
 (c) 0.68 kg 680 g
 (d) 0.07 kg 70 g

3. Convert the grams to kilograms

- (a) 6 070 g 6,07 kg
 (b) 9 200 g 9,2 kg
 (c) 320 g 0,32 kg
 (d) 850 g 0,85 kg

4. How many grams are in each of the following?

- (a) $\frac{2}{5}$ of a kg 400 g
 (b) $\frac{2}{4}$ of a kg 500 g
 (c) 0,2 of 10 kg 2 000 g

5. Solve problems in contexts involving mass

(a) A 500 g bag of flour costs R15,50. How much will 2,5 kg cost?

R77,50

(b) My brother has a mass of 25,43 kg and my sister is double that. What is my sister's mass?

My sister's mass is 50,86 kg

(c) A shopkeeper sold 15,26 kg of rice on Friday, Saturday sold 16,22 kg and 9,60 kg on Sunday. He took the rice from a 50 kg bag. How much rice was left in the bag?

8,92 kg of rice left

(d) My father eats 5 g of chocolate a day. How many grams will he eat in 2 weeks?

He will eat 70 g in two weeks

(e) A crate of apples has a mass of 56 kg.

i. If $\frac{3}{8}$ of the apples are rotten, how many kilograms can still be sold?

35 kg can still be sold

ii. Write the answer above in grams

35 000 g

1.3 Capacity / Volume

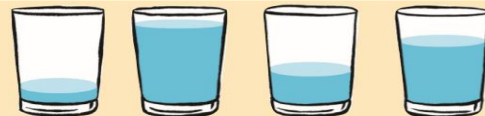
Capacity is the total amount of space inside a container.

Volume is the amount of space occupied by something.



When the container is filled up to the top, the volume of the liquid and capacity of the container will be same.

The four glasses have the same capacity, but they contain different volumes of water



- 1 litre = 1 000 ml
- $\frac{1}{2}$ litre = 500 ml
- $\frac{1}{4}$ litre = 250 ml
- $\frac{1}{8}$ of a litre = 125ml

1. Add the volume of the objects below and convert to litres.



2. Compare these numbers. Use $<$; $>$ or $=$ to replace the *

(a) $2\text{ l } 705\text{ ml} < 2\text{ l } 750\text{ ml}$

(b) $3\text{ l} > 2\text{ } 300\text{ ml}$

(c) $8\text{ } 080\text{ ml} = 8\text{ } 080\text{ ml}$

(d) $4\text{ l } 306\text{ ml} < 4\text{ l } 603\text{ ml}$

3. Arrange the following measurements from the smallest to the biggest:

(a) $4\text{ } 220\text{ ml}$; $0,68\text{ l}$; $0,5\text{ l}$; 98 ml

98 ml ; $0,5\text{ l}$; $0,68\text{ l}$; $4\text{ } 220\text{ ml}$

(b) $27,5\text{ l}$; $6\text{ } 470\text{ ml}$; $6\frac{3}{4}\text{ l}$; $6,09\text{ l}$

$6,09\text{ l}$; $6\text{ } 470\text{ ml}$; $6\frac{3}{4}\text{ l}$; $27,5\text{ l}$

(c) $7\frac{3}{4}\text{ l}$; $7\text{ } 900\text{ ml}$; $8\text{ l } 60\text{ ml}$; $9\text{ l},47\text{ ml}$

$7\frac{3}{4}\text{ l}$; $7\text{ } 900\text{ ml}$; $8\text{ l } 60\text{ ml}$; $9\text{ l},47\text{ ml}$

Problems in context involving capacity and volume:

1. Study the following diagram and answer questions.



(a) Which combination of spoons will be used to measure 30 ml of medicine?

15 ml; 7,5 ml, 5 ml and 2,5 ml or any other correct combination

(b) Which combination of spoons will be used to measure 20 ml of medicine accurately?

15 ml and 5 ml or any other correct combination

(c) Which combination of spoons will be used to measure 10 ml of medicine accurately?

7,5 ml and 2,5 ml or any correct combination

(d) If Tando drinks 7,5 millilitres of medicine from the 15-millilitre spoon. How many millilitres will be left?

7,5 ml will be left

2. Mr Jacobs uses water as follows weekly: 8 litres to wash the dishes; 50 litres to bath; 3,5 litres for cooking and 30 litres for drinking. How much water does he use weekly?

91,5 litres

3. The chef makes 10 litres of cocktail with granadilla juice, cranberry juice and lemonade. He uses 550 millilitres of cranberry juice and 2,5 litres of lemonade. How many litres of granadilla juice are in the 10 litres of fruit punch?

6,95 litres

4. Which is less: 9 bottles of 275 millilitres or 3 litres.

9 bottles of 275 millilitres is less than 3 litres

5. Patty makes 12 litres of fruit punch with granadilla juice, cranberry juice and lemonade. She uses $\frac{1}{4}$ millilitres of a cranberry juice and 2,5 litres of lemonade. What fraction of granadilla juice is in the 12-litre fruit punch?

9,5 ml = $9\frac{1}{2}$ ml of granadilla juice

6. I have a 5-litre bucket and a 250 ml cup.

(a) How many 250 ml cups will be needed to fill one bucket of water?

20 cups



(b) What fraction of a litre is 250 ml?

$\frac{1}{4}$

Consolidation

1. I bought a 4 700 meters and 2 550 meters of hosepipe from the store. How many meters did I buy altogether?

(a) Write the answer in kilograms.

7,250 kg

(b) Write the answer as a fraction.

$7\frac{1}{4}$ kg

2. Thato ran 2000 m in 1 hour. If he ran an hour longer at the same pace, how many kilometres would he have run?

4 km

3. Arthur has to travel 12 900 kilometres. He travelled 420 km on the first day of his journey, 830 km on the second day. How many more kilometres must he still travel to reach his destination?

11 650 km

4. In a marathon Peter ran 7, 4 km and Nikki ran 6320 meters. How much further did Peter run than Nikki?

1,08 km

5. Andile and Simi have ropes which when joined together are 2 572 m long. Andile's rope is 3 times longer than Simi's rope.

(a) How long is Andile's rope?

1 929 m

(b) How long is Simi's rope?

643 m

6. Busi weighs 112 kilograms. Her husband weighs 32 kilograms less than her mass. How much does her husband weigh?

80 kg

7. A farmer sells potatoes at R49,99 a bag. Each bag weighs 10 kg and he supplies five retail stores with 20 bags each

(a) How many kilograms will be received by each retail store?

200 kg

(b) How much will he be paid by each retail store?

R999,80

8. When my mother buys washing powder, she pays R45,65 for a 3 kg bag. How much does she pay for a 1 kg bag? **R15,22**
9. I was given a 4 kg bag of rice to take to my grandmother's house. On the way to my grandmother's house, the bag tore and a quarter of it spilled out. How many kilograms of rice will my grandmother have to cook? **3 kg**
10. One container holds 1, 6 litres, a second holds 1150 millilitres and a third holds 2 litres 460 millilitres. What is the three containers total capacity? Write the answer in litres? **5,21 ℓ**
11. Fifteen ladies drank tea in a restaurant. They drank 75 cups of tea altogether.
(a) How many cups did each lady drink, if they drank the same amount of tea? **Five cups**
12. How much tea did each lady drink in litres if each cup has a capacity of 250 ml? **1,250 ℓ**
13. Sandy drank 1 500 millilitres of water and Janine drank 750 millilitres.
(a) How much water did they drink altogether? **2 250 ml**
(b) Write the answer in litres. **2,250 ℓ**
(c) Write the answer as a fractions. **$1\frac{1}{4}$ ℓ**
14. In a bucket there are 5 litres of juice. How many learners can get 250 millilitres of juice? **20 learners**
15. The mass of 1 litre of water = 1 kg. Complete the following
(a) 2 litre of water **2 kg**
(b) 500 ml of water **0,5 kg**
(c) 125 ml of water **0,125 kg**
(d) 1,5 litres of water **1,5 kg**
(e) 4,55 litres of water **4,55 kg**

UNIT 2

CONSTRUCTION OF GEOMETRIC FIGURES

2.1 Revision

1. In Year 2 you learnt that angles are classified into types. Copy and complete the table. The first one has been done as an example for you.

ANGLE	SIZE OF AN NGLE	SKETCH OF AN ANGLE
Acute angle	Between 0° and 90°	
Right angle	90°	
Obtuse angle	Between 90° and 180°	
Straight angle	180°	
Reflex angle	Between 180° and 360°	
Revolution	360°	

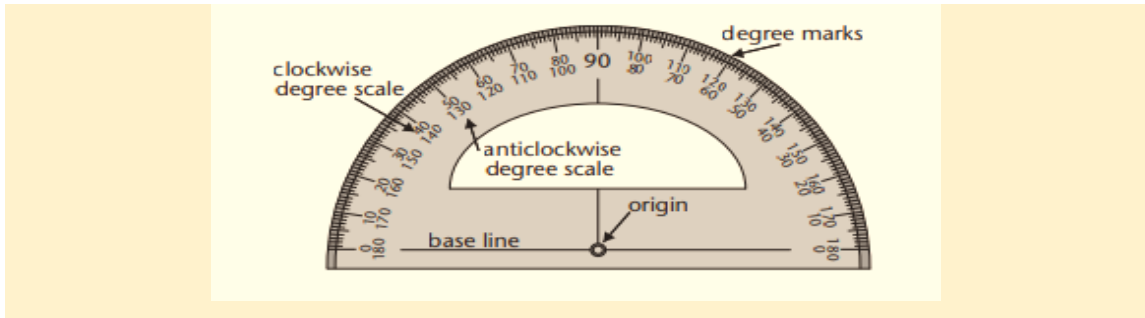
2. Give five everyday examples of **right angles or 90°** angles.

- (a) Corner of my teacher's table
- (b) Corner of a door
- (c) Corner of a window
- (d) Corner of an exercise book
- (e) Corner of my classroom etc.

[Any suitable answer is acceptable]

2.2 Measuring angles using a protractor.

Angles can be measured informally using a folded piece of paper; however, there is a special instrument for measuring angles. It is called a **protractor**. Look at the picture of a typical protractor with its parts labelled. Protractors can be big or small but they all measure angles in exactly the same way. The size of the protractor makes no difference to an angle size. The units of measurement for angles is a **degree ($^\circ$)**.



1. Study the pictures. What are these people using their protractors for?



Learner using a protractor to draw a triangle



Carpenter using a protractor to measure the angle of his roofing



Sailor man using a protractor to indicate direction in the sea

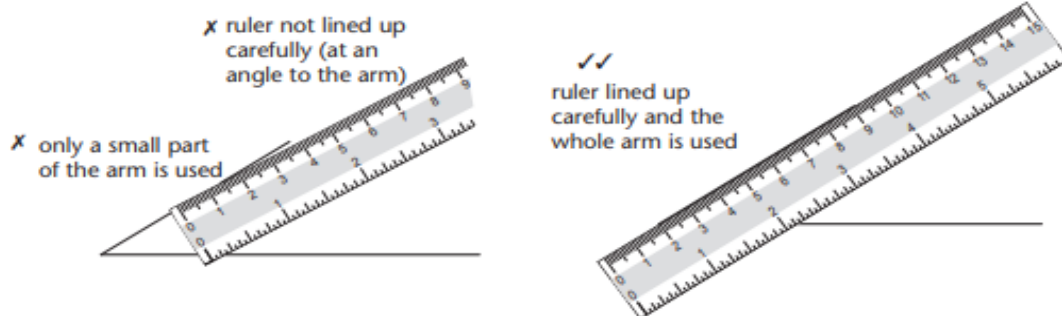
DBE Workbook 1
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Measuring angles using a protractor.

You will need a pencil, ruler, eraser and a protractor to measure angles.

Step 1: Check to see if the arms are long enough to measure. Are the angle arms long enough?

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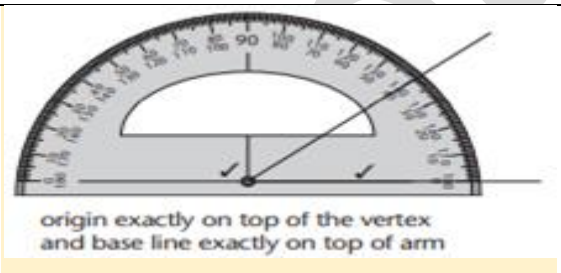
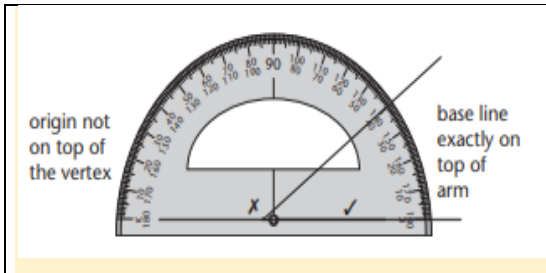
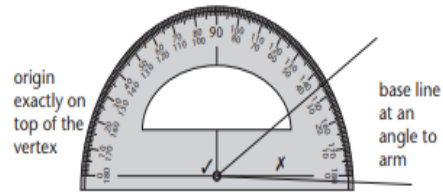
The angle arms must be a little longer than the distance from the origin of the protractor to its edge. If they are too short, use a sharp pencil and a ruler to make them longer. Be careful to line the ruler up with the arm.

Step 2: Line up the angle and your protractor

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Place your protractor on top of the angle. Make sure of the following:

- the origin is exactly on the vertex of the angle, and
- the base line is exactly on top of one of the arms of the angle.
- Keep adjusting the position of the protractor until the origin and the base line are exactly lined up.

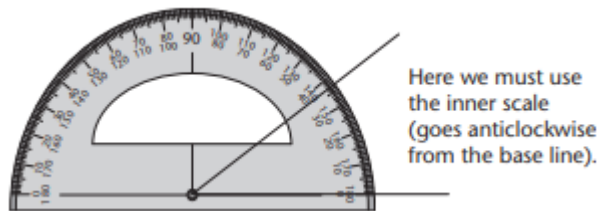


Once your protractor is in the correct place, keep a finger on the protractor to stop it from moving. If it moves... start again! You are now ready to make a measurement.

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Step 3: Measure the angle.

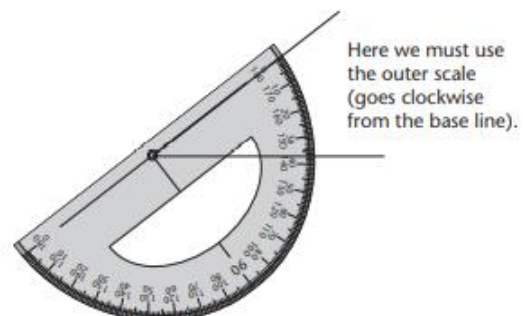
A protractor gives a clockwise degree scale and an anticlockwise degree scale. You choose the correct scale by



finding the one that starts with 0° on the angle arm. Look at where the other angle arm passes under the degree scale. That is where your measurement is.

You can also place the protractor on the angle using the other arm. Then the correct position looks like this:

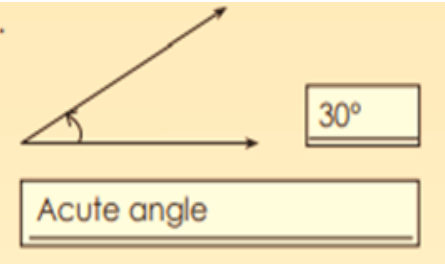
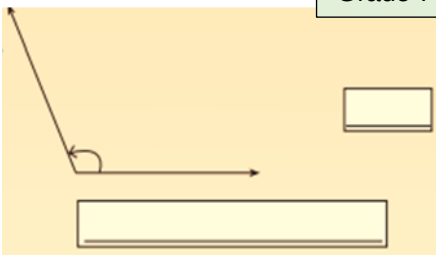
Here we must use the outer scale (goes clockwise from the base line). The angle in the pictures above is 37° .

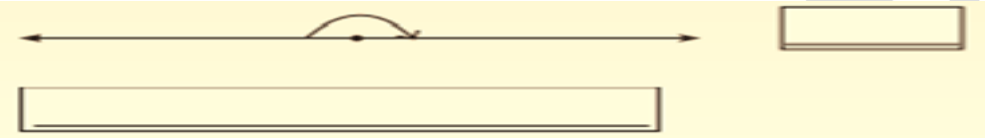


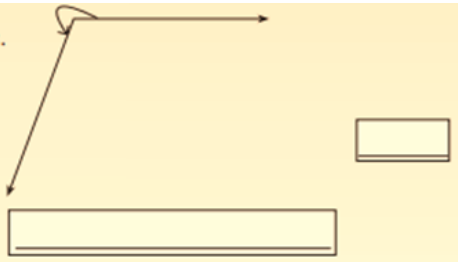
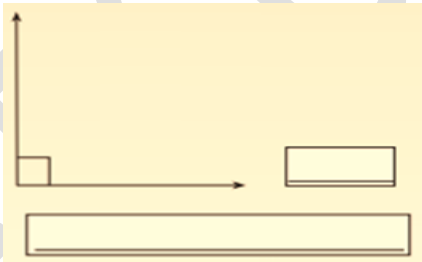
Do you agree? Do you see that there are two ways to measure an angle?

1. Measure and name each of the following angles. Remember to extend the arms of the angles to help measure.

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(a)  (b) 

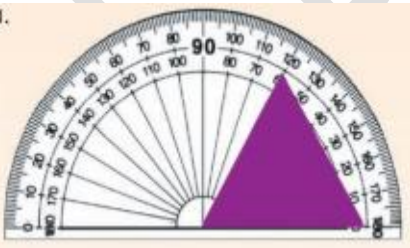
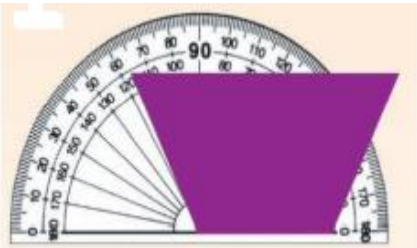
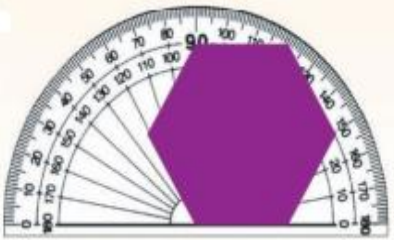
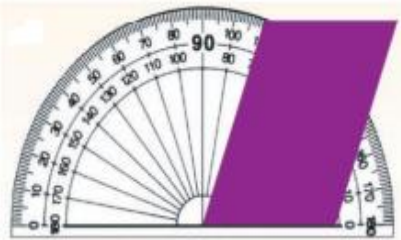
(c) 

(d)  (e) 

NB: A protractor must be used to measure the actual size of an obtuse and reflex angle

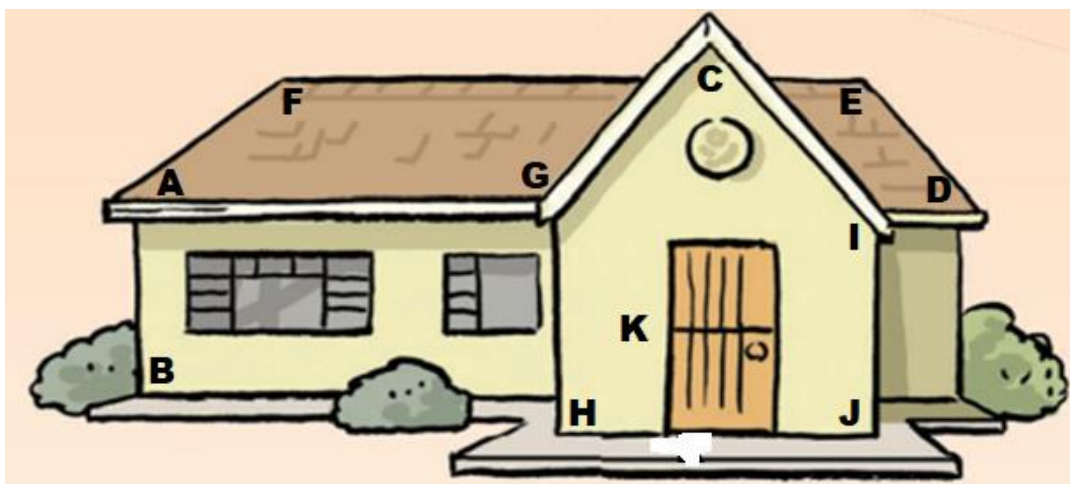
2. Write down the size of each of the interior angle that is indicated.

DBE Workbook 1
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No	Measured Angle	Size	No	Measured Angle	Size
(a)		60°	(b)		116°
(c)		120°	(d)		71°

3. Mark three (3) angles on the picture and measure them.

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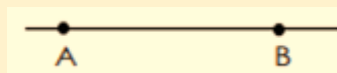
- | | | |
|------------------|--------------------|------------------|
| (a) Acute angle | (b) Right angle | (c) Acute angle |
| (d) Acute angle | (e) Obtuse angle | (f) Obtuse angle |
| (g) Obtuse angle | (h) Right angle | (i) Obtuse angle |
| (j) Right angle | (k) Straight angle | |

Mark any other correct angle identified by the learners

2.3 Constructions of angles

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Use a ruler, pencil and protractor to **construct** angles. An example of a 45° is given.



Step 1: Draw a line segment. Label it AB

Step 2: Place the protractor so that the origin (small hole) is over point A. rotate the protractor so that the base line is exactly along line AB



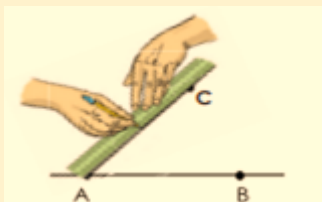
Step 3: Using (in this case) the inner scale, find the angle desired in Step 1. It is 45°



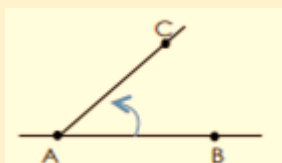
Step 4: Make a mark at this angle and remove the protractor.



Step 5: With the protractor or a ruler draw a straight line from A to the mark you just made. Label this point C.



Step 6: The line drawn makes an angle BAC that is 45° .



We can also represent it as $\angle CAB$ or $\widehat{BAC}$.

1. Draw a 45° angle ABC
2. Draw a 100° angle CDE
3. Draw 175° angle JKL

NB: Learners will need a pencil, ruler, eraser and protractor to draw the above angles.

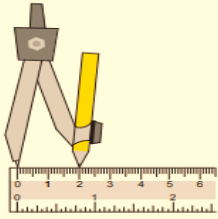
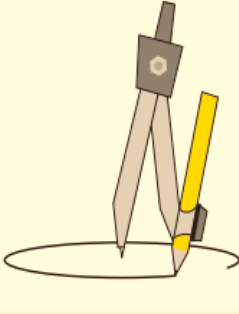
2.4 Construction of circles and parts of a circle

A circle is a two-dimensional shape made by drawing a curve that is always the same distance from the centre. We need a special instrument for drawing circles. It must have a pointy tip, like the centre pencil. It must also have a drawing tip, like the pencil you moved. If you can set the distance between these two tips, you can draw circles of any radius. This instrument is a pair of compasses, or often just a compass.

1. Use the following steps to help you draw a circle with a radius of 2 cm.

Step 1: Place a point on your exercise book using a lead pencil and label it **O** as shown. This point is the **centre** of your circle.



Step 2: Place the pointed tip on the zero line of your ruler. Carefully widen the angle between the arms. Move the pencil tip until it is exactly at 2 cm. Make sure that the pointed tip is still on zero. Be careful not to change the gap once it is set to 2 cm. This distance is the radius of the circle.	
Step 3: Gently push the pointed tip into point O. Push just deep enough into the paper to keep it in place.	
Step 4: Hold the handle between the forefinger and thumb of your writing hand. Keep your other hand out of the way. Use only one hand when you draw a circle with a compass.	
Step 5: Twist the handle between your thumb and forefinger. If you are right-handed, it is easiest to twist the compass clockwise. If you are left-handed, turn the compass anticlockwise. Let the pencil tip drag over the paper. Do not push down too hard on the pencil. Rather, push down lightly on the pointed arm as you draw. The pencil tip must move smoothly and easily.	

2. Draw circles with the following radii:

(a) 40 mm

(b) 3,5 cm

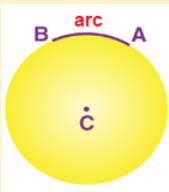
(c) $2\frac{1}{4}$ cm

NB: Learners will need a pencil, ruler, eraser and protractor to draw circles

Parts of a circle

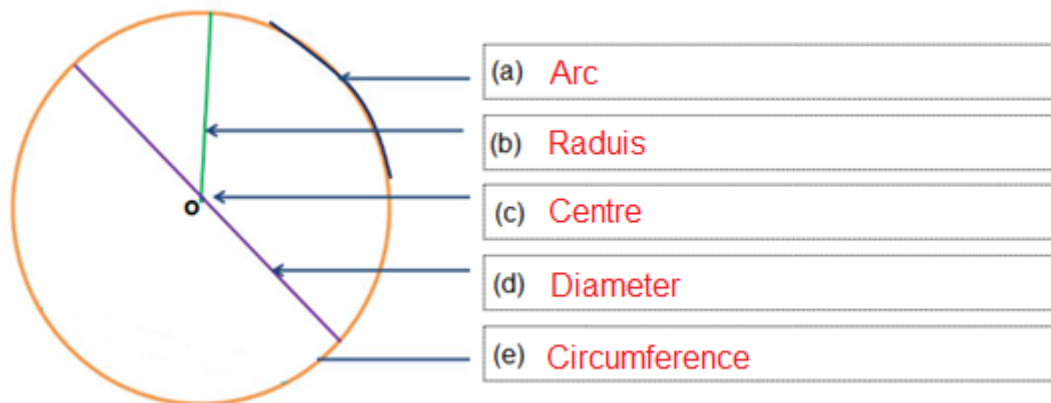
DBE Textbook Grade 7
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An **arc** is a small part of a circle. We use the term circumference when we refer to the distance around a circle or around any other curved shape.

	This is only part of a circle called an arc. 	This arc is a quarter circle. 	This arc is just more a semi-circle 	This arc is almost a full circle 	This is a full circle 
---	--	--	--	---	--

 The distance around the outside	 A centre is a point that is the same distance to every point on the circumference.
edge of a circle is called the circumference A radius is the distance from the centre to the circle or circumference. Radius = $\frac{1}{2}$ diameter	 A diameter is the distance from one side of a circle to the other going through the centre. Diameter = 2 x radius 

3. Label the following circle



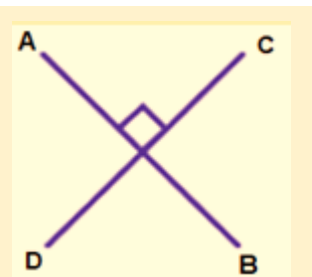
2.5 Construction of parallel and perpendicular lines

Perpendicular lines

Perpendicular lines are straight lines that meet each other at an angle of 90° . The sketch shows two perpendicular lines.

We say: CD is perpendicular to EF.

We write: $CD \perp EF$



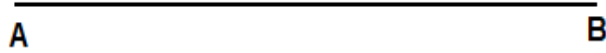
Constructing perpendicular lines

There are several methods we can use to construct parallel lines.

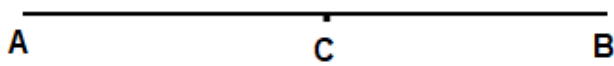
Method 1

1. Follow the steps to draw perpendicular lines. You will need a ruler, a pencil, an eraser and a protractor.

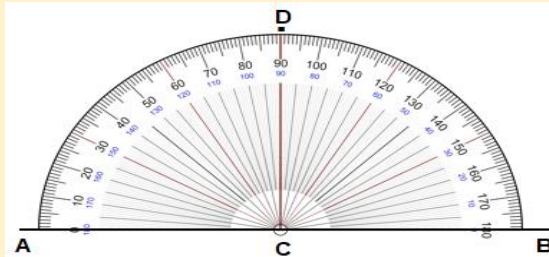
Step 1: Draw a line segment and name it AB.



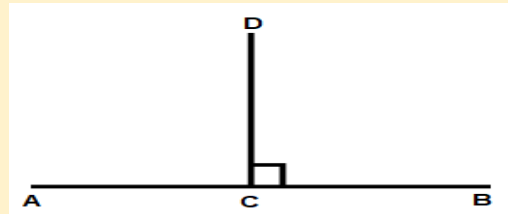
Step 2: Mark a point on line AB and name it C.



Step 3: Use your protractor. Put the origin on point C and mark a point on 90° and name it D.



Step 4: Join point C and D with a line.



AB is perpendicular to CD

$$AB \perp CD$$

2. Measure angle ACD, angle BCD and ACB to check how accurate your construction is.

$$\text{Angle ACD} = 90^\circ$$

$$\text{Angle BCD} = 90^\circ$$

$$\text{Angle ACB} = 180^\circ$$

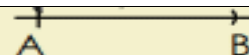
NB: Every learner will need a ruler, a pencil, an eraser and a protractor to draw the above mentioned angles.

Method 2

You will need a ruler, pencil, compass and an eraser.

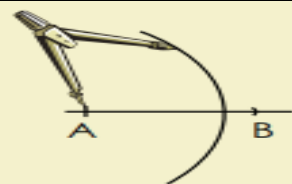
3. Follow the steps to construct perpendicular lines.

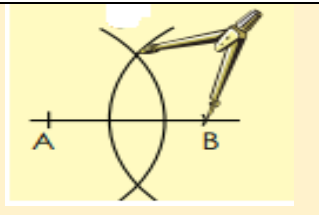
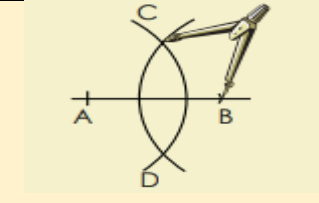
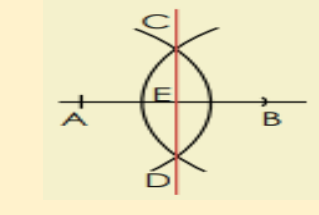
Step 1: Draw a line and mark point A and B on it.



Step 2:

Put the compass point on A, open it approximately two-thirds of the line length, and draw an arc.



Step 3: Do not change the width of the compass. Move the compass point to B and draw another arc which crosses the first (1) arc at two points.	
Step 4: Label the points where the arc intersect C and D.	
Step 5: Draw a line through point C and D bisecting line AB at E. AB is perpendicular to CD $AB \perp CD$	

4. Measure angle AEC and angle BED to check how accurate your construction is.

Angle AEC = 90°

Angle BED = 90°

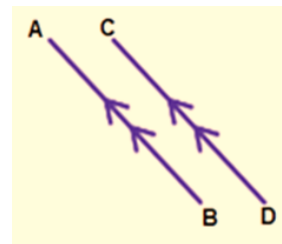
NB: Every learner will need a ruler, a pencil, an eraser and a protractor to draw the above mentioned angles.

Parallel lines

Parallel lines are straight lines that never touch each other and are always the same distance apart. They have the same direction. The sketch shows two parallel lines.

We say: AB is parallel to CD. We write: $AB \parallel CD$

The arrows on the middle of the lines show that the lines are parallel to each other.



Constructing parallel lines

There are different methods of constructing parallel lines. We will share three of those methods.

Method 1

5. Follow the steps to draw perpendicular lines. You will need a ruler, a pencil, an eraser and a protractor.

Step 1: Draw a line segment and name it AB.

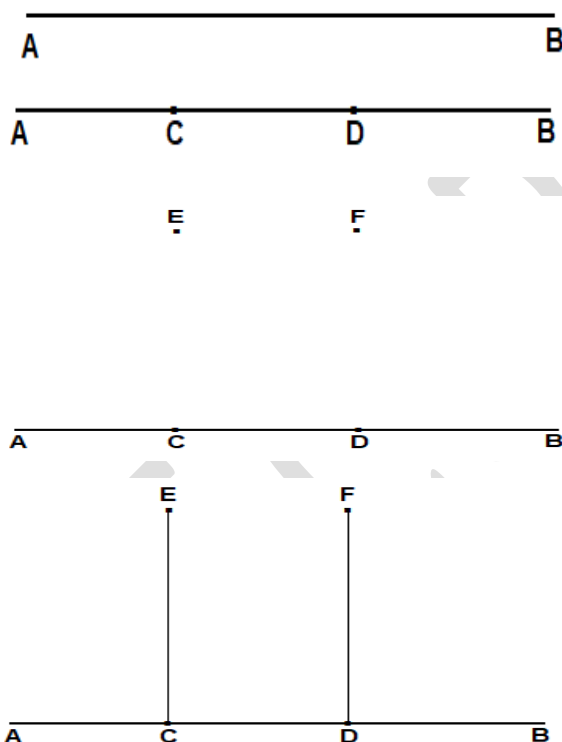
Step 2: Mark two (2) points on line AB and name them C and D.

Step 3: Use your protractor.

- Put the origin on point C, mark a point on 90° and name it E.
- Put the origin on point D, mark a point on 90° and name it F.

Step 4:

- Join points C and E with a line.
 - Join points D and F with a line
- CE is parallel to DF;
CE // DF



Method 2

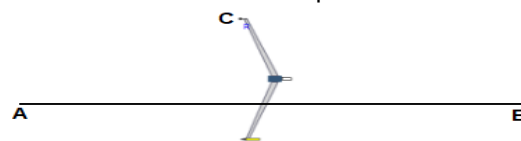
6. Follow the steps to draw perpendicular lines. You will need a ruler, a pencil, an eraser and a protractor.

Steps 1: Draw a line segment AB and a point C above the line.

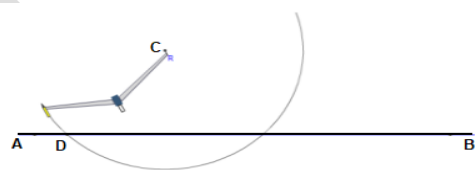


Steps 2: Place the compasses on point C and set its width to a little more than the distance to the line AB.

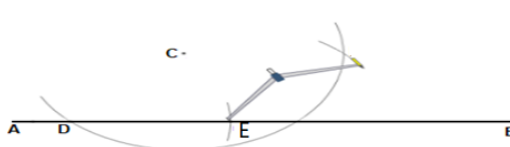
The exact distance is not important.



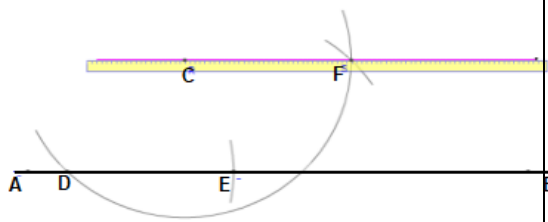
Step 3: Draw a wide arc from the right of C around so it crosses the line AB at two points. Label the left point D.



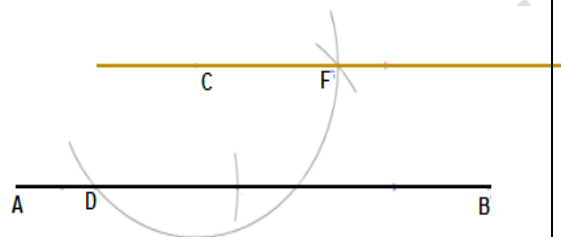
Step 4: Without adjusting the compasses' width, move the compasses to D and draw an arc across the line AB. Label this point E.



Step 5: Without adjusting the compasses' width, move the compasses to E and draw an arc across the large arc to the right of C. Label this point F.



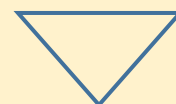
Step 6: Draw a straight line through points C and F.



Done. The line AB is parallel to the line CF or $AB \parallel CF$

2.6 Construction of triangles

In this section, you will learn how to construct triangles. You will need a pencil, a protractor, a ruler and a compass.



A triangle has three sides and three angles. We can construct a triangle when we know some of its measurements, that is, its sides, its angles, or some of its sides and angles.

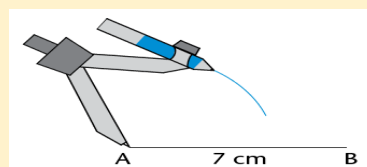
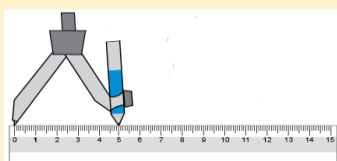
Constructing triangles when three (3) sides are given.

1. Study the following steps that describe how to construct $\triangle ABC$. Triangle ABC has side lengths of 3 cm, 5 cm and 7 cm.

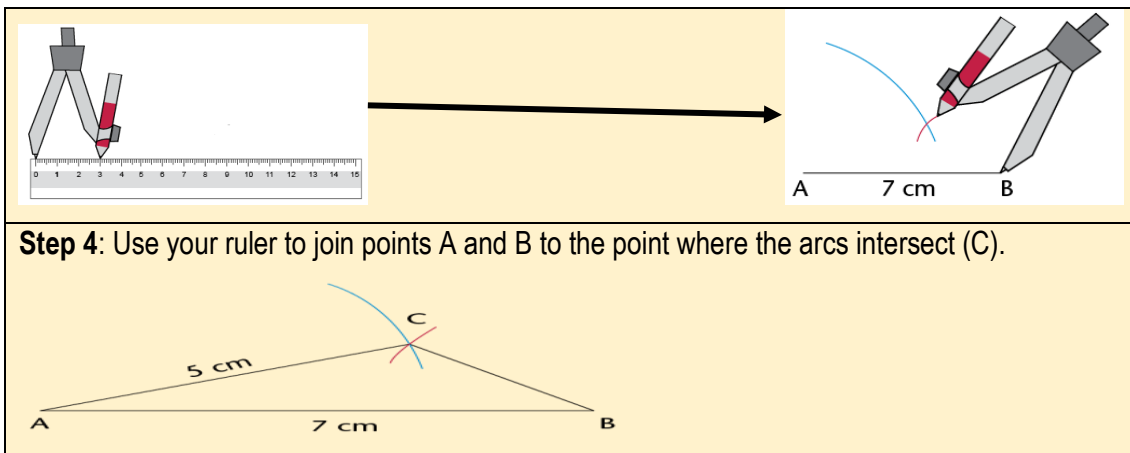
Step 1: Draw one side of the triangle using a ruler. It is often easier to start with the longest side.



Step 2: Set the compass width to 5 cm. Draw an arc 5 cm away from point A. The third vertex of the triangle will be somewhere along this arc.



Step 3: Set the compass width to 3 cm. Draw an arc from point B. Note where this arc crosses the first arc. This will be the third vertex of the triangle.



2. Work in your exercise book. Follow the steps in (1) to construct the following triangles:

- (a) $\triangle ABC$ with sides 6 cm, 7 cm and 4 cm
- (b) $\triangle KLM$ with sides 10 cm, 5 cm and 8 cm
- (c) $\triangle PQR$ with sides 5 cm, 9 cm and 11 cm

NB: Learners will need a pencil, a protractor, a ruler and a compass to construct the above triangles

2.7 Construction of quadrilaterals

A **quadrilateral** is a closed shape with four straight sides and four angles.

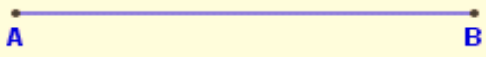
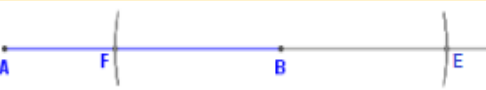
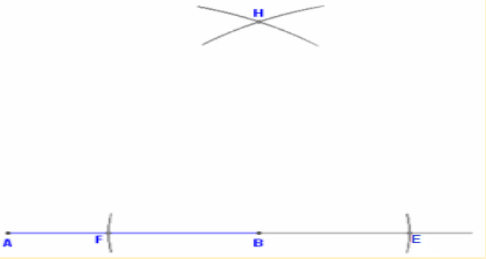
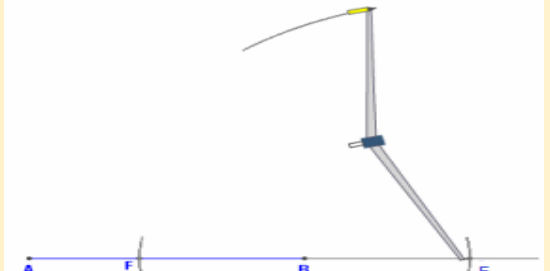
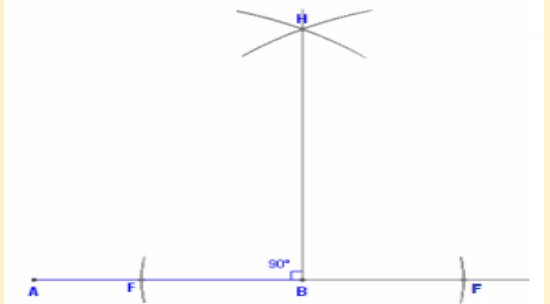
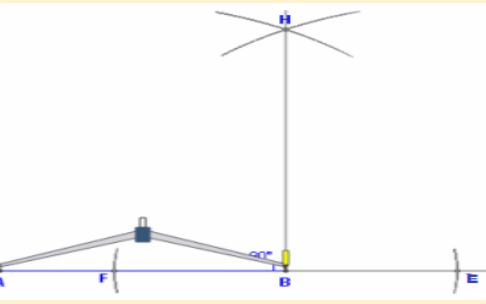
1. Match the following quadrilaterals and their descriptions.

(a) Rhombus	i. All sides are equal and opposite angles are equal
(b) Square	ii. A polygon with four equal sides and equal angles
(c) Rectangle	iii. Two pairs of opposite sides are equal and parallel. All angles are 90°
(d) Kite	iv. Two pairs of adjacent sides are equal. One pair of opposite angles equal
(e) Parallelogram	v. Two pairs of opposite sides are equal and parallel. Opposite angles equal
(f) Trapezium	vi. One pair of opposite sides are parallel

The focus of this section is construction of quadrilaterals using appropriate Mathematical instruments.

Constructing a square

2. Use the following steps to help you construct a square. You will need a ruler, compass, eraser and a sharp pencil.

Step 1: We start with a given line segment AB. This will become one side of the square.	
Step 2: Extend the line AB to the right.	
Step 3: Set the compasses on B and any convenient width. Scribe an arc on each side of B, creating the two points F and G.	
Step 4: With the compasses on E and any convenient width, draw an arc above the point B.	Step 5: Without changing the compasses' width, place the compasses on F and draw an arc above B, crossing the previous arc, creating point H 
	
We now create four sides of the square the same length as AB	
Step 6: Draw a line from B through H. This line is perpendicular to AB, so the angle ABH is a right angle (90°); This is the second side of the square	Step 7: Set the compasses on A and set its width to AB. This width will be held unchanged as we create the square's other three sides.
	
Step 8: Draw an arc above point A.	Step 9: Without changing the width, move the compasses to point B. Draw an arc across BH creating point C - a vertex of the square.

Step 10: Without changing the width, move the compasses to C. Draw an arc to the left of C across the exiting arc, creating point D - a vertex of the square.	Step 11: Draw the lines CD and AD
Done. ABCD is a square where each side has a length AB	

- Use a compass and a ruler to construct a square ABCD with a side length of 5 cm.

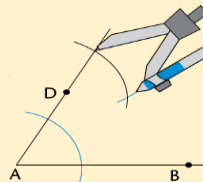
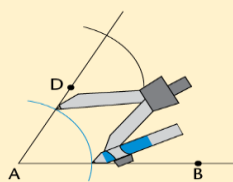
NB: Learners will need a pencil, a protractor, a ruler and a compass to construct the above square.

Constructing a parallelogram from parallel lines

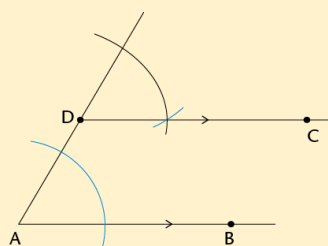
- Use the following steps to construct quadrilaterals. You will need a ruler, a pencil, an eraser and a pair of compasses.

Step 1: From line segment AB, mark a point D. This point D will be on the line that will be parallel to AB. Draw a line from A through D.	Step 2: Draw an arc from A that crosses AD and AB. Keep the same compass width and draw an arc from point D as shown.
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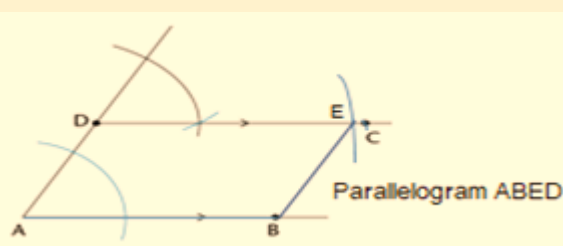
Step 3: Set the compass width to the distance between the two points where the first arc crosses AD and AB. From the point where the second arc crosses AD, draw a third arc to cross the second arc.



Step 4: Draw a line from D through the point where the two arcs meet. DC is parallel to AB.



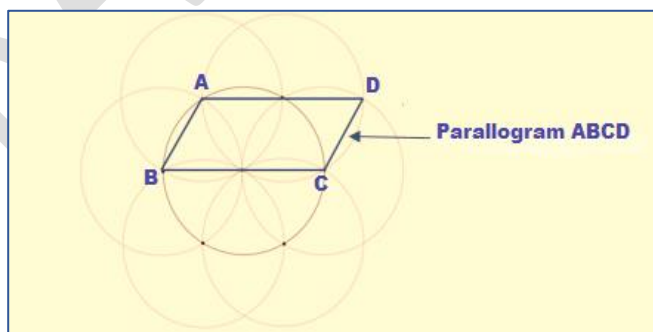
Step 5: Use the compass to measure the distance from A to B. Put your point on D and make an arc on DC and name it E. Join point B and E.



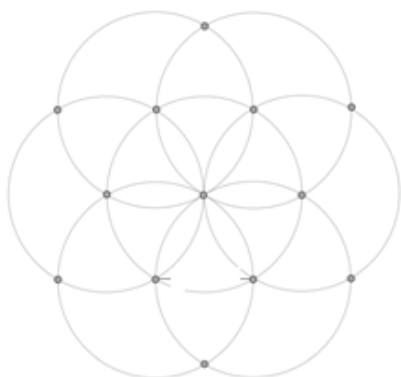
Constructing other quadrilaterals using circles on circles.

You will need a pencil and a ruler to draw quadrilaterals. A structure of circles is given.

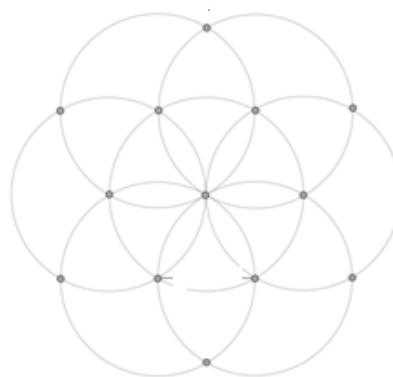
7. Use the following structure to help you draw quadrilaterals. You may draw more than one (1) quadrilateral on the given structure. See an example of a parallelogram.



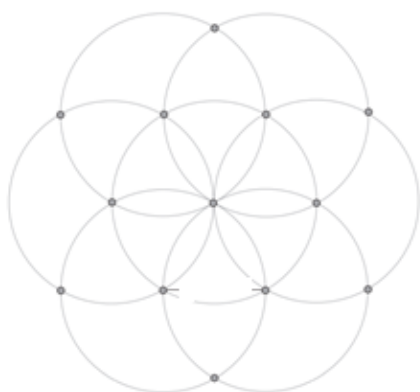
(a) Rectangle EFGH



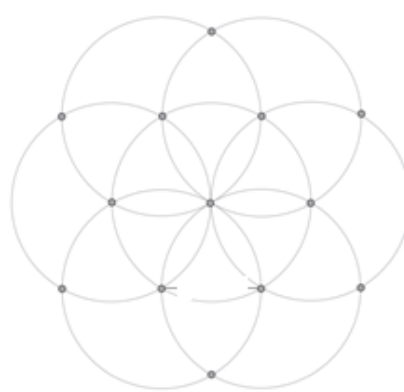
(b) Rhombus PQRS



(c) Trapezium JKLM



(d) Kite OPQR



NB: Learners will need a pencil and a ruler to make the constructions.

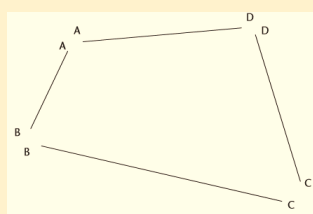
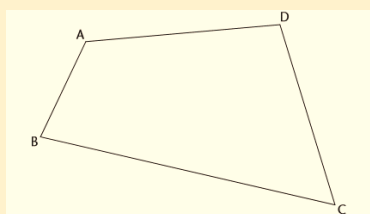
UNIT 3

GEOMETRY OF STRAIGHT LINES

3.1 Defining lines

Line Segment

Study the following quadrilateral. It has four sides. Each side of a quadrilateral is a line segment. A line segment has definite starting point and a definite ending point.



ABCD therefore has line segment AB, BC, CD and AD

Ray

The line below has a definite starting point and no definite ending point. The arrow shows that the line is continuing to infinity.



Straight line

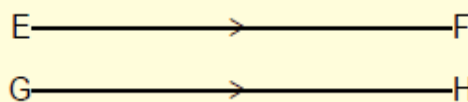
A straight line is a line with no definite starting point nor definite ending point, it will go on and on to infinity. The arrows on both ends indicate that the line does not stop.



Parallel lines

Two or more lines that are equidistant apart are called parallel lines.

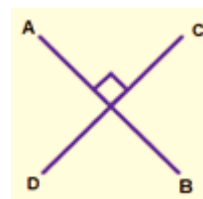
The symbol $\parallel$ represents parallel, therefore,
 $EF \parallel GH$



Perpendicular lines

When two lines intersect at right angles, they form perpendicular lines as shown on the diagram along side.

This is written as $AB \perp CD$, the symbol $\perp$ represents perpendicular



1. Draw ray TJ

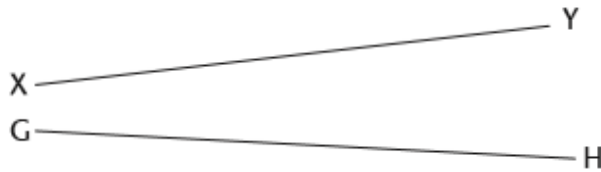


2. Did you draw the whole ray? Explain

Yes, it has a definite starting point and no ending point.

3. Do the following lines ever meet? Are they parallel lines? Explain.

- (a) No, they are not continuing



- (b) Yes, if the lines are extended, they will meet.



- (c) No, Point J is the definite starting point for JK while point S continues indefinitely.

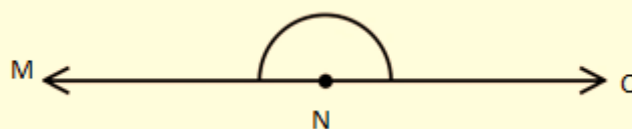


4. Match column A with Column B

Column A	Column B
(a) Line Segment	i.
(b) Perpendicular lines	ii.
(c) Parallel lines	iii.

Angles on a straight line

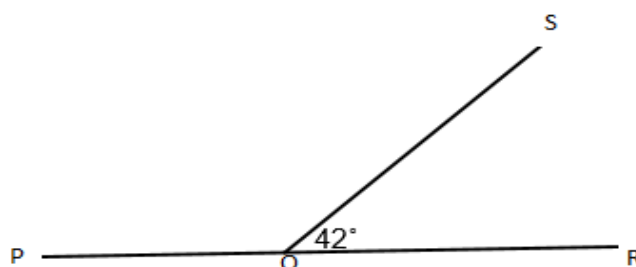
The sum of the angles on a straight line add up to 180°



$M\hat{N}O$ is a straight angle. A straight angle measures 180° .

5. Study the given diagrams and answer questions.

(a) Calculate the size of $P\hat{Q}S$ if $S\hat{Q}R$ is equal to 42°



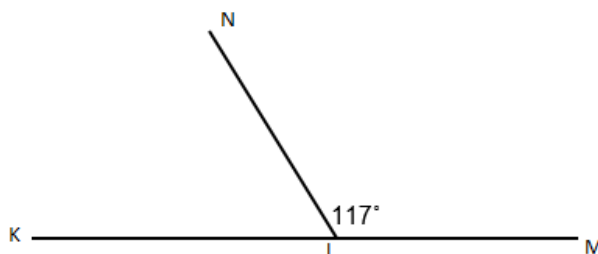
$P\hat{Q}S + S\hat{Q}R = 180^\circ$ (Sum of angles on a straight line are supplementary)

$P\hat{Q}S + 42^\circ = 180^\circ$

$P\hat{Q}S = 180^\circ - 42^\circ$

$P\hat{Q}S = 138^\circ$

(b) Calculate the size of $K\hat{L}N$ if $N\hat{L}M$ is equal to 117°



$K\hat{L}N + N\hat{L}M = 180^\circ$ (Sum of angles on a straight line are supplementary)

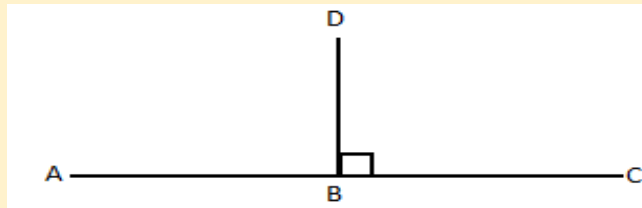
$K\hat{L}N + 117^\circ = 180^\circ$

$K\hat{L}N = 180^\circ - 117^\circ$

$K\hat{L}N = 63^\circ$

Intersecting lines

Two or more lines which share a common point are called intersecting lines. The common point is called the point of intersection. Line AC meet line BD at point B, therefore, B is called a point of intersection.



$\angle DBC$ is a right angle and it is equal to 90° . $\angle DBC$ lies next to $\angle ABD$ on a straight line AC. We therefore say $\angle DBC$ is adjacent to $\angle ABD$ and they add up to 180° . When angles add up to 180° we say they are supplementary.

$\angle ABD + \angle DBC = 180^\circ$ (Adjacent angles on a straight line AC)

$$\angle ABD + 90^\circ = 180^\circ$$

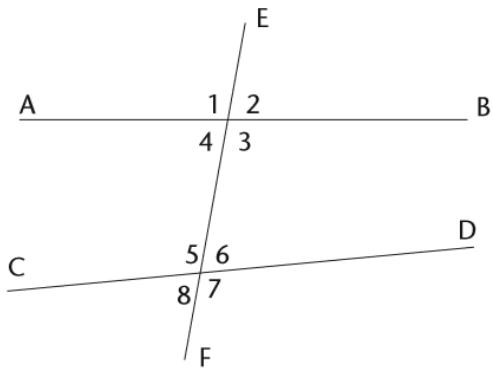
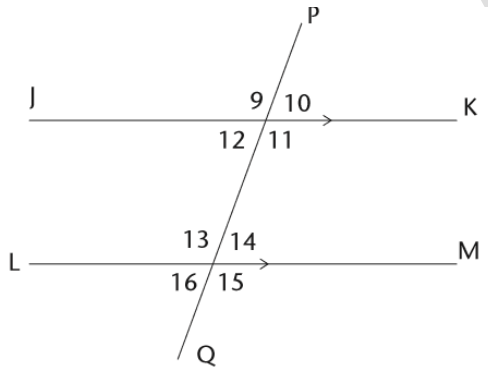
$$\angle ABD = 180^\circ - 90^\circ$$

$$\angle ABD = 90^\circ$$

6. For the following diagrams, use a protractor to measure all the four angles in each diagram

Figure 1	Figure 2
$\angle SUT = \underline{\hspace{2cm}}^\circ$ $\angle TUV = \underline{\hspace{2cm}}^\circ$ $\angle VUQ = \underline{\hspace{2cm}}^\circ$ $\angle QUS = \underline{\hspace{2cm}}^\circ$	$\angle ABC = \underline{\hspace{2cm}}^\circ$ $\angle CBD = \underline{\hspace{2cm}}^\circ$ $\angle DBE = \underline{\hspace{2cm}}^\circ$ $\angle BEA = \underline{\hspace{2cm}}^\circ$
$\angle SUT = \angle VUQ$ $\angle TUV = \angle QUS$	$\angle ABE = \angle CBD$ $\angle DBE = \angle ABC$
When two lines intersect, the opposite angles are equal. We call them vertically opposite angles	

7. Measure the angles in each diagram below and fill in the measurements in the following table.

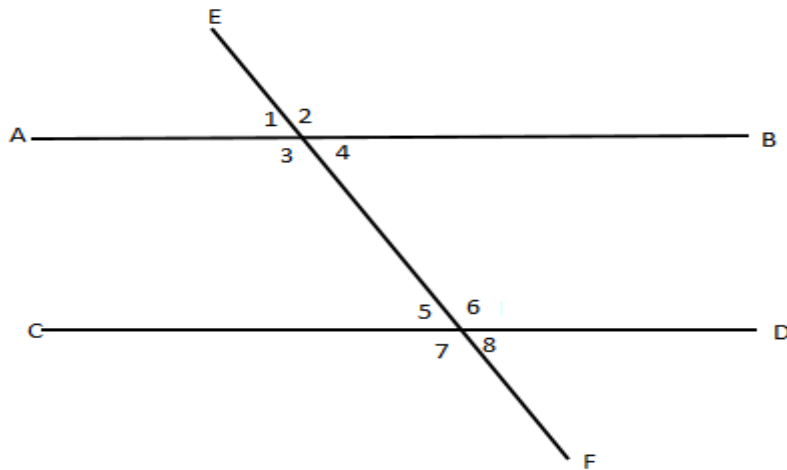
Non parallel lines cut by a transversal	Parallel lines cut by a transversal
	

Learners use protractors to measure the angles.

Angles	When two lines are not parallel	When two lines are parallel
Corresponding angles	$\hat{1} = \underline{\hspace{1cm}}^\circ$; $\hat{5} = \underline{\hspace{1cm}}^\circ$ $\hat{4} = \underline{\hspace{1cm}}^\circ$; $\hat{8} = \underline{\hspace{1cm}}^\circ$ $\hat{2} = \underline{\hspace{1cm}}^\circ$; $\hat{6} = \underline{\hspace{1cm}}^\circ$ $\hat{3} = \underline{\hspace{1cm}}^\circ$; $\hat{7} = \underline{\hspace{1cm}}^\circ$	$\hat{9} = \underline{\hspace{1cm}}^\circ$; $\hat{13} = \underline{\hspace{1cm}}^\circ$ $\hat{12} = \underline{\hspace{1cm}}^\circ$; $\hat{16} = \underline{\hspace{1cm}}^\circ$ $\hat{10} = \underline{\hspace{1cm}}^\circ$; $\hat{14} = \underline{\hspace{1cm}}^\circ$ $\hat{11} = \underline{\hspace{1cm}}^\circ$; $\hat{15} = \underline{\hspace{1cm}}^\circ$
Alternate angles	$\hat{4} = \underline{\hspace{1cm}}^\circ$; $\hat{6} = \underline{\hspace{1cm}}^\circ$ $\hat{3} = \underline{\hspace{1cm}}^\circ$; $\hat{5} = \underline{\hspace{1cm}}^\circ$	$\hat{12} = \underline{\hspace{1cm}}^\circ$; $\hat{14} = \underline{\hspace{1cm}}^\circ$ $\hat{11} = \underline{\hspace{1cm}}^\circ$; $\hat{13} = \underline{\hspace{1cm}}^\circ$
Co-interior angles	$\hat{4} + \hat{5} = \underline{\hspace{1cm}}^\circ$ $\hat{3} + \hat{6} = \underline{\hspace{1cm}}^\circ$	$\hat{12} + \hat{13} = 180^\circ$ $\hat{11} + \hat{14} = 180^\circ$

8. Complete: When two parallel lines are cut by a transversal, corresponding angles are **equal** and alternate angles are **equal** and the co-interior angles are **supplementary**.

9. Given the following diagram with AB parallel to CD.



(a) Name a pair of corresponding angles

$$\hat{2} = \hat{6} \text{ and } \hat{1} = \hat{5}; \hat{3} = \hat{7} \text{ and } \hat{4} = \hat{8}$$

(b) A pair of alternate angles

$$\hat{4} = \hat{5} \text{ and } \hat{3} = \hat{6}$$

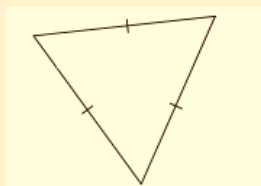
(c) A pair of Co-interior angles

$$\hat{4} \text{ and } \hat{6}; \hat{3} \text{ and } \hat{5}$$

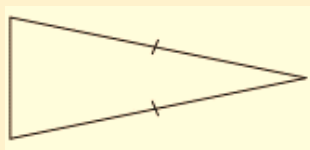
UNIT 4

GEOMETRY OF 2-D SHAPES

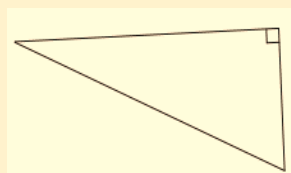
A triangle with three equal sides is an **equilateral triangle**.



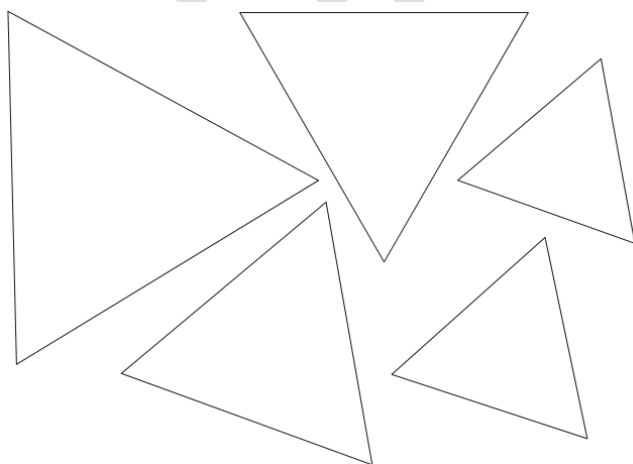
A triangle with two equal sides is an **isosceles triangle**.



A triangle with a right angle is called the **right-angled triangle**. The side opposite to the right angle is the **hypotenuse**.

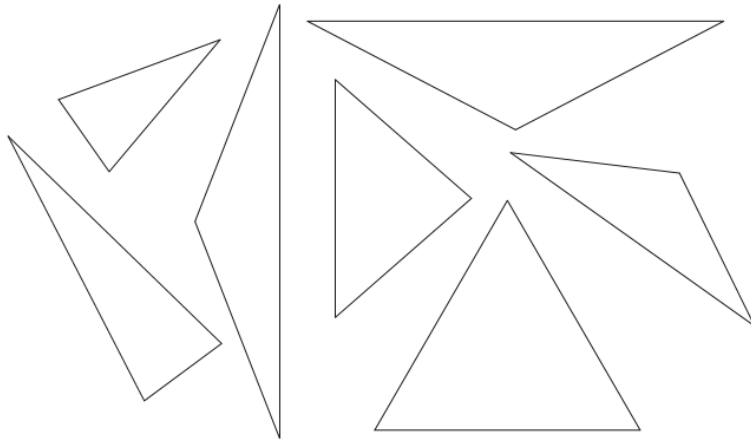


1. Study the following triangles and follow instructions.
 - (a) Measure the sides of each triangle below.



- i. Write what you notice about lengths of their sides.
All the three sides of each triangle are equal
- ii. Give the special name for the triangles.
Equilateral triangles

(b) Measure the sides of each triangle below and



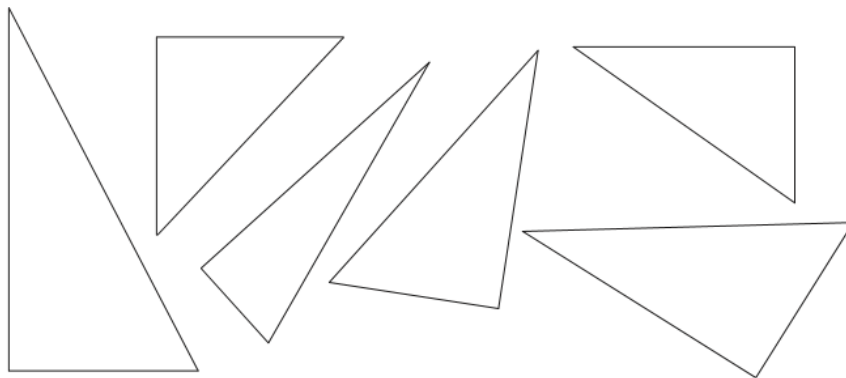
- i. Write what you do notice about lengths of their sides

The two adjacent sides are equal

- ii. Give the special name for the triangles

Isosceles triangle

(c) Measure the angles of each of the following triangles



- i. What is common about the sizes of these angles

They all have a right angle

- ii. Measure the sides of each triangle

NB: Learners need to use rulers to measure sides of triangles.

- iii. Identify the longest side and what do you notice about the position of the longest side.

Learners need to mark the longest side on the figures. The longest side is always opposite the right angle

- iv. Give the special name for the triangles

Right-angled triangle

2. Match the triangle in column A with the name in column B

Column A	Column B
	Right angled triangle
	Equilateral triangle
	Isosceles triangle

3. What do these triangular road signs mean? Draw the other two you can think of.



Pedestrian crossing

Children crossing / school ahead

Beware of cattle.

Any suitable diagram

Any suitable diagram



Slippery Road



speed hump



Road works ahead



Cross-Roads ahead

4. Study the picture of the Nelson Mandela bridge in Johannesburg. Identify parallel lines, perpendicular lines and line segments and label them on the diagram.



Learners identify the parallel lines, perpendicular lines and line segment by marking them on the picture.

TERM 4

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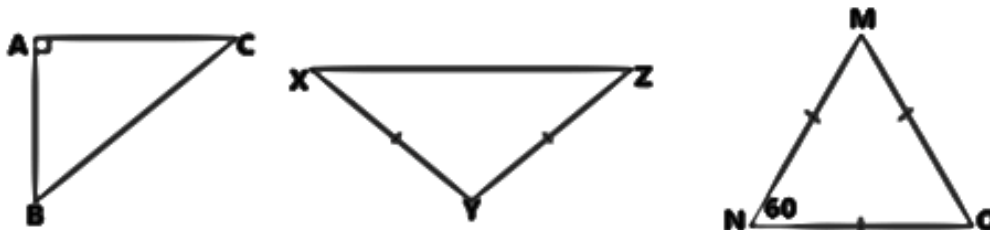
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UNIT 1

GEOMETRY OF 2-D SHAPES

1.1 Revision

2. Use the following triangles to answer the questions below:



- (a) Which triangle has only two sides that are equal? What is this type of triangle called?

Triangle XYZ

It is an isosceles triangle

- (b) If triangle XYZ is an isosceles triangle with the perimeter of 16 cm. If $XY = YZ = 4$ cm, calculate XZ.

$$\text{Perimeter} = XY + XZ + YZ$$

$$16 = 4 + 4 + YZ$$

$$16 - 8 = YZ$$

$$YZ = 8 \text{ cm}$$

- (c) Which triangle has all three sides equal? What is this type of a triangle called?

Triangle MNO: It is called an equilateral triangle

- (d) If triangle MNO is an equilateral triangle with angle MNO equals to 60° , what is the sum of angles NOM and OMN?

$\angle NOM + \angle OMN = 120^\circ$ (All angles of an equilateral triangle are equal)

- (e) Which triangle has an angle that is equal to 90° ? What is this type of a triangle called?

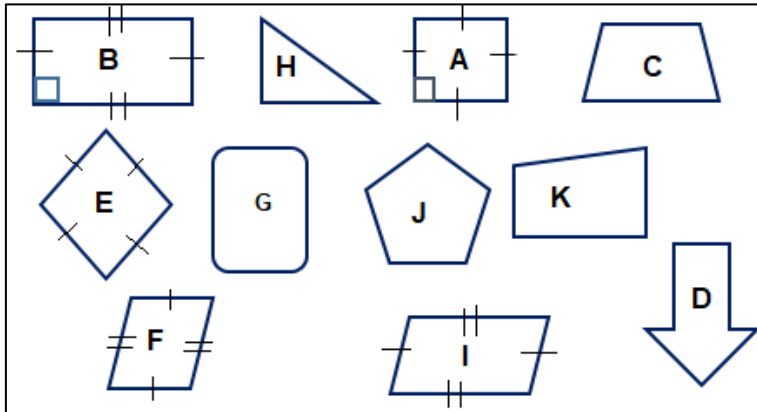
Triangle ABC: It is a right-angled triangle

- (f) If triangle ABC is a right-angled triangle with $AB = BC$, calculate angle BCA.

Triangle ABC: It is a right-angled triangle

1.2 Sort and classify shapes according to their properties.

1. Study the following shapes.



- (a) List the letters of shapes that are quadrilaterals.

A; B; C; F; E; K & I

- (b) Identify and write the letters of the quadrilaterals that are squares, rectangles and parallelograms in the shapes above.

A; B; F; E; & I

- (c) Write the letters of quadrilaterals that have:

- i. Two pairs of opposite sides that are parallel

A; B; F; E; & I

- ii. Perpendicular sides

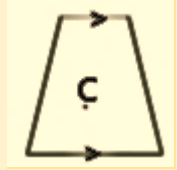
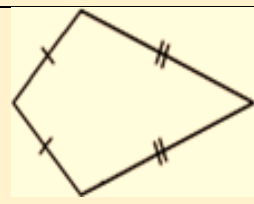
A & B

- iii. All four angles are right angles

A & B

1.3 Identifying quadrilaterals in terms of their sides and angles.

Quadrilateral is a polygon with four sides. Quadrilaterals can be classified according to the lengths of their sides, the number of parallel sides, perpendicular sides and the sizes of their angles. In this section you will be introduced to the rhombus, trapezium and the kite.

A rhombus is a quadrilateral with all four sides equal and the pairs of opposite sides parallel	
	A trapezium is a quadrilateral with only one pair of opposite sides parallel
A kite is a quadrilateral with two pairs of adjacent sides equal and opposite angles between the unequal sides equal	

1. Sort the following shapes according to their properties. Write letters for each Group

(a) Group 1: All four sides are equal

A & E

(b) Group 2: Opposite sides are equal

A; B; F; E & I

(c) Group 3: One pair of adjacent sides are equal

M

(d) Group 4: Perpendicular sides

A & B

(e) Group 5: All four angles are right angles

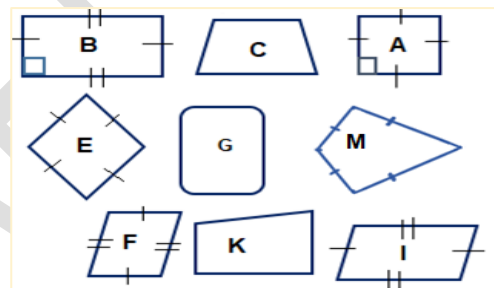
A & B

(f) Group 6: Two pairs of opposite sides parallel

A; B; E; F & I

(g) Group 7: One pair of opposite sides parallel

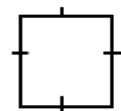
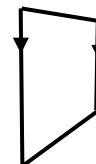
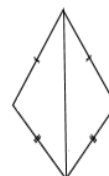
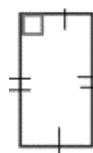
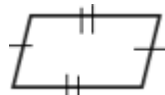
K & C



2. Match the shape in column A with the names in Column B

Answers Rhombus Parallelogram Rectangle Kite Trapezium Square

Column
A



Column
B

Kite

Trapezium

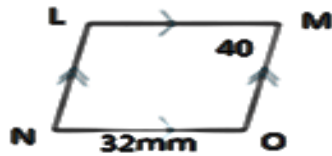
Rhombus

Square

Rectangle

Parallelogram

3. Given the quadrilateral LMNO, angle LMN = 40° , NO = 32mm, LO // MN, NO // ML



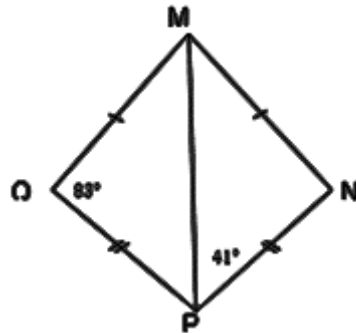
- (a) Name the quadrilateral
 (b) Write down the size of angle LNO
 (c) What is the length of side LM?

Rhombus

Angle LNO = 40°

LM = 32 mm (Opposite sides of a parallelogram are equal).

4. Given the quadrilateral below with angle MQP = 83°



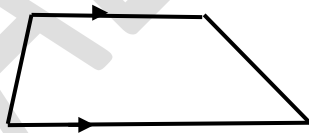
- (a) What is the measurement of angle MNP?

83° (Opposite angles between unequal sides of a kite are equal)

- (b) Write the name of the quadrilateral

Kite

5. Study the following quadrilateral and answer the questions below.



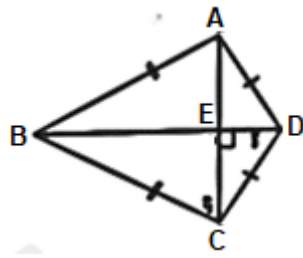
- (a) What is the sum of the interior angles?

The sum of the interior angles of a quadrilateral is 360° .

- (b) Give the name of the quadrilateral.

Trapezium

6. The quadrilateral ABCD has $AB = 3\text{cm}$:



(a) What will be the length of side BC?

Side $BC = 3\text{ cm}$

(b) Name the quadrilateral

A Kite

(c) What is the relationship between side AB and BC?

AB and BC are equal

(d) What is the size of angle AEB?

Angle $AEB = 90^\circ$

(e) What is the relationship between angle AEB and angle BEC?

Angle $AEB = BEC = 90^\circ$

7. Complete the table by filling in 'yes' or 'no' in the appropriate block:

Quadrilaterals	Equal side lengths			Parallel sides		Angles
	All four	Opposite only	Adjacent only	One pair	Two pairs	
	Yes	-	-	-	Yes	Yes
	No	Yes	No	No	Yes	No
	No	Yes	No	No	Yes	Yes
	Yes	-	-	-	Yes	No
	No	No	No	Yes	No	No
	No	No	Yes	No	No	No

8. Study each of the following pictures below and give the name of the quadrilateral that was used to make its outline.

(a)	(b)	(c)	(d)	(e)
				
Square	Rectangle	Kite	Square or Rhombus	Trapezium

1.4 Describing and naming parts of a circle.

A circle has different parts and the names are:

The **centre** of the circle is the point exactly in the middle of the circle.

The **circumference** is the outline or border around the outside of the circle

The **radius** of a circle is the distance from the centre of the circle to any point on the circumference

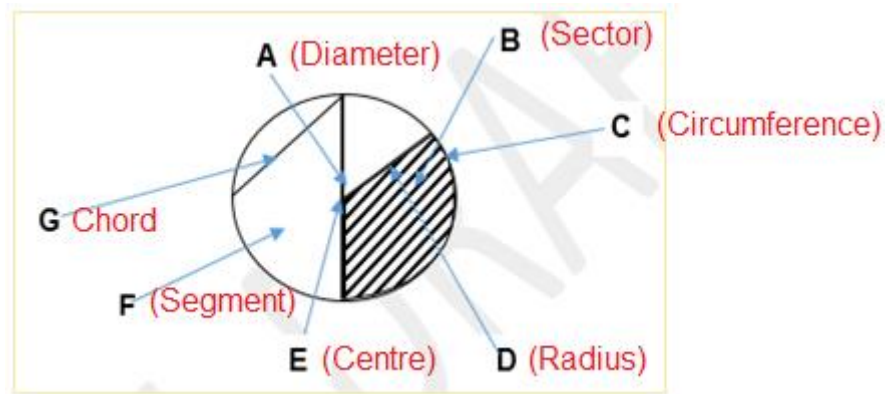
The **diameter** of a circle is the distance from one side of the circumference, through the centre of the circle to the other side. It cuts the circle in half. The diameter is twice the length of the radius

A **chord** is a line that joins any two points on a circumference of the circle.

A **segment** is a section of the circle created by a chord. The curved outside edge of the segment is an arc.

A **sector** is a section of the circle created by two radii. The curved outside edge of the sector is an arc.

1. Name the parts of the circle in the letters below:



2. Lerato is having a birthday party. She cuts her circular cake with a knife. First she cuts it exactly in half and she cuts it into slices for her guests.



- (a) What is the name of the line cutting the circular cake in half?
Diameter
- (b) Name the part of the circle represented by one slice of a cake.
Sector
- (c) What do you call the outline the cake?
Circumference

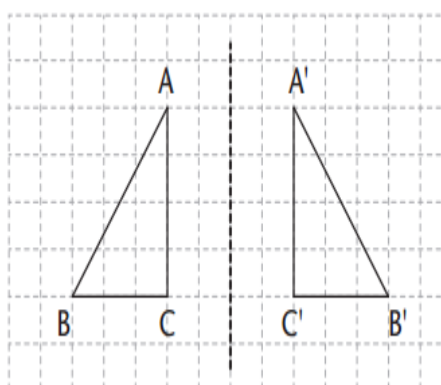
UNIT 2

TRANSFORMATIONS

2.1 Recognize, describe and perform translations, reflections and rotations with geometric figures and shapes.

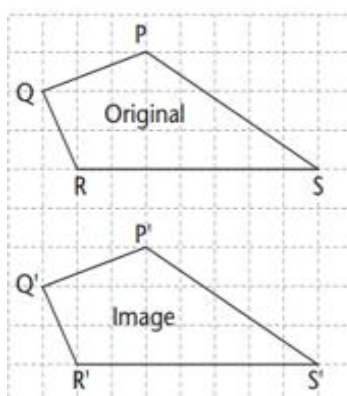
1. Identify and describe the type of transformation performed on the diagram below.

(a)



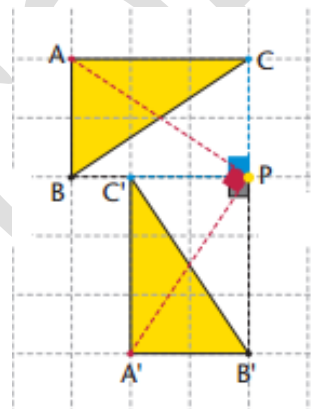
Triangle ABC is **reflected** about the line of reflection

(b)



Quadrilateral PQRS is **translated** 5 units downwards

(c)

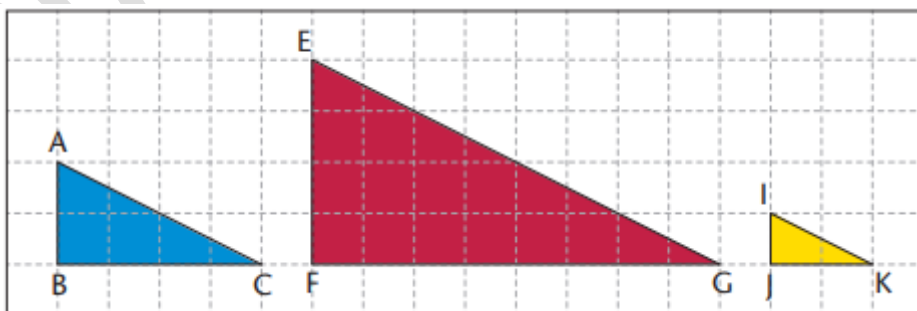


Triangle ABC is **rotated** 90° anti-clockwise round the centre of rotation P

2.2 Enlargements and reductions

Enlarging a figure means that we make it bigger in a specific way. **Reducing** a figure means that we make it smaller in a specific way. Enlarging or reducing figures is also called **resizing**.

1. Study the following triangles and answer the questions that follow



(a) How many times is:

FG longer than BC? **2 times** JK shorter than BC? **2 times**
 EF longer than AB? **2 times** IJ shorter than AB? **2 times**
 EG longer than AC? **2 times** IK shorter than AC? **2 times**

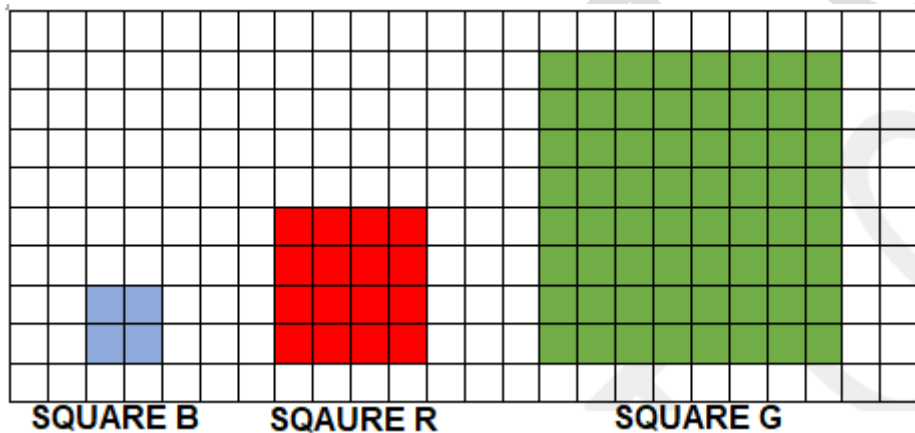
(b) Is $\triangle EFG$ an enlargement of $\triangle ABC$? Explain your answer.

$\triangle EFG$ is an enlargement of $\triangle ABC$ because each dimension has been doubled. [Any suitable answer]

(c) Is $\triangle IJK$ a reduction of $\triangle ABC$? Explain your answer

$\triangle IJK$ is a reduction of $\triangle ABC$ because each dimension has been halved. [Any suitable answer]

2. Study the following squares and use them to answer questions.



(a) Complete the following table.

Square B	Square R	Square G
Length = 2 units	Length = 4 units	Length = 8 units
Width = 2 units	Width = 4 units	Width = 8 units

(b) The length of square B is $\frac{1}{2}$ times the length of square R.

The width of the square B is $\frac{1}{2}$ times the width of square R.

Square R is enlarged $\frac{1}{2}$ times square B

(c) The length of square G is **2** times the length of square R.

The width of square G is **2** times the width of square R.

Square G is enlarged **2 times square R**

- (d) The length of square G is 4 times the length of square B.
The width of the square G is 4 times the width of square B.
Square B is reduced 4 times square B

When the lengths of **all the sides** of a figure are **multiplied by the same number** to produce a second figure, the second figure is an **enlargement** or **reduction** of the first figure. The number by which the sides are multiplied to produce an enlargement or reduction is called the **scale factor**.

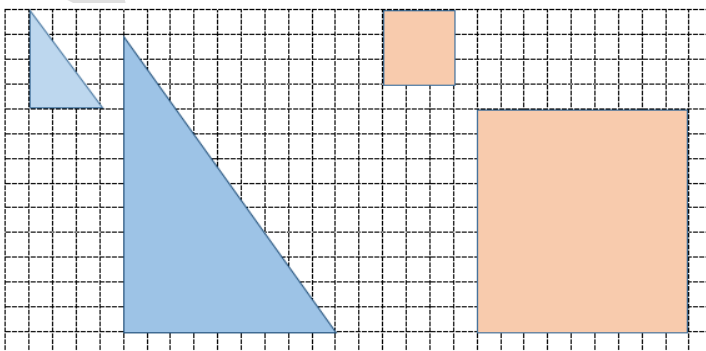
The scale factor

- When the scale factor is 1, the image is the same size as the original.
- When the scale factor is less than 1, the image is a reduced. For example, if the scale factor is $\frac{1}{2}$ or 0,5, each side of the image is half the length of its corresponding side in the original figure.
- When the scale factor is greater than 1, the image is enlarged. For example, if the scale factor is 2, each side of the image is double the length of its corresponding side in the original figure.

3. State whether the following scale factors will produce a larger or smaller image:

- | | |
|-------------------|---------------|
| (a) 6 | larger image |
| (b) 0,75 | smaller image |
| (c) 4,3 | larger image |
| (d) $\frac{4}{5}$ | smaller image |

4. On a grid paper, enlarge the triangle and square with a scale factor of 2



2.3 Describe patterns

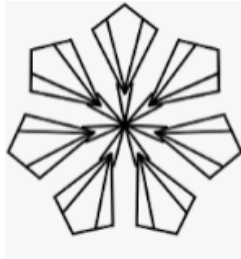
1. Describe how the following fabric patterns have been developed.

(a)



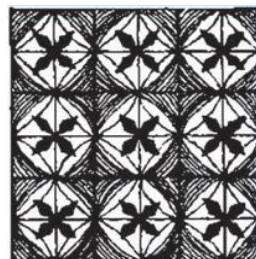
Reflection

(b)



Rotation

(c)



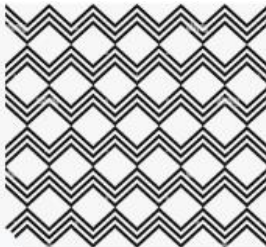
Translation

(d)



Reflection or translation

(e)



Reflection

(f)



Enlargement

(g)



Reflection

(h)



Rotation

https://www.google.com/imgres?imgurl=https%3A%2F%2Fmedia.istockphoto.com%2Fvectors%2Fhand-drawn-doodle-african-national-hut-ndebele-tribal-dwelling-simple-vector-id1334815709%3Fk%3D20%26m%3D1334815709%26s%3D612x612%26w%3D0%26h%3D1petu16xtl3ciHBtuICSNEHQa-7hNiu2y2fSL8ZoWdM%3D&imgrefurl=https%3A%2F%2Fwww.istockphoto.com%2Fillustrations%2Fndebele&tbnid=85Njt3vkWgeaxM&vet=12ahUkFwit3Q0Q5f4AhlIV1IIKH6WCrMOMvuneeIIARChAn_i&docid=611_nyGtAIE7viM&w=612&h=612&q=ndebele%20trihe%20house%20patterns&v

UNIT 3

GEOMETRY OF 3-D OBJECTS

4.1 Revision

You will need to use three-dimensional (3D) empty containers for these activities.

1. Name the 3D object and label it using the words face; vertex and edge.

Rectangular prism



4.2 Building 3-D models using nets

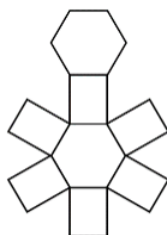
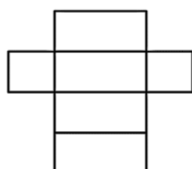
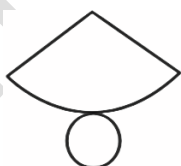
In mathematics, a **net** is a flat pattern that can be folded to form a 3D object. Different 3D objects have different nets. Sometimes the same 3D object can have different nets.

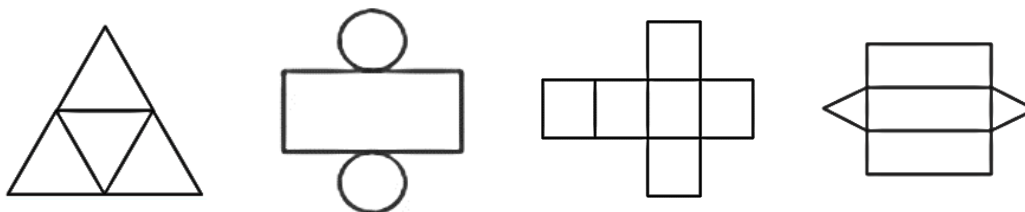
Learners work in groups to do the following activities.

1. Use the containers brought to class to do the following:
 - (a) Cut out the 3D object containers and flatten them
 - (b) Identify the shape of each face or surface.
 - (c) Draw the net of each objects.

NB: The outcome of this activity will depend on the 3D objects brought to class.

2. Make geometric objects using the nets provided. You will need some sticky tape and a pair of scissors. Examples of nets have been provided but more nets can be added. Nets of real life objects can be brought to class for folding.



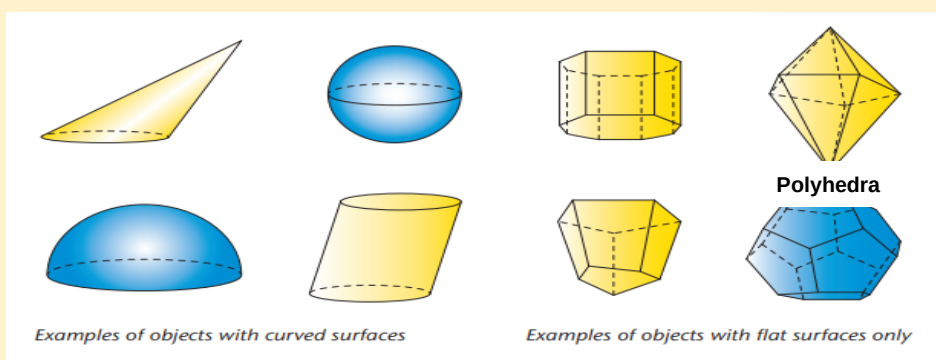


- Sort the geometric objects built in (2) using the nets. Explain how each group is formed. [Any suitable outcome and answer is accepted]
- Study the following nets and say if each of them will form a closed geometric object. Name the object that can be made.

(a) Closed object – Cube	(b) Not a closed object	(c) Closed object – Triangular prism
(d) Not a closed object – circles will be bigger than the rectangle.	(e) Not a closed object. Two faces of a will overlap	(f) Not a closed object. Two faces of a will overlap

4.3 Describe, sort and compare polyhedra

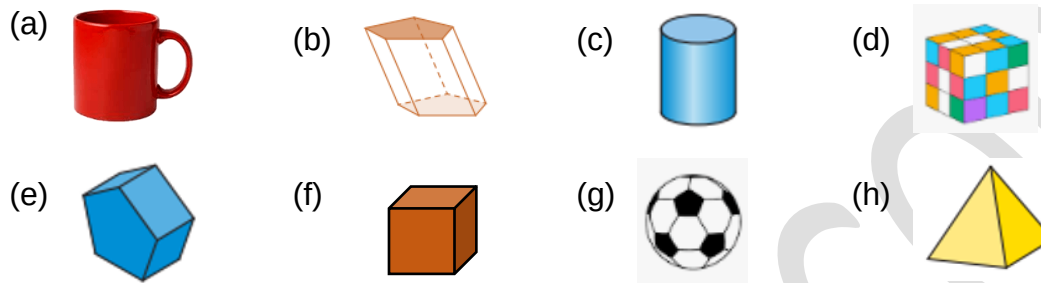
There are two main groups of objects with three dimensions (length, width and height), namely those with curved surfaces and those with flat surfaces. Spheres (balls), cylinders and cones are examples of objects with curved surfaces. Objects with only flat surfaces are called **polyhedra**.



A **polyhedron** is a three-dimensional object made of flat surfaces only. It has no curved surfaces. It consists of faces, edges and vertices.

1. Which of the following objects are polyhedra?

b, d, e, f and h are polyhedra. g depends on the material used.

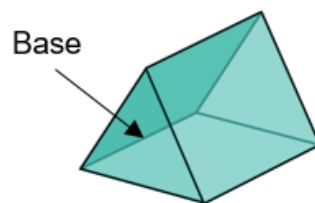


Prisms and **pyramids** are two special groups of polyhedra.

Prisms

A **prism** is a polyhedron with two faces that are congruent and parallel polygons. These faces are called bases and they are connected by lateral faces that are parallelograms. Prisms are named using their base. The two opposite congruent faces are the base.

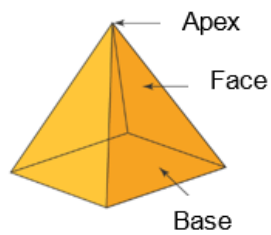
Example: Triangular prism



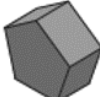
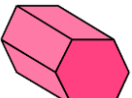
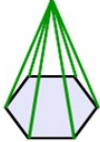
Pyramids

A **pyramid** has only one base. The lateral faces of a pyramid are always triangles. These triangles meet at the same vertex at the top. This vertex is called the **apex** of the pyramid. Pyramids are named using their base. The base is not always the face it is sitting on.

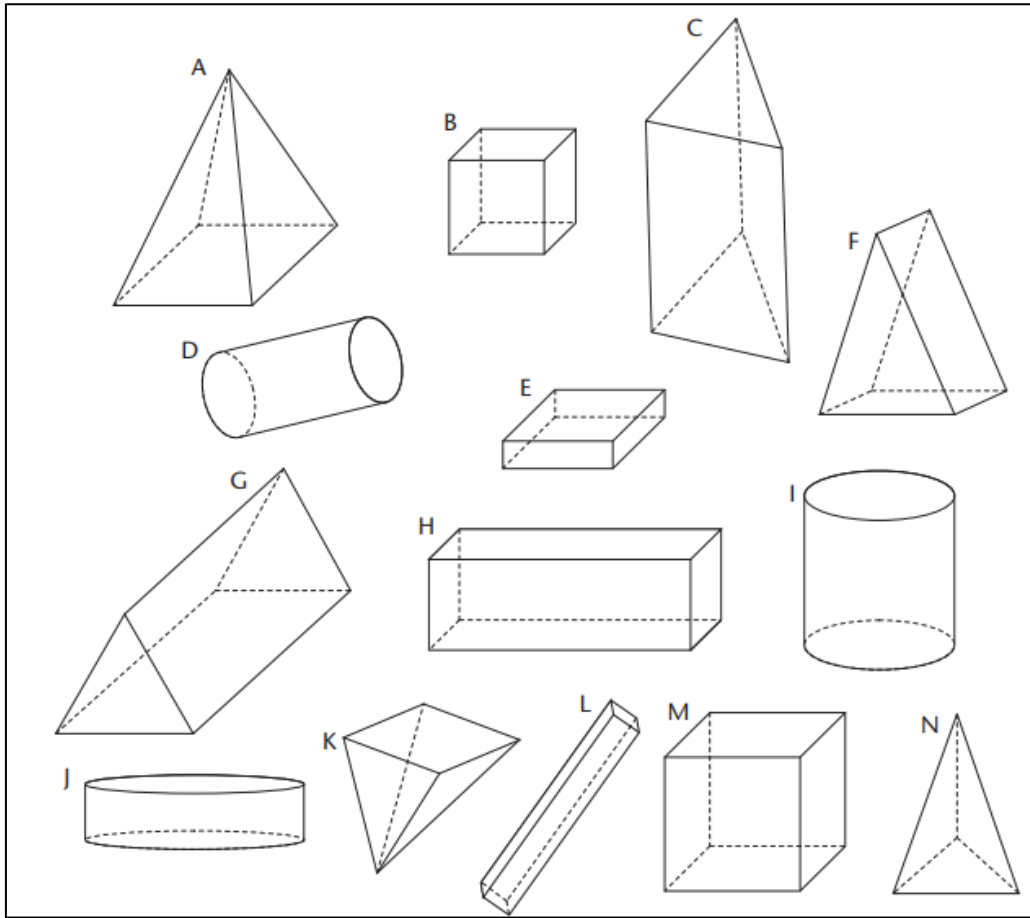
Example: Square-based pyramid



2. Copy and complete the following table.

Object	Pyramid or prism	Shapes that makes up the faces	Number of faces	Number of vertices	Number of edges
	Prism	Square $\times$ 6	6	8	12
	Prism	Rectangle $\times$ 6	6	8	12
	Prism	Triangle $\times$ 2 Rectangle $\times$ 3	5	6	9
	Prism	Pentagon $\times$ 2 Rectangle $\times$ 5	7	10	15
	Pyramid	Square $\times$ 1 Triangle $\times$ 4	5	5	8
	Prism	Hexagon $\times$ 2 Rectangle $\times$ 6	8	12	18
	Pyramid	Hexagon $\times$ 1 Triangle $\times$ 6	7	7	12

3. Study the following 3D objects.



(a) Sort the given 3D objects into the three groups, namely prisms, pyramids and cylinders. Write down the correct letter under each group.

Prisms	Pyramids	Cylinders
B; C; E; F; G; H; L; M	A; K; N	D; I; J

(b) Separate the prisms into three groups (cubes, rectangular prisms and triangular prisms). Write the letter under each group.

Cubes	Rectangular Prisms	Triangular Prisms
B; M	E; H; L	C; F; G

UNIT 4

AREA AND PERIMETER

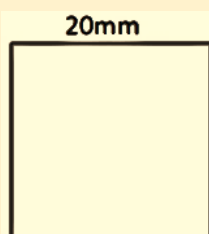
4.1 Calculate the perimeter of regular and irregular polygons

Perimeter of polygons is the total distance around the shape.

Regular polygons

Regular polygons have all the sides and angles equal. The perimeter of regular polygons is found by adding all the sides of a regular polygon or multiplying one side of the regular polygon by the number of its sides.

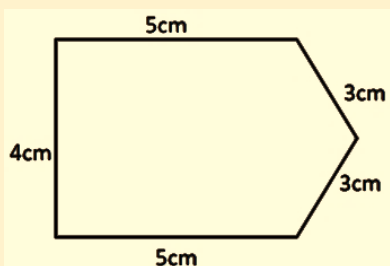
A square is an example of a regular polygon. Its perimeter is found by multiplying the length of one side by 4 as shown in the example.



$$\begin{aligned}\text{Perimeter of a square} &= 4 \times \text{length of its side (4s)} \\ &= 4 \times 20 \text{ mm} \\ &= 80 \text{ mm}\end{aligned}$$

Irregular polygons

Irregular polygons have unequal sides. The perimeter of irregular polygons is found by adding all the sides of the irregular polygon as shown in the example.

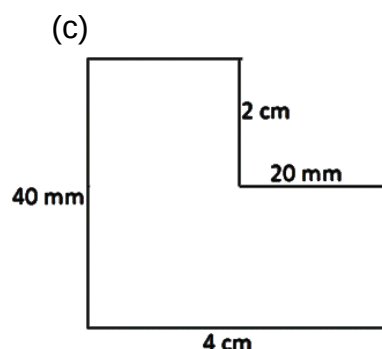
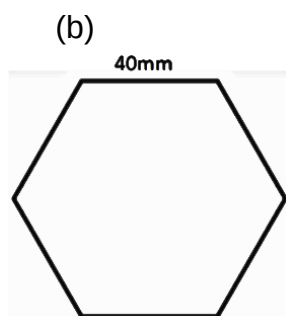
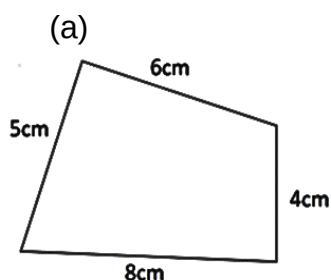


$$\begin{aligned}\text{Perimeter} &= \text{sum of all the lengths} \\ &= 5 \text{ cm} + 3 \text{ cm} + 3 \text{ cm} + 5 \text{ cm} + 4 \text{ cm} \\ &= 20 \text{ cm}\end{aligned}$$

Conversions:

To convert	Do this	To convert	Do this
cm to mm	$\times 10$	mm to cm	$\div 10$
m to cm	$\times 100$	cm to m	$\div 100$
km to m	$\times 1000$	m to km	$\div 1000$

1. Calculate the perimeter of the following polygons. Leave your answer in centimetres.



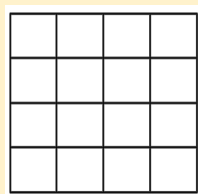
(a) Perimeter (a) = 6 cm + 4 cm + 8 cm + 5 cm = 23 cm

(b) Perimeter (b) = $6 \times 40 \text{ mm} = 240 \text{ mm} = 24 \text{ cm}$

(c) Perimeter (c) = 40 mm + 20 mm + 2 cm + 20 mm + 20 mm + 4 cm
 $= 40 \text{ mm} + 20 \text{ mm} + (2 \text{ cm} \times 10) + 20 \text{ mm} + 20 \text{ mm} + (4 \text{ cm} \times 10)$
 $= 40 \text{ mm} + 20 \text{ mm} + 20 \text{ mm} + 20 \text{ mm} + 20 \text{ mm} + 40 \text{ mm}$
 $= 160 \text{ mm}$
 $= 16 \text{ cm}$

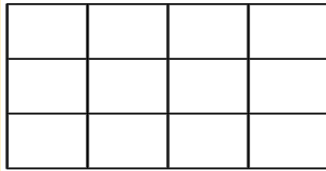
4.2 Use appropriate formulae to calculate the perimeter of squares and rectangles

Perimeter is the total length around a shape. To find perimeter of a shape, you add together the lengths of all its sides or edges. To calculate the perimeter of a square, we add all the sides of a square. The formula to calculate the perimeter of a square is $P = 4s$ (where s represents side).



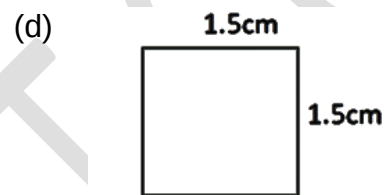
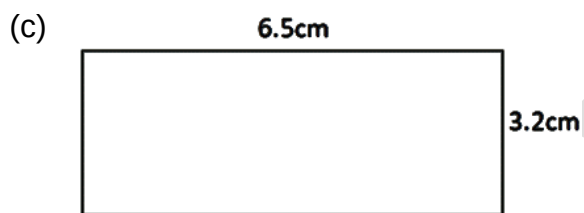
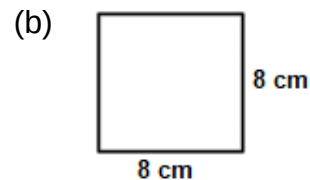
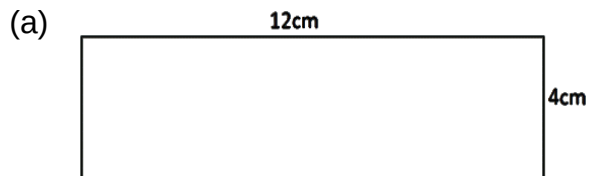
$$\begin{aligned} \text{Perimeter} &= 4s \\ &= 4 (4 \text{ cm}) \\ &= 16 \text{ cm} \end{aligned}$$

The formula to calculate the perimeter of a rectangle is $P = 2l + 2b$ or $P = 2(l + b)$. In the rectangle below, if the side of one block is 1cm, then the perimeter of the figure will be:



$$\begin{aligned}
 P &= 2l + 2b \\
 &= 2(l + b) \\
 &= 2(4 \text{ cm} + 3 \text{ cm}) \\
 &= 14 \text{ cm}
 \end{aligned}$$

1. Calculate the perimeter of the following quadrilaterals with the given lengths and breadths



(a) Perimeter $= 2l + 2b = 2(12\text{cm}) + 2(4 \text{ cm}) = 24 \text{ cm} + 8 \text{ cm} = 32 \text{ cm}$

(b) Perimeter $= 4s = 4 \times 8 \text{ cm} = 32 \text{ cm}$

(c) Perimeter $= 2l + 2b = 2(6,5\text{cm}) + 2(3,2 \text{ cm}) = 13 \text{ cm} + 6,4 \text{ cm} = 19,4 \text{ cm}$

(d) Perimeter $= 4s = 4 \times 1.5 \text{ cm} = 6 \text{ cm}$

- (e) If the perimeter of the square is 32 cm. What is the length of each side?

Perimeter $= 4s$

$32 \text{ cm} = 4 \times s$

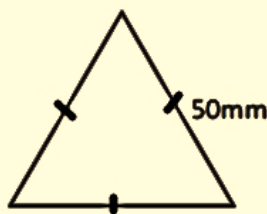
$\frac{32\text{cm}}{4} = s$

$s = 8 \text{ cm}$

4.3 Use appropriate formulae to calculate the perimeter of triangles and circles

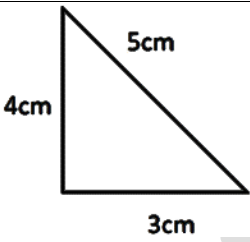
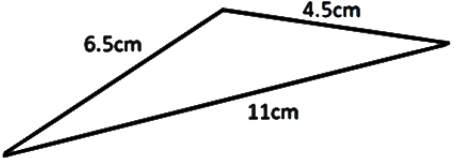
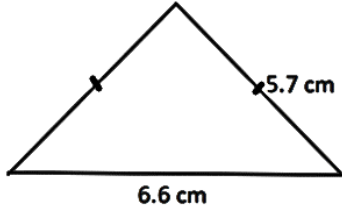
Perimeter of a triangle

To calculate the perimeter of a **triangle**, we add all the sides of a triangle. The formula to calculate perimeter of a triangle is $P = s + s + s$



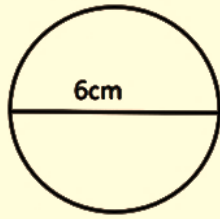
$$\begin{aligned}\text{Perimeter} &= s + s + s \\ &= 50 \text{ mm} + 50 \text{ mm} + 50 \text{ mm} \\ &= 150 \text{ mm}\end{aligned}$$

1. Calculate the perimeter of the following triangles given the lengths of the sides.

(a)  $\begin{aligned}P &= s + s + s \\ &= 5 \text{ cm} + 3 \text{ cm} + 4 \text{ cm} \\ &= 12 \text{ cm}\end{aligned}$	(b)  $\begin{aligned}P &= s + s + s \\ &= 4.5 \text{ cm} + 11 \text{ cm} + 6.5 \text{ cm} \\ &= 22 \text{ cm}\end{aligned}$
(c)  $\begin{aligned}P &= s + s + s \\ &= 5.7 \text{ cm} + 5.7 \text{ cm} + 6.6 \text{ cm} \\ &= 18 \text{ cm}\end{aligned}$	

Perimeter of a circle

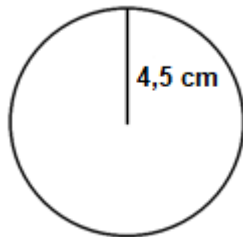
To find the perimeter of a circle, we take the whole distance around the circle. The formula to calculate the perimeter of a circle is $P = 2\pi r$ (Perimeter of a circle is called the circumference)



$$\begin{aligned}
 P &= 2\pi r \\
 &= 2\pi(3 \text{ cm}) \\
 &= 2 \times 3,14 \times 3 \text{ cm} \\
 &= 18,84 \text{ cm}
 \end{aligned}$$

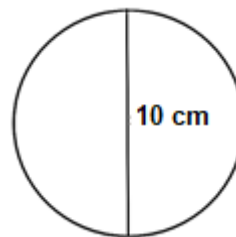
2. Calculate the perimeter of the following circles.

(a)



$$\begin{aligned}
 C &= 2\pi r \\
 &= 2\pi(4.5 \text{ cm}) \\
 &= 2 \times 3,14 \times 4,5 \text{ cm} \\
 &= 28,3 \text{ cm}
 \end{aligned}$$

(b)



$$\begin{aligned}
 C &= 2\pi r \\
 &= 2\pi \times \frac{d}{2} \\
 &= 2 \times 3,14 \times \frac{10\text{cm}}{2} \\
 &= 2 \times 3,14 \times 5 \text{ cm} \\
 &= 31.4 \text{ cm}
 \end{aligned}$$

(c) What is the perimeter of the circle with radius 14,5 cm?

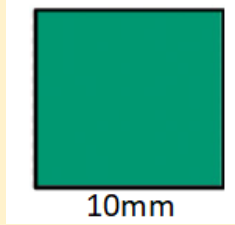
$$\begin{aligned}
 C &= 2\pi r \\
 &= 2 \times 3,14 \times 14,5 \text{ cm} \\
 &= 91,1 \text{ cm}
 \end{aligned}$$

(d) What is the circumference of a pizza if one straight side of each slice equals 3,9 cm?

$$\begin{aligned}
 C &= 2\pi r \\
 &= 2 \times 3,14 \times 3,9 \text{ cm} \\
 &= 24,5 \text{ cm}
 \end{aligned}$$

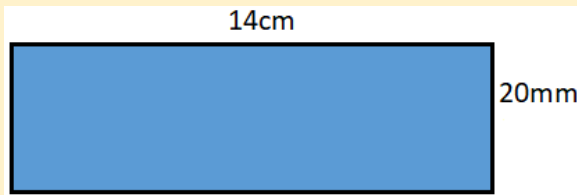
4.4 Use appropriate formulae to calculate the area of squares and rectangles

Area of the square is calculated by the formula: $\text{Area} = \text{length} \times \text{length} = l^2$



$$\begin{aligned} A &= l \times l \\ &= 10 \text{ mm} \times 10 \text{ mm} \\ &= 100 \text{ mm}^2 \end{aligned}$$

Formula to calculate the area of a rectangle is $\text{Area} = \text{length} \times \text{breadth}$



$$\begin{aligned} A &= l \times b \\ A &= 14 \text{ cm} \times 20 \text{ mm} \\ &= 14 \text{ cm} \times 2 \text{ cm} \\ &= 28 \text{ cm}^2 \end{aligned}$$

1. Calculate the area of the following rectangles and squares.

(a) $\begin{aligned} A &= l \times b \\ &= 12.5 \text{ cm} \times 2.5 \text{ cm} \\ &= 31.3 \text{ cm}^2 \end{aligned}$	(b) $\begin{aligned} A &= l^2 \\ &= 50 \text{ mm} \times 5 \text{ cm} \\ &= 5 \text{ cm} \times 5 \text{ cm} \\ &= 25 \text{ cm}^2 \end{aligned}$
--	--

- (a) Given the area of a square 49 m^2 calculate its length.

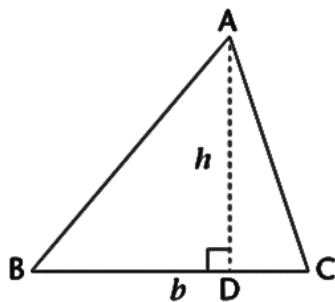
$$\begin{aligned} A &= l^2 \\ 49 \text{ cm}^2 &= l^2 \\ \sqrt{49} &= l \\ l &= 7 \text{ cm} \end{aligned}$$

- (b) Calculate the breadth of a rectangular table if the length is 2 m and the Area is 32 m.

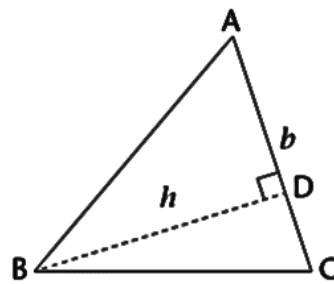
$$\begin{aligned} A &= l \times b \\ 32 \text{ m}^2 &= 2 \text{ m} \times b \\ b &= \frac{32}{2} = 16 \text{ m} \end{aligned}$$

4.5 Use appropriate formulae to calculate the area of triangles and circles

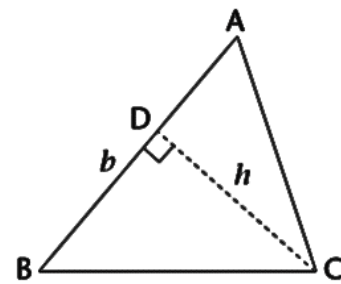
The formula to calculate **area of a triangle** is $A = \frac{1}{2} \times b \times h$ (where b represents the base and h the height). The height is a line segment drawn from the vertex perpendicular to the base.



AD = height
BC = base

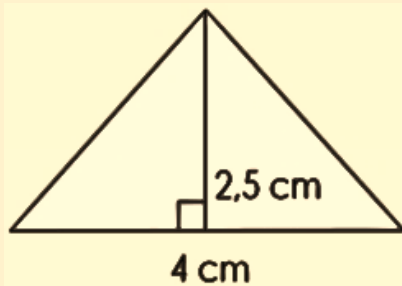


BD = height
AC = base



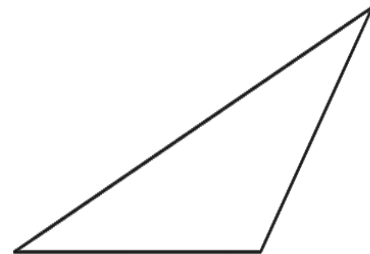
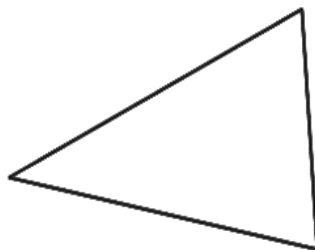
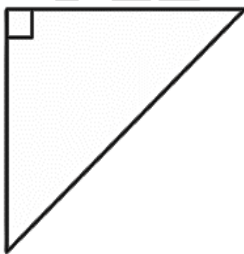
DC = height
AB = base

Example: Calculate the area of the triangle below.

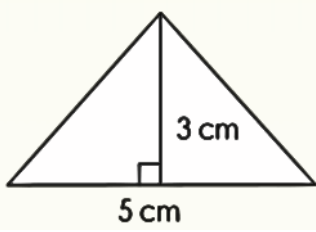
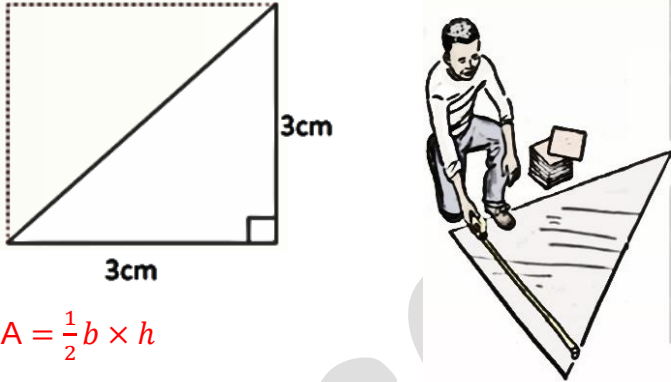


$$\begin{aligned} A &= \frac{1}{2} \times b \times h \\ A &= \frac{1}{2} \times 4 \text{ cm} \times 2.5 \text{ cm} \\ &= \frac{1}{2} \times 10 \text{ cm}^2 \\ &= 5 \text{ cm} \end{aligned}$$

- Copy the following triangles and draw any height in each. Label the base (b) and height (h) of each triangle



2. Calculate area of the following triangles given the lengths and breadth.

(a)  $A = \frac{1}{2}b \times h$ $= \frac{1}{2} \times 5 \text{ cm} \times 3 \text{ cm}$ $= 7,5 \text{ cm}^2$	(b)  $A = \frac{1}{2}b \times h$ $= \frac{1}{2} \times 3 \text{ cm} \times 3 \text{ cm}$ $= 4,5 \text{ cm}^2$
---	--

(c) Calculate the area of a triangle with base 3,2 cm and height 9,1 cm. Round your answer to one decimal place.

$$A = \frac{1}{2}b \times h$$

$$= \frac{1}{2} \times 3,2 \text{ cm} \times 9,1 \text{ cm}$$

$$= 14,56 \text{ cm}^2$$

$$= 14,6 \text{ cm}^2$$

(d) If the area of a triangle QRS is given as 47,88 cm² and height 6,2 cm; calculate its base. Round off your answer to one decimal place.

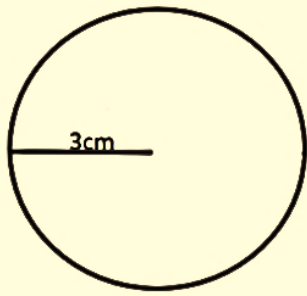
$$A = \frac{1}{2}b \times h$$

$$47,88 \text{ cm}^2 = \frac{1}{2}b \times 6,2 \text{ cm}$$

$$b = 15,45 \text{ cm}$$

Area of a circle

The formula to calculate the area of circles is $A = \pi r^2$ (where r is the radius of the circle and $\pi = 3,14$). The radius of the circle below is 3cm. The area of the circle is:

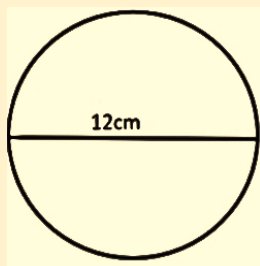


$$\begin{aligned}
 A &= \pi r^2 \\
 &= 3,14 (3 \text{ cm})^2 \\
 &= 3,14 \times 9 \text{ cm}^2 \\
 &= 28,26 \text{ cm}^2
 \end{aligned}$$

The formula to calculate area of a circle with a given diameter is $A = \pi \left(\frac{d}{2}\right)^2$

NB: Diameter of a circle is double its radius.

Example: Calculate the area of the following circle.



$$\begin{aligned}
 A &= \pi \left(\frac{d}{2}\right)^2 \\
 A &= \pi \left(\frac{12}{2}\right)^2 \\
 A &= \pi (6 \text{ cm})^2 \\
 &= 113,04 \text{ cm}^2
 \end{aligned}$$

1. Calculate the area for the following circles if:

(a) Radius is 2.8 cm

$$A = \pi r^2 = 3,14 \times (2,8 \text{ cm})^2 = 3,14 \times 7,84 \text{ cm}^2 = 24,6 \text{ cm}^2$$

(b) Radius is 4 cm

$$A = \pi r^2 = 3,14 \times (4 \text{ cm})^2 = 3,14 \times 16 \text{ cm}^2 = 50,2 \text{ cm}^2$$

(c) Radius is 10.1 cm

$$A = \pi r^2 = 3,14 \times (10,1 \text{ cm})^2 = 3,14 \times 102,01 \text{ cm}^2 = 230,3 \text{ cm}^2$$

(d) Diameter is 16 cm

$$A = \pi \left(\frac{d}{2}\right)^2 = 3,14 \times \left(\frac{16}{2}\right)^2 = 201 \text{ cm}^2$$

(e) Diameter is 140 cm

$$A = \pi \left(\frac{d}{2}\right)^2 = 3,14 \times \left(\frac{140}{2}\right)^2 = 3,14 \times 4\,900 = 15\,386 \text{ cm}^2$$

4.6 Problem solving involving perimeter and area of polygons

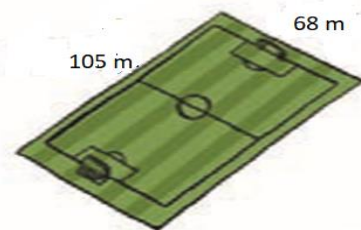
1. Thabo wants to build a 6 m by 5 m kraal for his cattle. Calculate the perimeter of the kraal to be built and the Area of the land to be used for the kraal.



$$P = 2l + 2b = 2(6 \text{ m}) + 2(5 \text{ m}) = 22 \text{ m}$$

$$A = 6 \text{ m} \times 5 \text{ m} = 30 \text{ m}^2$$

2. The youth of a village is requesting for a soccer pitch from the community. They require a pitch with a length of 105 m and a width of 68 m. Assist the youth by calculating perimeter and area they will use for the pitch.



$$\begin{aligned} P &= 2(l + b) \\ &= 2(105 + 68) \\ &= 2(173) \\ &= 346 \text{ m} \end{aligned}$$

3. Nosipho plants a vegetable patch measuring 12 m x 8 m. What is the area of the vegetable patch? She plants carrots on half of the patch, and tomatoes and potatoes on a quarter of the patch each. Calculate the area covered by each type of vegetable?

$$\text{Area (Vegetable patch)} = l \times b = 12 \times 8 = 96 \text{ m}^2$$

$$\text{Area (Carrot)} = 96 \text{ m}^2 \div 2 = 48 \text{ m}^2$$

$$\text{Area (Tomatoes)} = 48 \text{ m}^2 \div 2 = 24 \text{ m}^2$$

$$\text{Area (Potatoes)} = 24 \text{ m}^2 \text{ (the same as the tomatoes patch)}$$

4. Mr Mchunu must tile a classroom floor measuring 4 m x 4 m. The blue tiles he uses each measures 20 cm x 20 cm. How many tiles does Mr Mchunu need? The tiles are sold in boxes containing 20 tiles each. How many boxes should he buy?



$$\text{Area (classroom)} = 4 \text{ m} \times 4 \text{ m} = 16 \text{ m}^2$$

$$A \text{ (each tile)} = 20 \text{ cm} \times 20 \text{ cm} = 0.2 \text{ m} \times 0.2 \text{ m} = 0.4 \text{ m}^2$$

$$\text{Number of tiles needed} = 16 \text{ m}^2 \div 0.4 \text{ m}^2 = 40 \text{ tiles}$$

Mr Mchunu will need two boxes

5. Lerato wants to bake a cake with diameter 25 cm for her brother's birthday. Calculate the area and perimeter of this cake.



$$A = \pi \left(\frac{d}{2}\right)^2 = 3,14 \times \left(\frac{25}{2}\right)^2 = 490,6 \text{ cm}^2$$

$$\text{Perimeter} = 2\pi r = 2 \times 3,14 \times 12,5 = 78,5 \text{ cm}$$

The area of the cake will be 490,6 cm² and the perimeter will be 78,5 cm

6. Palesa ordered a 78,5 cm² cookie cake for her birthday. She requested the icing to have a sun covering the centre and to be 5 cm in diameter. How much of Palesa's cookie cake wasn't covered by icing?

$$\text{Area of sun covering of icing is} = \pi \left(\frac{5}{2}\right)^2 = 3,14 \times (2,5)^2 = 19,625 \text{ cm}^2$$

$$\text{Uncovered Area} = 78,5 \text{ cm} - 19,625 \text{ cm} = 58,9 \text{ cm}^2$$

7. A fence was built around this community. Answer the questions and write your answers in kilometres and then metres.



(a) How long is the fence?

$$P = 2(l + b) = 2(5 \text{ km} + 4 \text{ km}) = 18 \text{ km}$$

The length of the fence will be 18 km

(b) Calculate the area of this residence.

$$A = l \times b = 5 \times 4 = 20 \text{ km}^2$$

The area of this residence is 20 km²

FORMULA SHEET

Perimeter of a square	$P = 4s$
Perimeter of a rectangle	$P = 2l + 2b = 2(l + b)$
Perimeter of a triangle	$P = s + s + s$
Circumference of a circle	$C = 2\pi r$
Area of a square	$A = l^2$
Area of a rectangle	$A = l \times b$
Area of a triangle	$A = \frac{1}{2}b \times h$
Area of a circle	$A = \pi r^2$

UNIT 5

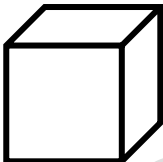
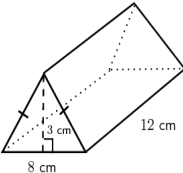
SURFACE AREA AND VOLUME

5.1 Using appropriate formulae to calculate the surface area of prisms and cylinders

The surface area of a prism is the sum of the areas of its faces. The formula for calculating the area of a two-dimensional shape is length $\times$ breadth. The unit for area is units².

Recognising and naming prisms

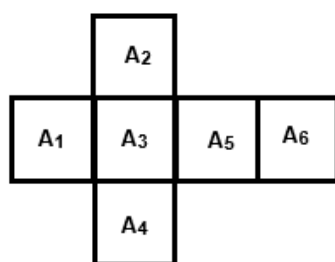
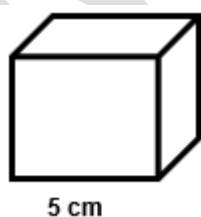
1. Match the picture of prisms with the correct names.

Answers	Cube	Rectangular prism	Cylinder	Triangular prism
Drawing of the object				
Name of the object	Cylinder	Triangular prism	Cube	Rectangular prism

Calculating the total surface area of a cube

A cube is a special prism with 6 congruent faces (equal in all respects). The area of the face of a cube is s^2 , therefore the total surface area of a cube is $6 \times s^2$.

The following example illustrates a picture of a cube and its net and shows how the surface area of a cube is calculated.

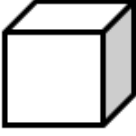
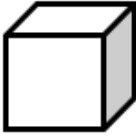
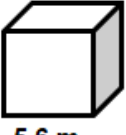
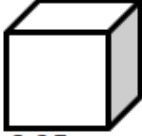


A₁ stands for area of face 1

The cube has 6 faces with side length 5 cm

$$\begin{aligned}\text{Total Surface Area} &= 6 \times (5 \text{ cm} \times 5 \text{ cm}) \\ &= 6 \times 25 \text{ cm}^2 \\ &= 150 \text{ cm}^2\end{aligned}$$

2. Calculate the surface area of the cubes in (a – d) below, then convert the answers as required.

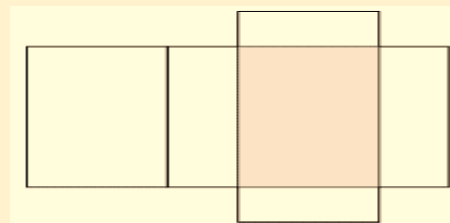
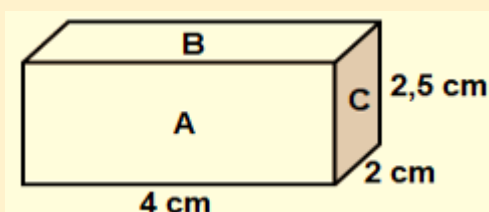
No.	Diagram an object	Total Surface Area	Answer in mm ²	Answer in cm ²	Answer in m ²
(a)	 3 cm	$6 \times (3 \text{ cm} \times 3 \text{ cm})$ $= 54 \text{ cm}^2$	5 400	54	0,0054
(b)	 19 mm	$6 \times (19 \text{ mm} \times 19 \text{ mm})$ $= 2 166 \text{ mm}^2$	2 166	21,66	0,002166
(c)	 5,6 m	$6 \times (5,6 \text{ m} \times 5,6 \text{ m})$ $= 168,45 \text{ m}^2$	168 540 000	1 685 400	168,54
(d)	 2,35 m	$6 \times (2,35 \text{ m} \times 2,35 \text{ m})$ $= 33,14 \text{ m}^2$	331 135 000	331,4	33,14

Calculating the total surface area of a rectangular prism

A rectangular prism has 6 rectangular faces. Opposite faces of a prism are congruent i.e. they are equal in all respects.

If A_1, A_2, B_1, B_2, C_1 and C_2 are faces of a rectangular prism and $A_1 = A_2, B_1 = B_2$ and $C_1 = C_2$, then the total surface area = 2 area of A_1 + area of B_1 + area of C_2 .

The following example shows a diagram of a rectangular prism and its net and illustrate how the total surface area of a rectangular prism is calculated.



Area of rectangle A ₁ = 2 × (4 cm × 2,5 cm) = 2 (10 cm) = 20 cm ²	Area of rectangle B ₁ = 2 × (4 cm × 3 cm) = 2 (12 cm) = 24 cm ²	Area of rectangle C ₁ = 2 × (2,5 cm × 3 cm) = 2 (7,5 cm) = 15 cm ²
--	---	---

$$\therefore \text{Total Surface Area} = 20 \text{ cm}^2 + 24 \text{ cm}^2 + 15 \text{ cm}^2 = 59 \text{ cm}^2$$

3. Calculate the surface area of the following rectangular prisms with the following dimensions:

(a) Height = 2,4 cm ; Length = 5 cm ; Width = 3,1 cm

$$SA = 2[(5 \text{ cm} \times 2,4 \text{ cm} + 3,1 \text{ cm} \times 2,4 \text{ cm} + 5 \text{ cm} \times 3,1 \text{ cm})] = 34 \text{ cm}^2$$

(b) Width = 2,2 mm ; Length = 7 mm ; Height = 1,5 mm

$$= 2(7 \text{ mm} \times 1,5 \text{ mm} + 2,2 \text{ mm} \times 1,5 \text{ mm} + 7 \text{ mm} \times 2,2 \text{ mm}) = 29,2 \text{ mm}^2$$

(c) Width = 4,5 m ; Length = 6 m ; Height = 3,7 m

$$= 2(4,5 \text{ m} \times 6 \text{ m} + 3,7 \text{ m} \times 4,5 \text{ m} + 4,5 \text{ m} \times 6 \text{ m}) = 70,65 \text{ m}^2$$

(d) Width = 1,8 cm ; Length = 4 cm ; Height = 2,3 cm

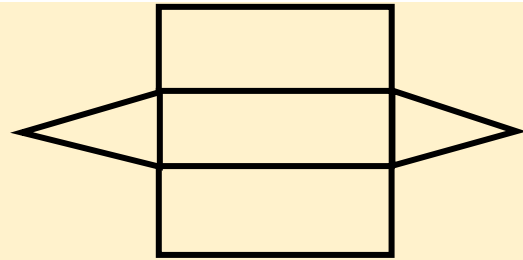
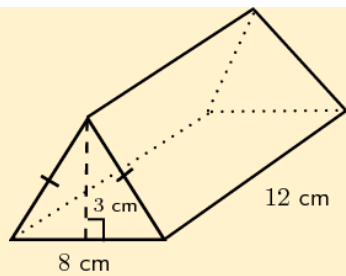
$$= 2(1,8 \text{ cm} \times 4 \text{ cm} + 2,3 \text{ cm} \times 1,8 \text{ cm} + 4 \text{ cm} \times 1,8 \text{ cm}) = 18,54 \text{ cm}^2$$

Calculating the surface area of a triangular prism

A triangular prism has 3 rectangular faces and 2 triangular faces. In a regular triangular prism, the pair of triangular faces are congruent and the three rectangular faces are congruent.

The surface area of a triangular prism can be calculated as follows: 2 (area of a triangular face) + (area of the three rectangular faces)

The example below illustrates a picture of a triangular prism and its net and shows how the surface area of a triangular prism is calculated. The perpendicular height of the triangular face is 3 cm.



$SA = 2 \text{ (area of triangle)} + \text{(area of the three rectangles)}$

$$\begin{aligned} SA &= 2 \times \frac{1}{2} (8 \text{ cm} \times 3 \text{ cm}) + [(8 \text{ cm} \times 12 \text{ cm}) + 2 (5 \text{ cm} \times 12 \text{ cm})] \\ &= 24 \text{ cm}^2 + 96 \text{ cm}^2 + 2 (60 \text{ cm}^2 + 60 \text{ cm}^2) \\ &= 240 \text{ cm}^2 \end{aligned}$$

4. Calculate total surface area of triangular prisms with the following dimensions:

(a) height of 5 cm; base of 24 cm and $H = 13$ cm

$$770 \text{ cm}^2$$

(b) height of 8 cm; base of 30 mm and $H = 17$ mm

$$56,64 \text{ cm}^2$$

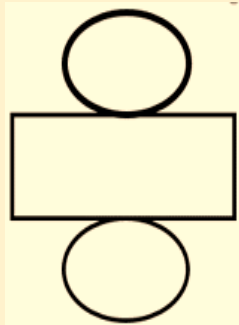
(c) height of 6 cm; base of 16 cm and $H = 10$ cm

$$456 \text{ cm}^2$$

Calculating the surface area of a cylinder

A **cylinder** is a 3D object with two congruent circular bases. The area of the circular base is πr^2 and the area of the circumference of the circular base is $2\pi r$, therefore the total surface area $= 2\pi r^2 + 2\pi r h$.

The example below illustrates a picture of a cylinder and its net and shows how the surface area of a cylinder is calculated. The height of a cylinder = 25 cm and the radius = 10 cm.



$$\begin{aligned} SA &= 2\pi r^2 + 2\pi r h \\ &= 2\pi(10^2) + 2\pi(10)(25) \\ &= 2 \times 3,14 (10^2) + 2 \times 3,14 \times 10 \times 25 \\ &= 628 + 1\,570 \\ &= 2\,198 \text{ cm}^2 \end{aligned}$$

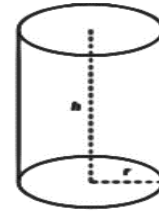
5. The closed cylinder has a base with the diameter of 14 cm and the height of 16 cm.

(a) Calculate the area of the circular surface.

$$2\pi rh = 2 \times 3,14 \times 7 \times 16 = 703,72 \text{ cm}^2$$

(b) Calculate the total surface area.

$$\begin{aligned} \text{Total Surface Area} &= 2\pi r^2 + 2\pi rh = 2 \times 3,14 \times 7^2 + 2 \times 3,14 \times 7 \times 16 \\ &= 1\,011,08 \text{ cm}^2 \end{aligned}$$



6. Study the cylinder below and calculate:

(a) the surface area, of the circular faces.

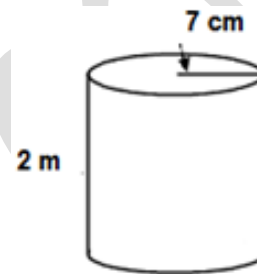
$$\text{Total Surface Area} = 2\pi r^2 = 307,7 \text{ cm}^2$$

(b) the total surface area of the cylinder.

$$\text{Total Surface Area} = 395,64 \text{ cm}^2$$

(c) the new surface area of the cylinder if the height dimension is doubled.

$$483,56 \text{ cm}^2$$

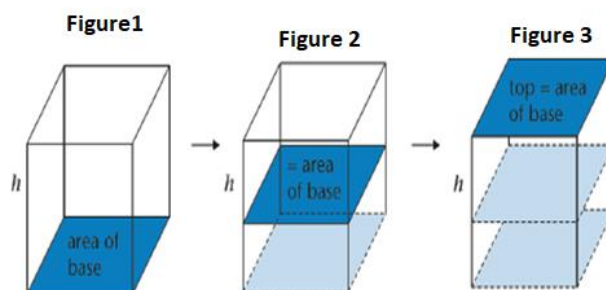


5.2 Use appropriate formulae to calculate volume of 3-D objects (cube, rectangular and, triangular prism)

Volume and Capacity of 3-D objects.

A prism is a 3-dimensional figure that has two parallel, congruent bases connected by lateral faces. The lateral faces of a prism are **always parallelograms and are usually rectangles**.

The figures below show that the base area of a prism does not change, even if the object is moved between the lateral faces.

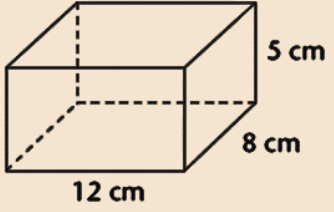
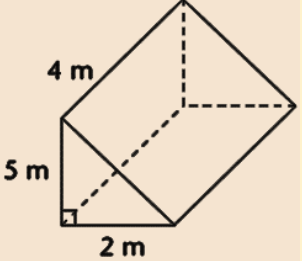
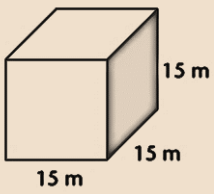


The **volume** of an object is the amount of space it takes up. The formula for calculating volume is area of base $\times$ height. Volume is measured in cubic units

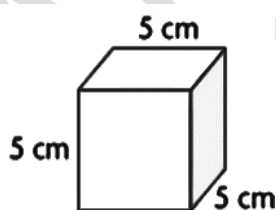
e.g. cubic centimetres (cm³) and cubic metres (m³). If volume is measured in cubic centimetres (cm³) the measure of capacity is millilitres.

Volume is the amount of space an object takes up and **capacity** is how much a container can hold.

Calculating volume for prisms

Name	Diagram	Calculation
Rectangular prism		$\begin{aligned}\text{Volume} &= (l \times w) \times h \\ &= 12 \text{ cm} \times 8 \text{ cm} \times 5 \text{ cm} \\ &= 420 \text{ cm}^3\end{aligned}$
Triangular prism		$\begin{aligned}\text{Volume} &= \left(\frac{1}{2} \text{ base} \times h_b\right) \times H_P \\ &= \frac{2}{2} \text{ cm} \times 5 \text{ cm} \times 4 \text{ cm} \\ &= 20 \text{ cm}^3\end{aligned}$
Cube		$\begin{aligned}\text{Volume} &= (s \times s) \times s \\ &= 15 \text{ cm} \times 15 \text{ cm} \times 15 \text{ cm} \\ &= 3\,375 \text{ cm}^3\end{aligned}$

1. Calculate the volume of the cube



$$\begin{aligned}\text{Volume of the cube} &= s \times s \times s = s^3 \\ &= 5 \text{ cm} \times 5 \text{ cm} \times 5 \text{ cm} = 125 \text{ cm}^3\end{aligned}$$

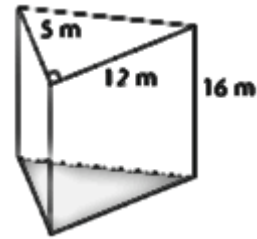
2. Which of the cubes given below has a bigger volume? Explain.

(a) A cube with a volume of 1m³ OR A cube with a volume of 1 000 000 cm³

They have the same volume. 100 cm makes a metre, so 100 cm × 100 cm × 100 cm = 1 000 000 cm³ which is equivalent to 1 m × 1 m × 1 m = 1 m³.

3. The triangular prism is to be dunked in water.
(a) Calculate the volume of a triangular pyramid.

$$\begin{aligned}\text{Volume} &= \frac{1}{2} \text{ base} \times h_b \times h_p \\ &= \frac{1}{2} \times 12 \text{ m} \times 5 \text{ m} \times 16 \text{ m} \\ &= 480 \text{ m}^3\end{aligned}$$

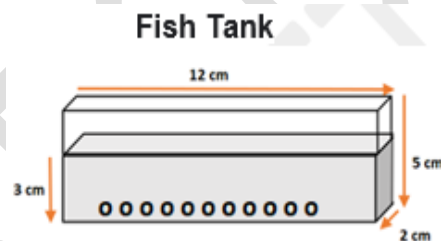


- (b) How much water will the prism displace?

$$480 \text{ m}^3 = 480\,000 \text{ l}$$

Therefore 480 000 l will be displaced by the prism

4. Sandi has a fish tank that is 12 cm long, 5 cm high and 2 cm wide. The bottom of the tank is filled with marbles and the tank is full to its capacity. When the marbles are removed the water level drops to a height of 3 cm



- (a) Calculate the volume displaced by the marbles.

$$\begin{aligned}\text{Volume displaced by the marbles will be} &= \text{base area} \times \text{height} \\ &= 24 \text{ cm}^2 \times 2 \text{ cm} = 48 \text{ cm}^3\end{aligned}$$

- (b) The volume of the water when the marbles are removed.

$$\begin{aligned}\text{Volume when the marbles are removed} &= \text{base area} \times \text{height} \\ &= 24 \text{ cm}^2 \times 3 \text{ cm} = 72 \text{ cm}^3\end{aligned}$$

- (c) What is the difference between the volume displaced by the marbles and the volume after the marbles have been removed?

$$\begin{aligned}\text{Volume displayed by the marbles} - \text{Volume after marbles are removed} \\ &= 72 \text{ cm}^3 - 48 \text{ cm}^3 \\ &= 24 \text{ cm}^3\end{aligned}$$

5. A textbook with a shape of a cube has a volume of $638,25 \text{ cm}^3$. What is the maximum number of similar textbooks that can fit into a box that has a capacity of $2,5 \text{ m}^3$? Round off your answer to the nearest whole number.

$$1 \text{ m}^3 = 10^6 \text{ cm}^3$$

$$2,5 \text{ m}^3 = 2,5 \times 10^6 \text{ cm}^3$$

$$\therefore \text{Number of textbooks} = \frac{2,5 \times 10^6}{638,25} = 3\,917 \text{ books}$$

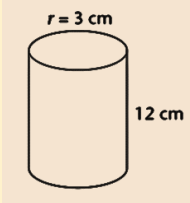
3 917 books can fit into the boxes that has capacity of $2,5 \text{ m}^3$

5.3 Use appropriate formulae to calculate volume of cylinders

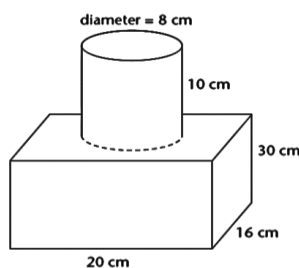
To calculate the volume of a cylinder, multiply the area of the base by the height of the cylinder. The base of a cylinder is circular, therefore:

$$\text{Volume of a cylinder} = \text{Area of base} \times h = \pi r^2 \times h$$

π is an irrational number and does not have an exact value. In general, the **value of π** is considered as 3,14 or $\frac{22}{7}$ for various mathematical calculations.

Name	Diagram	Calculation
Cylinder		Volume of a cylinder $= \text{Area of base} \times h$ $= \pi r^2 \times h$ $= 3,14 \times 9 \times 12$ $= 339,12 \text{ cm}^3$

1. Calculate the volume of the object in the following figure. Use a calculator and round off all answers to two decimal places.

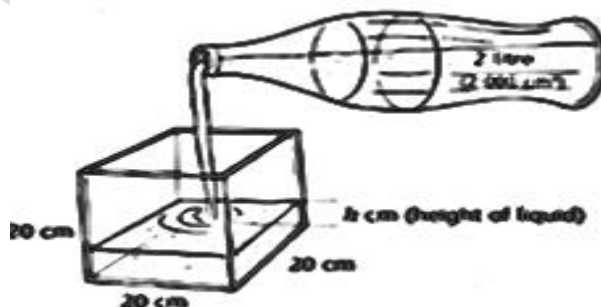


$$V_{\text{cylinder}} = \pi r^2 \times h = 3,14 \times 4 \times 4 \times 10 \text{ cm}^3 = 502,40 \text{ cm}^3$$

$$V_{\text{prism}} = \text{Base area} \times h = (16 \times 20) \times 20 \text{ cm}^3 = 9\,600 \text{ cm}^3$$

$$V_{\text{object}} = 502,40 \text{ cm}^3 + 9\,600 \text{ cm}^3 = 10\,102,40 \text{ cm}^3$$

2. A liquid is poured from a full 2 l bottle into a cubic glass tank with the side measurements of 20 cm by 20 cm by 20 cm.



- (a) What is the volume of the liquid when it is in the bottle?

2 000 cm³ or 2 l

(b) What is the capacity of the bottle?

2 000 cm³ or 2 ℓ

(c) What is the volume of the liquid after it is poured into the tank?

2 000 cm³ or 2 ℓ

(d) What is the capacity of the tank?

$20 \times 20 \times 20 = 8\,000\text{ cm}^3$ or 8 ℓ

(e) How high does the liquid go in the tank?

Note: Question 2 above shows the following:

Volume of liquid in the tank

= Volume of the liquid in the bottle

= $20\text{ cm} \times 20\text{ cm} \times h$ (liquids height in tank)

= 2 000 cm³

The height of liquid in the tank

= $2000\text{ cm}^3 \div (20\text{ cm} \times 20\text{ cm})$

= 5 cm

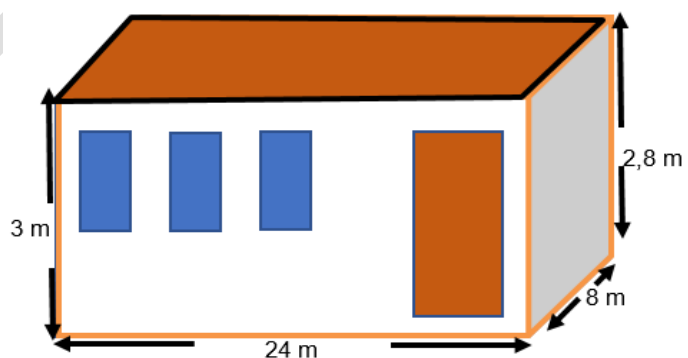
5.4 Problem solving involving surface area and volume.

Case study 1

A church congregation decided to build a hall with four walls that have four rectangular windows, two circular windows and one door. You are responsible for coordinating the project of painting the walls of the hall.

The height of the door is 2,2 m and its area is 2.86m²

Advise them on the amount of paint needed. Use the questions below.



1. There are four rectangular windows and one door at the front and no windows on the back and the left walls.

- (a) Calculate the width of the door in metres, if the area is $2,86 \text{ m}^2$ including the frames using the formula for the area of a rectangle: $\text{Area} = \text{Length} \times \text{Breadth or Height} \times \text{Breadth}$

Width of the door $2,86 \text{ m} \div 2,2 \text{ m} = 1,3 \text{ m}$

- (b) Calculate the area of the back wall with a width/breadth and a height of 24 m and $2,8 \text{ m}$, using the formula of a rectangle: $A = B \times H$.

Area of the back wall $= 67,2 \text{ m}^2$

- (c) Calculate the area of the left wall with a width and a height of 8 m and 3 m using the formula: $A = L \times B$

Area of the left wall $= 8 \text{ m} \times 3 \text{ m} = 24 \text{ m}^2$

2. The height of the rectangular window is $1,2 \text{ m}$, the width is $0,45 \text{ m}$ and the length is 800 cm including frames.

- (a) Calculate the area of each rectangular window

$A = 3,60 \text{ m}^2$

- (b) Write the answer in centimetres

$A = 36\,000 \text{ cm}^2$

- (c) Calculate the total area of the four rectangular windows. $A = 14,40 \text{ m}^2$

3. The left wall has two circular windows, each with a radius of $1,25 \text{ m}$

- (a) Calculate the area of one window using the formula for the area of a circle:

πr^2

Area of one circular window $= 4,91 \text{ m}^2$

- (b) Calculate the total area of the two circular windows.

TA of two circular windows $= 9,82 \text{ m}^2$

4. A decision has been taken to start with the outside of the hall.

- (a) Calculate the total surface area (outside walls) that must be painted.

TSA of the four walls $- (\text{Area of the windows} + \text{Area of the door})$

$= 182,4 \text{ m}^2 - (24,22 \text{ m}^2 + 2,86 \text{ m}^2)$

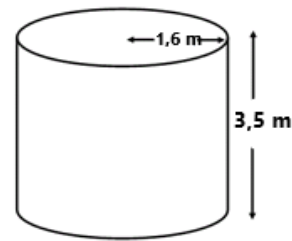
$= 155,32 \text{ m}^2$

- (b) If a litre of paint covers 6 m^2 , calculate the number of litres of paint that must be bought for the inside and the outside walls.

25,88 litres

Case Study 2

The diagram shows the size of a pre-school water tank installed to collect rain water for toilet use. The governing body of the school has decided to repaint the tank to match the walls of the school.



1. Calculate the circumference of the tank base using the formula: $C = 2\pi r$

$$C = 2\pi r = 2 \times 3,14 \times 1,6 \text{ m} = 10,048 \text{ m}^2$$

2. If two litres of paint is used to paint the top and bottom how much paint is needed for the outside of the tank?

$$\text{Area of the top and bottom} = 2(\pi r^2) = 2(3,14 \times 1,6 \text{ m} \times 1,6 \text{ m}) = 16,08 \text{ m}^2$$

$$\text{Area of the side of the tank} = \text{Circumference} \times 3,5 = 35,18 \text{ m}^2$$

Four and half litres will be needed to paint the area outside the tank.

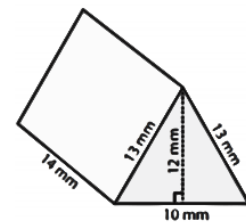
3. If the tank is three quarters full, the volume of water in the tank is 3 750 litres.

(a) What is the capacity of the tank? 5 000 litres

(b) How much water can two tanks of the same size hold? 10 000 litres

Case study 3

The diagram shows the size of the triangular gift box donated by the local supermarket to the home of the aged every year. Study the picture and answer the questions below.



1. How much cardboard will be needed to make one gift box?

Total Surface Area of 1 box

$$= 10 \times 14 + 2(13 \times 14) + 10 \times 12) \text{ mm}^2$$

$$= 624 \text{ mm}^2$$

2. How much cardboard will be needed to make 10 gift boxes?

$$\text{Total Surface Area of 10 boxes} = 10 \times 624 = 6\,240 \text{ square millimetres}$$

NOTE: More cardboard will be needed since an allowance needs to be made for off-cuts and flaps.

3. If a 15 m by 15 m roll of cardboard is ordered, how many gift boxes can be made?

Area of the board = $15\text{ m} \times 15\text{ m} = 225\text{ m}^2$

Area of each gift box = $626\text{ mm}^2 = 0,000624\text{ m}^2$

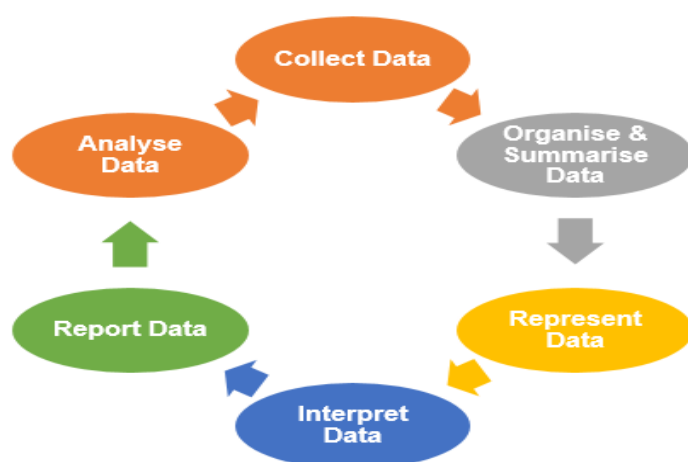
The number of boxes that can be made = $225\text{ m}^2 \div 0,000624\text{ m}^2 = 360\text{ boxes}$

UNIT 6

DATA HANDLING

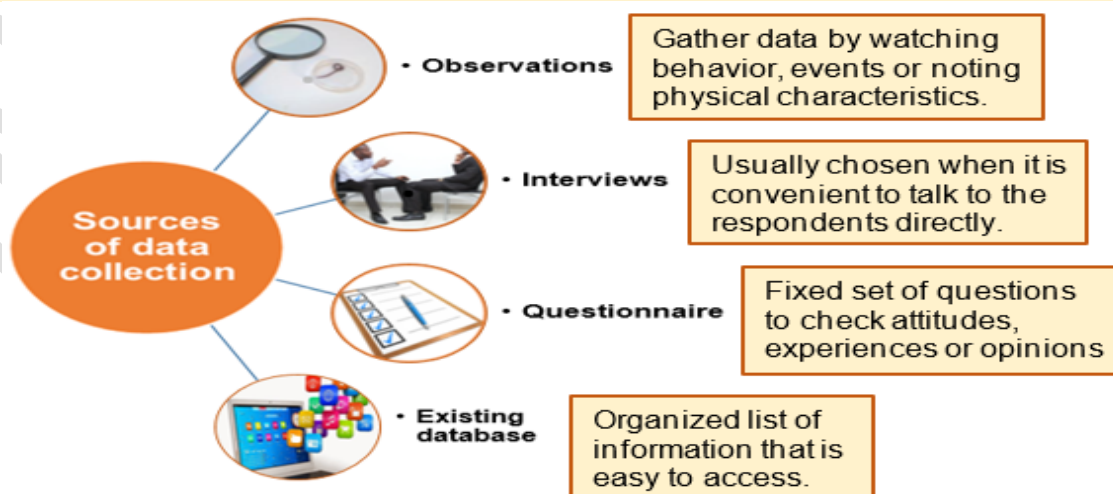
Data handling is a term used to describe certain ways of trying to make sense of large collections of observations (data) about things in the real world. Data can be about many different things, for example, people's opinions on politics or the success rates of treating people with a certain kind of medicine. We use data to help us make decisions and solve problems about the world around us.

A complete data cycle involves the following:



6.1 Collecting data

Data collection is a systematic process of gathering and measuring information. The method of collecting data will help us to identify the most appropriate tool to be used when collecting the data. There are various sources of data collection.



Once one has a research idea in mind, one can start planning how to collect the data. This requires one to consider:

- questions one has to ask
- where to collect the data
- how to collect the data?
- who to collect the data from [whole population or sample]

Population refers to the whole group you are asking the question
Sample refers to a small number of the group that you think will represent the whole

Here are some research questions.

1. Copy the statements below and use a **P** to show which statement describes the population and an **S** to show which statement describes a sample of the population.

- (a) What percentage of plants in the vegetable patch is affected by disease?

All the plants in the vegetable garden	P
Every fourth or fifth plant in the vegetable patch.	S

- (b) How often do teenagers recycle plastic?

About 40 teenagers in the community	S
Every teenager in South Africa	P

- (c) How many hours of sleep do 10-year-olds in my community get per night?

All 10-year-olds in the community	P
About ten 10-year-olds in the community	S

Questionnaires

A **questionnaire** is a sheet with questions used to collect data from people. Each respondent in the sample completes a questionnaire. The questions on the sheet can be structured differently, for example:

- The questions may require “yes” or “no” answers.
- A selection of answers (multiple-choice answers) may be provided for respondents to choose from.
- The **respondents** may enter their own views or information on the questionnaire.

A respondent is a person from whom you collect data

The type of responses you need, for example a simple “yes” or “no” or more detailed information, depends on the data you intend to collect.

Example 1

Do you help with chores at home?

☐ Yes ☐ No

Example 2

Which of these chores do you help with?

☐ cleaning dishes ☐ washing clothes
☐ sweeping/vacuuming ☐ making beds

Example 3

How old are you?

☐ 5–8 years ☐ 9–11 years
☐ 12–15 years ☐ 16–19 years

Note in example 3 how all the ages from 5 to 19 are covered, but without any overlaps.

Example 4

Tick the box if you have:

- | | |
|---|---|
| 1 ☐ Running water inside your home | 6 ☐ A cell phone |
| 2 ☐ Electricity inside your home | 7 ☐ Access to a computer |
| 3 ☐ A radio at home | 8 ☐ Access to the internet |
| 4 ☐ A TV at home | 9 ☐ Access to a library |
| 5 ☐ A telephone at home | |

1. You think most people in your school get to school by bus. You want to investigate this by means of a survey. A multiple-choice questionnaire can be used to record your data.

Design a simple questionnaire for your survey, using multiple-choice questions. Your data must also include:

- (a) Grade
- (b) Gender
- (c) Transport use to come to school.

TRANSPORT SURVEY FOR YEAR 3 LEARNERS

We want to determine the most popular means of transport used by learners to school.

Please assist by answering a few questions.

Gender (Tick the correct box)	Female		Male	
--------------------------------------	---------------	--	-------------	--

Which year are you? (Tick the correct box)

Year 1	Year 2	Year 3	Year 4

Which type of transport do you use mostly to get to school? (Tick the correct box)

Walk	Bicycle	Bus	Car	Other

2. Write a suitable question with multiple-choice responses to find out the following information:

- (a) Gender
- (b) Age
- (c) What do teenagers spend their money on?
- (d) How much time do spend on homework every day?

NB: Any suitable questionnaire will be acceptable

3. Use the multiple-choice question that you developed in Question 2 to collect the data. Keep the results for the next section.

6.2 Organising data

Organizing data is a systematic way of bringing data together in a way that makes it easier to read. Tally marks (I) can be used to record data in different categories as shown in the callout. Tallies are grouped in to five to make it easier to count them quickly.

Tally marks
A count of three = III
A count of five = IIII
A count of seven = IIII II

Example

Litter that is thrown all over the neighborhood troubles a community. Year 4 class learners were asked to find out the type of litter available in their community and

identify where the litter was located. The following observation sheet was used by Tshiamo to collect information.

AN OBESRVATION SHEET ON LITTER IN THE COMMUNITY			
NAME: Tshiamo Barnes			CLASS: Year 3 b
TYPE OF LITTER	WHERE LOCATED	TYPE OF LITTER	WHERE LOCATED
Plastic	Streets	Chips paper	Schools
Sweet wraps	Shops	Bottles	Streets
Chips wraps	Shops	Paper	Playgrounds
Bottles	Playgrounds	Cans	Streets
Cans	Playgrounds	Sweet wraps	Streets
Paper	Shops	Cans	Playgrounds
Paper	Schools	Sweet wraps	Playgrounds
Bottles	Streets	Paper	Schools
Plastic	Shops	Sweet wraps	Playgrounds
Paper	Street	Chips wraps	Playgrounds
Chips wraps	Schools	Sweet wraps	Schools
Chips wraps	Shops	Cans	Playgrounds

1. Use the information provided to construct a tally table for the type of litter identified.

Litter identified	Tally	Frequency
Plastic	//	2
Sweet wraps	///	5
Chips wraps	///	5
Bottles	///	3
Paper	///	5
Cans	///	4

2. Use the information provided to construct a tally table for the location where litter was identified.

Where litter was found	Tally	Frequency
Streets	/// /	6
Shops	///	5
Playground	/// ///	8
Schools	///	5

3. Copy and complete the combined tally table for the information given. The first five (5) has been filled for you.

Type of litter \ Where found	Street	Shops	Playground	Schools	FREQUENCY
Plastic	I	/			2
Sweets wraps	/	I	//	/	5
Chips wraps		//	/	//	5
Bottles	//		I		3
Cans	/		///		4
Paper	/	/	/	//	5
TOTAL	6	5	8	5	24

4. Add the bottom totals together. What do you get? 24
5. Add the vertical totals together. What do you get? 24

Can you see that the two totals are the same? We use the fact that the two totals are the same to check that we have recorded the numbers correctly.

6. Use the completed frequency table to answer the following questions.

(a) How many items of litter were identified?

There were 6 items identified

(b) Where was the most litter found?

Most litter was found at the playground.

(c) Where were the cool drink cans found?

Cool drink cans were found on the street and on the playground.

(d) Where were most sweet papers found?

Playground

(e) Where was the scrap paper found?

At the school

(f) How many cool drink cans were found in the playground?

There were three cans

(g) Where would be the best place to put an extra rubbish bin? Give a reason for your answer.

At the playgrounds since most litter was collected there.

Frequency table for large amounts of data.

A frequency table has rows and columns. When the set of data values is spread out, it is difficult to set up a frequency table for every data value as there will be too many rows.

The best way to summarise large amounts of data in a table or graph is to group the possible options together into certain intervals or categories. When analyzing marks, instead of having 100 rows of data of marks out of 100, one may limit it to five (5) rows by grouping scores together like this: scores from **0 – 20; 21 – 40; 41 – 60; 61 – 80 and 81 – 100.**

Example

Summarise marks obtained by 30 learners in an examination.

Name	Marks	Name	Marks	Name	Marks
Simon	60	Lerato	55	Zola	53
Elise	47	Siyabulela	65	Tshiamo	49
Manana	79	Alicia	45	Edward	91
Laura	64	Betty	30	Tefo	60
Piet	59	Susan	85	Dimakatso	63
Nhlanhla	77	Sibongile	25	John	46
Naledi	51	Jakob	73	Mark	45
Patricia	19	Zollie	47	Sipho	33
Zebon	90	Koos	79	Azwindini	85
Julius	60	Dumisile	16	Penelope	34

Frequency Table

Marks	Tallies	Frequency	From this table it is easy to see that most learners scored between 41% and 60% for the exam. Six learners failed the exam, because they scored between 0% and 40% and four learners got distinctions , because they scored between 81% and 100%
0 – 20	II	2	
21 – 40	IIII	4	
41 – 60	IIII IIII II	13	
61 – 80	IIII II	7	
81 – 100	IIII	4	

1. The number of calls from motorists per day for roadside service was recorded for a month. The results were as follows:

28	70	100	81	122	40	75	217	145
104	130	187	97	120	113	75	86	90
123	80	68	100	90	174	82	120	194
109	140	170	120					



Set up a frequency table for this set of data values, using grouped data, grouped in five groups with intervals of 40.

Number of calls	Tally	Frequency
0 – 39	I	1
40 – 79	IIII	5
80 – 119	IIII IIII II	12
120 – 159	IIII III	8
160 – 199	IIII	4
200 – 239	I	1

2. Consider the following data set.

36	57	31	45	78	15	56
88	37	57	3	41	48	42
63	66	18	32	59	77	62
32	60	73	49	82	77	37
39	92	22	42	54	25	97

- (a) Sort the data in increasing (ascending) order.

3; 15; 18; 22; 25; 31; 32; 32; 36; 37; 37; 39; 41; 42; 42; 45; 48; 49; 54; 57;
57; 56; 59; 60; 62; 63; 66; 73; 77; 77; 78; 82; 88; 92; 97

- (b) Make a frequency table for this data with classes: 0 – 19; 20 – 39; 40 – 59; 60 – 79; and 80 – 99.

Classes	Tally	Frequency
0 – 19	///	3
20 – 39	//// ////	9
40 – 59	//// //// /	11
60 – 79	//// ///	8
80 – 99	////	4

- (c) Make another frequency table for this data with classes: 0 – 50 and 51–99.

Classes	Tally	Frequency
0 – 50	//// //// //// ///	18
51 – 99	//// //// //// //	17

- (d) Make a frequency table for this data with classes 0 – 9; 10 – 19; 20 – 29; . . . 80 – 89 and 90 – 99.

Classes	Tally	Frequency
0 – 9	/	1
10 – 19	//	2
20 – 29	//	2
30 – 39	//// //	7
40 – 49	//// /	6
50 – 59	////	5
60 – 69	////	4
70 – 79	////	4
80 – 89	//	2
90 – 99	//	2
TOTAL		35

- (e) Discuss the advantages and disadvantages of each of the frequency table.

Any suitable answer

(f) What do you learn from each?

Any suitable answer.

(g) Overall, which one is the most informative? Why?

b is the most informative because it has manageable classes. The information is represented much better than the other two frequencies.

(h) Write a short paragraph in which you discuss what your frequency tables reveal.

They summarise the raw data sets as displayed.

3. The data shows the mass of 40 students in a class to the nearest kg. Construct a frequency table for the data using appropriate class intervals.

55	60	75	49	52	70	48	64	66	76
57	58	65	62	71	73	54	57	76	61
55	69	71	61	53	59	51	78	63	56
64	63	76	63	67	72	78	62	76	71

(a) How will we group these numbers into class intervals? What do you suggest? Discuss with others.

Any suitable answer.

(b) Now look at this method.

Smallest value = 48

Highest value = 78

Range = highest value – smallest value

$$= 78 - 48 = 30$$

Range is the difference between the smallest value and the highest value.

(c) Now we decide that we want five class intervals.

Therefore: $28 \approx 30$ (rounded off to the next 10)

$$\frac{30}{5} = 6$$

Now we can construct a table with three columns, and then write the date groups or class intervals in the first column.

INTERVAL	TALLIES	FREQUENCY
45 – 51	///	3
52 – 58	/// ///	9
59 – 65	/// /// ///	13
66 – 72	/// /	6
73 – 79	/// ///	9
TOTAL		40

6.3 Summarising data

Measures of central tendency and measures of spread and dispersion

A measure of central tendency is a single value that attempts to show a central position of a set of data. There are three types of measures of central tendency i.e. mode, median and mean.

Mode

The mode is the data value that appears most often in a set of data. No calculation is needed to find the mode. You just find the value that appears most frequently. If no number appears more than the other numbers, then there is no mode. It is important to arrange data in ascending or descending order in order to easily identify the mode.

Example.

What is the mode in the following data set?

150; 152; 143; 146; 135; 145; 151; 139; 141; 161; 158; 148; 144; 146; 155; 159; 165; 149; 139; 153; 146.

Let us arrange the data from small to biggest.

It is very important to cancel each number you have used when ordering numbers. Three have been cancelled so that numbers are not omitted.

mode

135; 139; 139; 141; 143; 144; 145; ~~146~~; ~~146~~; ~~146~~; 148; 149; 150; 151; 152; 153; 155; 158; 159; 161; 165

The value that appears the most is 146. Therefore, the mode for this data set is 146.

4. What is the mode of these numbers: 75; 78; 75; 71; 78; 25; 75; 29?

25; 29; 71; 75; 75; 75; 78; 78

Therefore, the mode is 75

Median

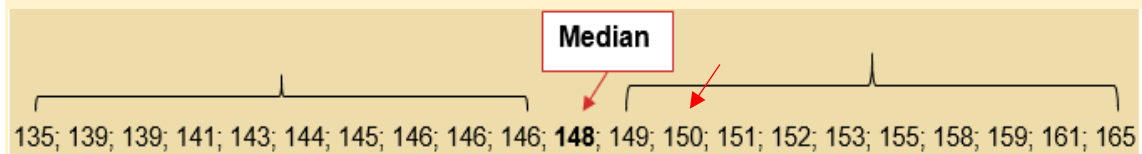
The median is the middle number in the data set. To determine the median, you have to write all the numbers in the data set from the smallest to the highest and the number in the middle will be your median. If there is more than one number in the middle (i.e. if the data contains an even number of data values) add the two numbers in the middle and divide the answer by 2.

Example

What is the median in the following data set?

150; 152; 143; 146; 135; 145; 151; 139; 141; 161; 158; 148; 144; 146; 155; 159; 165; 149; 139; 153; 146.

Let us arrange the data from small to biggest.



Therefore, the median for this data set is 148

5. What is the median of these numbers: 75; 78; 75; 71; 78; 25; 75; 29?

25; 29; 71; 75; 75; 75; 78; 78

Therefore, the median is $\frac{75+75}{2} = 75$

Mean

The mean or the average is the most common measure of central tendency that is used, but it can be easily influenced by high or low numbers in the data set. It is calculated by adding all the values together and dividing by the number of values in the data set.

Example

What is the mean of the following data set?

135; 139; 139; 141; 143; 144; 145; 146; 146; 146; 148; 149; 150; 151; 152; 153;
155; 158; 159; 161; 165

$$\text{Mean} = \frac{\text{sum of the values}}{\text{number of values}}$$

=

$$\frac{135+139+139+141+143+144+145+146+146+146+148+149+150+151+152+155+158+159+161+165}{21}$$

$$= \frac{3\,125}{21}$$

$$= 148,8$$

The mean or average is not always a whole number

Therefore, the mean of this data set is 148,8.

6. What is the mean of these numbers: 18; 12; 10, 10?

$$\frac{10 + 10 + 12 + 18}{4}$$

$$\frac{50}{4} = 12,5$$

Therefore, the mean is 12,5

Range

The range of a data set is the difference between the highest value and the lowest value. For example, the range of the data 18; 12; 10, 10, 11 is: $18 - 10 = 8$.

The bigger the range, the more the data is spread out. The smaller the range, the more the data is clustered around similar values

7. Five (5) children have heights of 138 cm; 135 cm; 140 cm; 139 cm; and 141 cm. What is the range of their heights?

The range is $= 141 \text{ cm} - 135 \text{ cm} = 6 \text{ cm}$

8. Sipho wrote seven Mathematics tests and got scores of 68, 71, 71, 84, 53, 62 and 67. What is the mean, median and the mode of these scores?

Mean

$$\frac{53 + 62 + 67 + 68 + 71 + 71 + 84}{7} = \frac{476}{7} = 68$$

Therefore, the mean is 68

Median

53; 62; 67; **68**; 71; 71; 84

Therefore, the median is 68

Mode

53; 62; 67; 68; **71; 71**; 84

Therefore, the mode is 71

9. Use the following data set to calculate the range, mode, median and the mean.

3; 13; 7; 5; 21; 23; 39; 23; 40; 23; 14; 12; 56; 23; 29

Range

3; 5; 7; 12; 13; 14; 21; 23; 23; 23; 23; 29; 39; 40; 56

Range = $56 - 3 = 53$

Therefore, the range is 53

Mode:

3; 5; 7; 12; 13; 14; 21; **23; 23; 23; 23**; 29; 39; 40; 56

Therefore, the mode is 23

Median

3; 5; 7; 12; 13; 14; 21; **23**; 23; 23; 23; 29; 39; 40; 56

Therefore, the median is 23

Mean

$$\frac{3 + 5 + 7 + 12 + 13 + 14 + 21 + 23 + 23 + 23 + 23 + 29 + 39 + 40 + 56}{15} = \frac{331}{15} = 22,06$$

Therefore, the mean is 22,06

10. The mean of three (3) numbers is 8. Two of the numbers are 11 and 7. What is the third number? ■

$$\frac{\blacksquare + 7 + 11}{3}$$

= 8

Therefore, the third number is 6 because $24 \div 3 = 8$

11. The cost of five (5) cakes is R28, R19, R45; R45 and R15. What is the median cost?

R15 ; R19; **R28**; R45; R45

= R28

Therefore, the median is R28

12. What is the median of these numbers: 2,4 ; 2,8 ; 2,3 ; 2,9 ; 2,9?

2,3 ; 2,4 ; **2,8** ; 2,9 ; 2,9

= 2,8

Therefore, the median is 2,8

6.4 Representing data

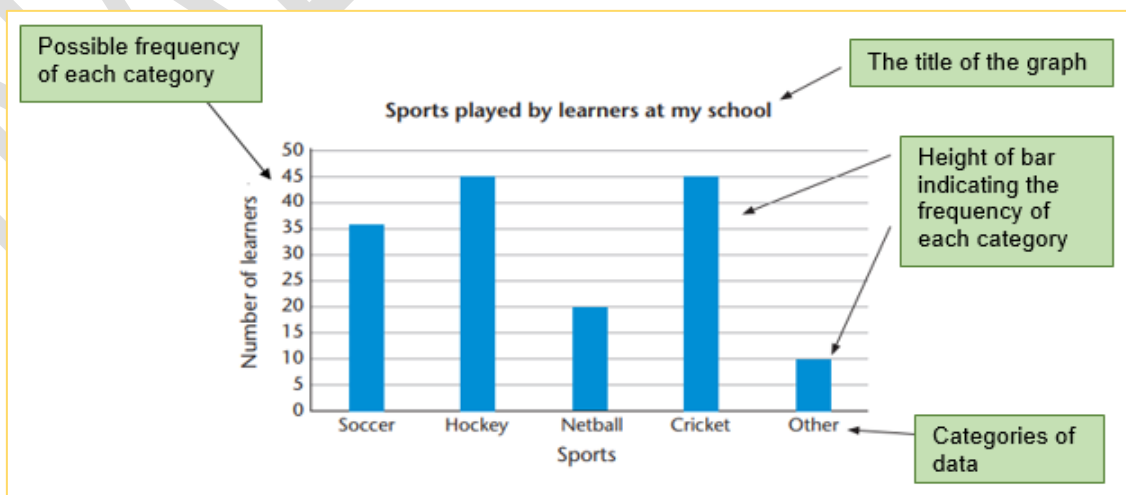
Data can be represented using different types of graphs. They help with displaying and reporting data and make it easier to report patterns and relationships. We will be working with three graphs i.e. bar graphs, double bar graphs and pie charts.

Bar graph

A bar graph usually shows categories (or classes) of data along the horizontal axis, and the frequency of each category along the vertical axis.

A bar graph is a visual display used to compare the frequency of occurrence of different characteristics of data.

This type of display allows us to compare groups of data and make generalisations about the data. Features of bar graphs are shown.



Steps in drawing a bar graph

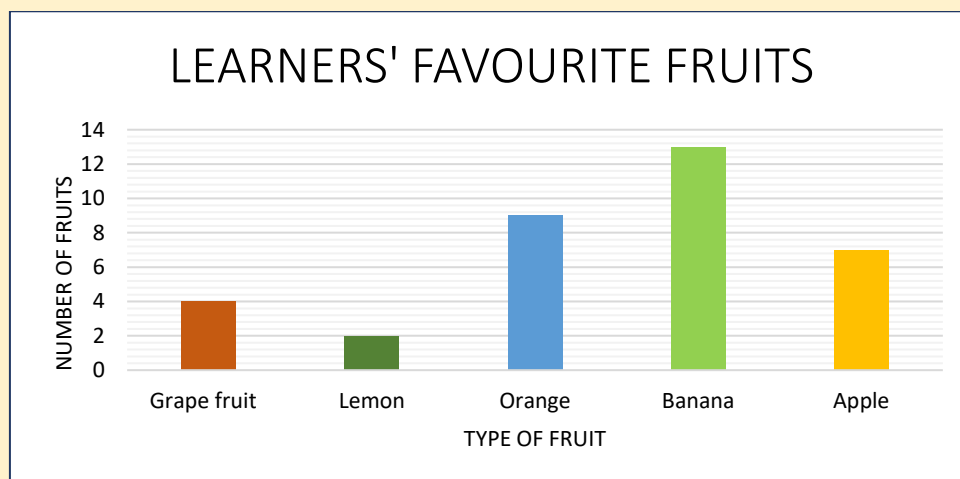
- To draw a bar graph you will have to start with your frequency table.
- From the frequency table, decide on the range and scale of the vertical axis and the horizontal axis
- Draw the vertical and horizontal axes and label them
- Write the title of the graph at the top
- Mark the data on the graph for each data group and draw the bar
- Add the colour or shading of the bar to the key (legend), if required to show.

Example

1. Use the following frequency table to represent data in a bar graph

Favourite fruit	Tallies	Frequency
Grape fruit		4
Lemon		2
Orange		9
Banana		13
Apple		7

Solution:



2. Answer the following questions based on the bar graph.

(a) Which fruit do most learners prefer?

Banana

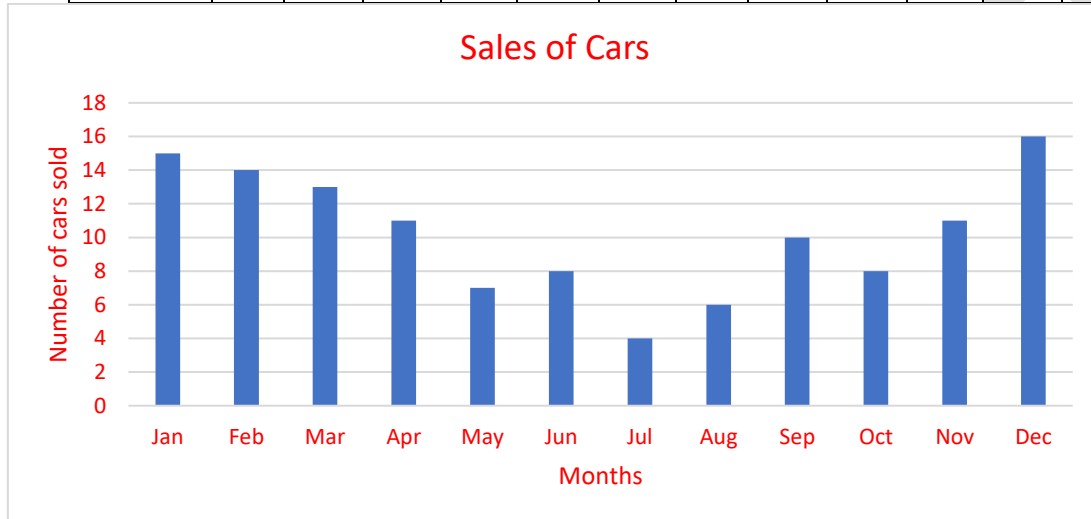
(b) What is the least favourite fruit?

Lemon

(c) If there are no bananas in the shop, which fruit will, learners most likely buy? **Orange**

3. The following table shows the sales of cars per month at a second-hand car dealership. Create a bar graph for the data.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Cars sold	15	14	13	11	7	8	4	6	10	8	11	16



- (a) During which month were most cars sold? **December**
- (b) During which month were least cars sold? What could be the possible reason for less sales.

December. Any suitable answer is acceptable for the reason e.g. Stokvels, bonuses.

- (c) How many cars were sold in May and June altogether?

7 cars + 8 cars

15 cars sold in May and June altogether

- (d) How many cars were sold during the year?

$15 + 14 + 13 + 11 + 7 + 8 + 4 + 6 + 10 + 8 + 11 + 16 = 123$

There were 123 cars that were sold that year

Double bar graph

A double bar graph shows two sets of data for each category (or class). For example, the double bar graph below shows data collected from girls for each category, and data collected from boys for each category

Example

The following frequency table shows the number of men and women that visited the Snake Park. Construct a side-by-side double bar graph to represent the information.

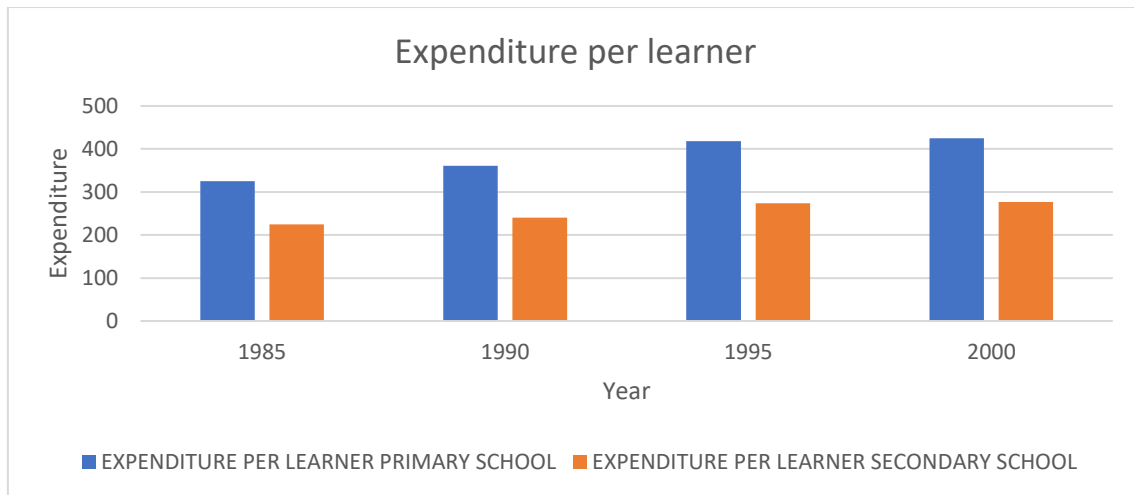
VISITORS TO THE SNAKE PARK

MONTH	JAN	FEB	MAR	APR	MAY	JUNE
MEN	300	150	550	250	100	200
WOMEN	200	170	430	300	250	50



4. The following table represents the expenditure per learner for primary and high schools. Draw a double bar graph.

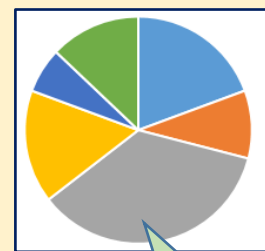
EXPENDITURE PER LEARNER		
YEAR	PRIMARY SCHOOL	SECONDARY SCHOOL
1985	325	225
1990	361	240
1995	418	274
2000	425	277



Pie chart

A pie chart is a circular graph in which the circle is divided into sectors. Each sector visually represents an item in a data set to match the amount of the item as a percentage or fraction of the total data set. (the whole circle) Pie charts are useful to compare different parts of a whole amount. They are often used in media to show clear and important differences, but they cannot show shape and spread of data

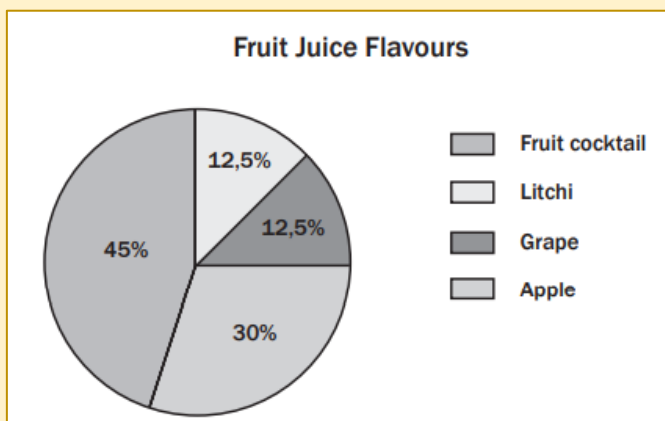
This pie chart shows six categories of data.



Sector

Example

The pie chart below shows a survey of the different types of the favourite fruit juice flavours that are normally bought by a group of 120 learners from Nduku High school during their lunchtime.



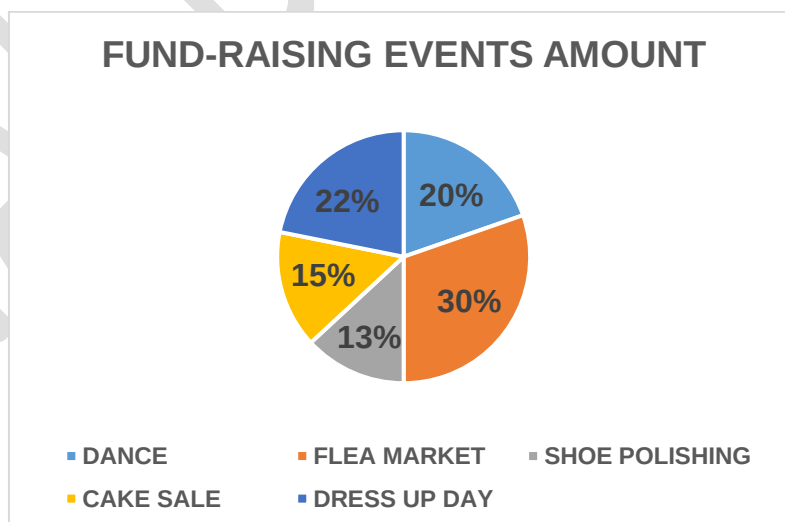
The key (or legend) shows the category that each colour represents

- Calculate how many learners chose each type of juice.
 - 45% of 120 learners = 54 learners who chose fruit cocktail.
 - 30% of 120 = 36 learners who chose apple.
 - 12,5% of 120 learners = 15 learners who chose grape.
 - 12,5% of 120 = 15 learners who chose litchi
- In what way does the pie chart work better than a bar graph to represent this data?
 - The pie chart is a simple, visual representation that works well for representing percentages. A pie chart allows us to see at a glance the relative proportions of the learners who prefer each flavour.
- What information would a bar graph give you that this pie chart does not?
 - The number of learners who prefer each flavour.

Study the following.

5. Pauline is the treasurer of the Year 3 class at the Sukazi High School. Her class raised money for activities through various events. The total amount raised was R 1 830. Pauline used a pie graph to show the amount of money each event raised.

Study the graph and answer the following question.



- (a) What percentage of the total money was raised at the flea market?

30 %

(b) How much money was raised at the flea market?

$$\frac{30}{100} \times R1\ 830 = R549 \text{ was raised at the flea market}$$

(c) What percentage of the total money was raised during the dress up day?

$$\frac{20}{100} \times R1\ 830 = R366$$

(d) How much money was raised at the dance?

$$\frac{22}{100} \times R1\ 830 = R402,60$$

(e) Which event gave them more money?

The Flea market gave them more money.

(f) Calculate the difference between the money raised at the cake sale and on shoe polishing.

Cake sale	Shoe polishing	Therefore, the difference is
$\frac{15}{100} \times R1\ 830$ = R274,50	$\frac{13}{100} \times R1\ 830$ = R237,90	$R274,50 - R237,90 = R36,60$

6.5 Interpreting and reporting data

Data **analysis** and **interpretation** is a process of assigning meaning to the collected information and determining the conclusion, significance and implications of the findings. There are four steps to data interpretation i.e. collect data required; develop findings; develop conclusions and develop recommendations.

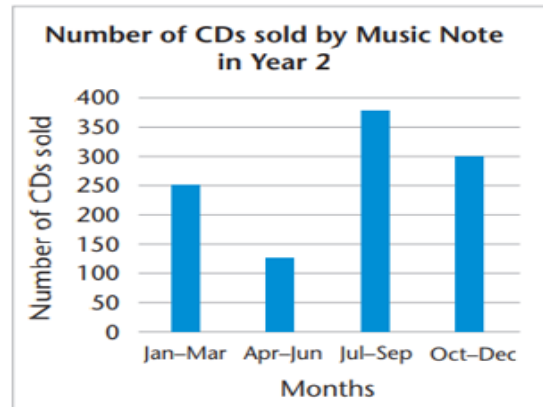
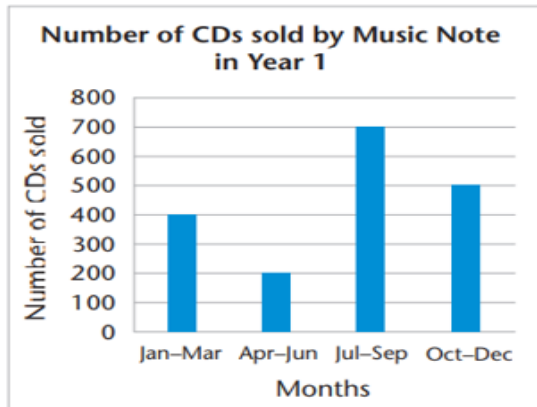
How do you interpret data?

There are four steps to data interpretation.

- Collect the information you need.
- Develop findings
- Develop conclusions
- Develop recommendations.

Example

Study the following graphs and answer the questions that follow.



(a) What does each of the graphs show?

The graphs show the number of CD's sold by Music Note in Year 1 and 2.

(b) How many CDs were sold from July to September of Year 1?

700 CDs were sold.

(c) How many CDs were sold from July to September of Year 2?

375 CDs were sold.

(d) The heights of the bars indicate that Music Note sold more CDs in October to December of Year 2 than in the same months of Year 1. Is this the case?

No, 500 CDs were sold in Oct – Dec in Year 1 and 300 CDs were sold during that period in Year 2. The scales used in the two graphs are not the same so the graph for Year 2 shows some bias.

(e) How many CDs were sold altogether in Year 1?

1 800 CDs were sold in Year 1.

(f) How many CDs were sold altogether in Year 2?

1 075 CDs were sold in Year 2.

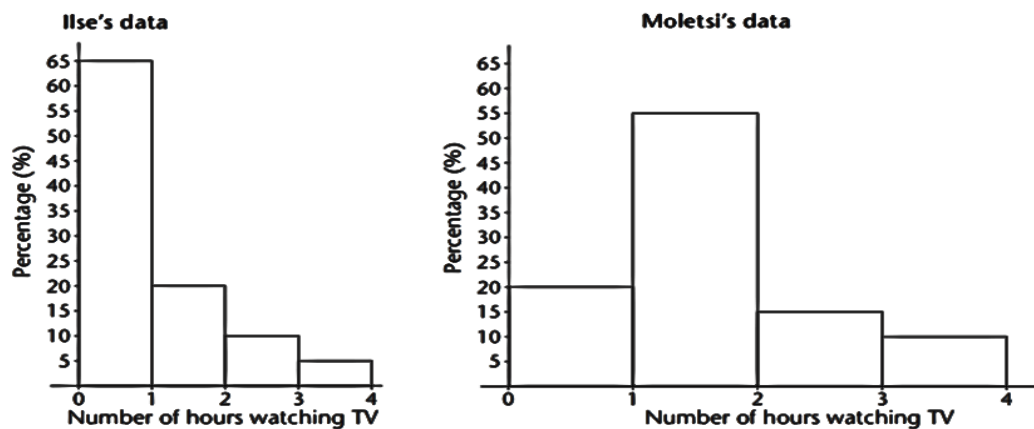
(g) Explain why the heights of the bars seem to indicate that Music Note sold more or less the same number of CDs in both years, which is not true.

The scales used for Year 1 and Year 2 graphs are not the same. In Year 1, the scale is 0 – 100 and for Year 2, the scale is 0 – 50.

1. Study the following graphs and answer the questions that follow.

Ise and Moletsi wanted to find out more about the number of hours people spend watching TV on a particular public holiday. Ise did her survey on the public holiday from 13:00 to 15:00. She visited a supermarket and asked adult respondents to complete her questionnaire. Moletsi did his survey on

the same day from 17:00 to 19:00. He went from door to door in his neighbourhood and asked the children to complete his questionnaire.



- (a) According to Ilse's data, how long did most people spend watching TV on the public holiday?

1 hour

- (b) According to Moletsli's data, how long did most people spend watching TV on the public holiday?

2 hours

- (c) Write a paragraph to summarise and compare Ilse's data and Moletsli's data.

Ilse's data shows that most people (65%) spent up to one hour watching TV on the public holiday, and only 5% spent between three- and four-hours watching TV. Moletsli's data shows that fewer people (20%) spent up to one hour watching TV, and most people (55%) spent between one- and two-hours watching TV.

- (d) How could the time when the data was collected have affected the data?

Ilse collected her data earlier in the day than Moletsli did. If she collected the data later in the afternoon, people may have had more time to watch TV on that day, and the data could possibly have shown that people spent more time watching TV on that particular public holiday.

- (e) How could the place where the data was collected have affected the data?

Ilse collected data from people at a supermarket. These respondents would have had less time at home to watch TV on that day. Moletsi's respondents were all at home at the time of data collection, so they would most likely have had more opportunity to watch more TV.

- (f) How could the people from whom data was collected have affected the data?

Adults and children probably spend different amounts of time watching TV. Ilse's respondents were adults, who would perhaps generally have less time to watch TV than children. Moletsi's respondents were children, who would perhaps generally spend more time watching TV than their parents.

1. The following table shows the Mathematics marks of Class A and Class B

Class A	Class B
94, 42, 23, 67, 67, 68, 13, 53, 44, 34, 64, 69, 50, 31, 91, 40, 10, 30, 49, 61	74, 26, 65, 45, 71, 77, 58, 35, 39, 45, 68, 45, 57, 62, 29, 55, 23, 56, 38, 36, 50, 64, 58, 32, 42

- (a) Find the range of each set of data.

Class A = $94 - 10 = 84$ and Class B = $77 - 23 = 54$

- (b) What can you say about the two classes by looking at the range of marks?

Class A has a wider range of marks, meaning that the spread of marks is wider.

- (c) Copy and complete the table by calculating the mean (average) Mathematics mark for each class. You may use your calculator. Class
- | Total marks | Number of marks | Mean |
|-------------|-----------------|------|
| Class A | Class B | |

Class	Total marks	Number of marks	Mean
Class A	1000	20	50
Class B	1250	25	50

- (d) Compare the two sets of data using the means.

Both data sets have the same mean. From the means, it looks as though the learners in the two classes are performing at the same level.

- (e) Copy and complete the table by finding the median for each class. Class
Marks from highest to lowest Middle position Median Class A Class B

Class	Marks from Highest to lowest	Middle position	Median
Class A	94; 91; 69; 68; 6; 67; 64; 61; 53; 50; 49; 44; 42; 40; 34 31; 30; 23; 13; 10	10 and 11	$\frac{50+49}{2} = 49,5$
Class B	77; 74; 71; 68; 65; 64; 62; 58; 58; 57; 56; 55; 50; 45; 45; 45; 42; 39; 38; 36; 35; 32; 29; 26; 23	13	50

- (f) Compare the two sets of data using the medians.

From the medians, it looks as though the learners in the two classes are performing at more or less the same level.

- (g) Copy and complete the table by finding the mode for each class. Class
Highest frequency Mode Class A Class B

Class	Highest frequency	Mode
Class A	2	67
Class B	3	45

- (h) Compare the two sets of data using the mode.

From the modes, it looks like Class A is performing much better than Class B.

- (i) Which of the following do you think best represents each set of data: mean, median or mode? Explain your answer

The mode describes only the most frequent values in the set, and does not take into account all the values. The mean and median take into account all the values, so these probably represent the data better than the mode does

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